

ELECTROMAGNETISM

Y. Ding
November 6, 2025

CONTENTS

1	Introduction	6
1.1	Electromagnetism	6
1.2	Vector Calculus	7
1.2.1	Dot Products and Cross Products	7
1.2.2	Gradient, Divergence, and Curl	7
1.2.3	Product Rules	8
1.2.4	Second Derivatives	8
1.2.5	Integral Theorems	9
1.2.6	Curvilinear Coordinates	9
1.2.7	Useful Formulas	11
1.2.8	One More Useful Thing	11
2	Electrostatics	12
2.1	The Electric Field	12
2.1.1	Coulomb's Law	12
2.1.2	The Electric Field	12
2.2	The Divergence and Curl of Electric Fields	13
2.2.1	Electric Flux and Gauss's Law	13
2.2.2	The Curl of Electric Field	14
2.2.3	Useful Formulas and Results	14
2.3	Electric Potential	15
2.3.1	Introduction to Potential	15
2.3.2	Poisson's Equation and Laplace's Equation	16
2.3.3	The Potential of a Localized Charge Distribution	16
2.3.4	Boundary Conditions	17
2.3.5	Useful Formulas and Results	18
2.4	Work and Energy in Electrostatics	18
2.4.1	The Work It Takes to Move a Charge	18
2.4.2	The Energy of a Point Charge Distribution	18
2.4.3	The Energy of a Continuous Charge Distribution	19
2.5	Conductors	19
2.5.1	Basic Properties	20
2.5.2	Induced Charges	20

2.5.3	Surface Charge and the Force on a Conductor	21
2.5.4	Capacitors	21
2.5.5	Useful Formulas and Results	23
3	Potentials	24
3.1	Laplace's Equation	24
3.1.1	Laplace's Equation in Three Dimensions	24
3.1.2	Boundary Conditions and the First Uniqueness Theorem	25
3.1.3	Conductors and the Second Uniqueness Theorem	26
3.1.4	The Third Uniqueness Theorem	27
3.2	The Method of Images	27
3.2.1	The Classic Image Problem	27
3.2.2	Induced Surface Charge	28
3.2.3	Force and Energy	28
3.3	Separation of Variables	29
3.3.1	General Ideas	29
3.3.2	Cartesian Coordinates	29
3.3.3	Spherical Coordinates	33
3.4	Multipole Expansion	36
3.4.1	Approximate Potentials at Large Distances	36
3.4.2	The Monopole and Dipole Terms	37
3.4.3	Dipoles in an Electric Field	38
3.4.4	Comments on the Choice of Origin for Multipole Expansion	39
4	Electric Fields in Matter	41
4.1	Polarization	41
4.1.1	Induced Dipoles	41
4.1.2	Polarization	42
4.2	The Field of a Polarized Object	42
4.2.1	Bound Charges	42
4.2.2	Physical Interpretation of Bound Charges	44
4.3	The Electric Displacement	44
4.3.1	Gauss's Law in the Presence of Dielectrics	44
4.3.2	A Note on Electric Displacement	45
4.3.3	Boundary Conditions	45
4.4	Linear Dielectrics	45
4.4.1	Susceptibility, Permittivity, Dielectric Constant	45

4.4.2	Boundary Value Problems with Linear Dielectrics	47
4.4.3	Energy in Dielectric Systems	47
4.4.4	Forces on Dielectrics	48
5	Magnetostatics	49
5.1	The Lorentz Force Law	49
5.1.1	Magnetic Fields	49
5.1.2	Magnetic Forces	49
5.1.3	Currents	50
5.1.4	The Continuity Equation	51
5.2	The Biot-Savart Law	51
5.2.1	Steady Currents	51
5.2.2	The Magnetic Field of a Steady Current	52
5.3	The Divergence and Curl of Magnetic Field	52
5.3.1	The Divergence	52
5.3.2	The Curl	53
5.3.3	Ampère's Law	53
5.3.4	Comparison of Magnetostatics and Electrostatics	55
5.4	Magnetic Vector Potential	55
5.4.1	The Vector Potential	56
5.4.2	Boundary Conditions	57
5.4.3	Multipole Expansion of the Vector Potential	57
6	Magnetic Fields in Matter	58
7	Electrodynamics	59
7.1	Electromotive Force	59
7.1.1	Ohm's Law	59
7.1.2	Electromotive Force	60
7.1.3	Motional EMF	60
7.2	Electromagnetic Induction	62
7.2.1	Faraday's Law	62
7.2.2	Inductance	63
7.2.3	Energy in Magnetic Fields	64
7.3	Maxwell's Equations	65
7.3.1	Fixing Ampère's Law	65
7.3.2	Maxwell's Equations	66

8	Conservation Laws	67
8.1	Poynting's Theorem	67
8.2	Momentum	68
8.2.1	Maxwell's Stress Tensor	68
8.2.2	Conservation of Momentum	69
9	Electromagnetic Waves	71
9.1	Waves in One Dimension	71
9.1.1	The Wave Equation	71
9.1.2	Boundary Conditions	72
9.1.3	Polarization	73
9.2	Electromagnetic Waves in Vacuum	73
9.2.1	The Wave Equation for $\mathbf{E}$ and $\mathbf{B}$	73
9.2.2	Monochromatic Plane Waves	74
9.2.3	Energy and Momentum in Electromagnetic Waves	74
9.3	Electromagnetic Waves in Matter	75
9.3.1	Propagation in Linear Media	75
9.3.2	Reflection and Transmission at Normal Incidence	76
9.3.3	Reflection and Transmission at Oblique Incidence	77
9.4	Absorption and Dispersion	80
9.4.1	Electromagnetic Waves in Conductors	80
9.4.2	Reflection at a Conducting Surface	81
9.4.3	Dispersion	82
9.5	Guided Waves	82
9.5.1	Wave Guides	82
9.5.2	TE Waves in a Rectangular Wave Guide	84
9.5.3	The Coaxial Transmission Line	84
10	Potentials and Fields	85
10.1	The Potential Formulation	85
10.1.1	Scalar and Vector Potentials	85
10.1.2	Gauge Transformations	85
10.1.3	Coulomb Gauge and Lorenz Gauge	86
10.1.4	Lorentz Force Law in Potential Form	87
10.2	Continuous Distributions	87
10.3	Point Charges	89
10.3.1	Liénard-Wiechert Potentials	89

10.3.2	The Fields of a Moving Charge	90
11	Radiation	94
11.1	Power Radiated by a Point Charge	94
11.1.1	What is Radiation?	94
11.1.2	The Larmor Formula	94
11.1.3	The Liénard Formula	95
11.2	The Radiation Reaction	95
A	Theorems, Proofs and Derivations	96
A.1	Theorems	96
A.2	Electromagnetic Waves	97
A.2.1	Fresnel's Equations	97
A.2.2	Guided Waves	97
A.2.3	TE_{mn} Mode	97
A.3	The Potential Formulation	98
A.3.1	The Lorenz Gauge	98
A.3.2	Rate of Change of Energy in Potential Form	98
A.4	Liénard's Generalization	99

1 INTRODUCTION

1.1 Electromagnetism

Mechanics is about forces and how objects react to them. We can roughly divide mechanics into 4 parts according to their characteristics, specifically the speed and the size of the regime we are interested in:

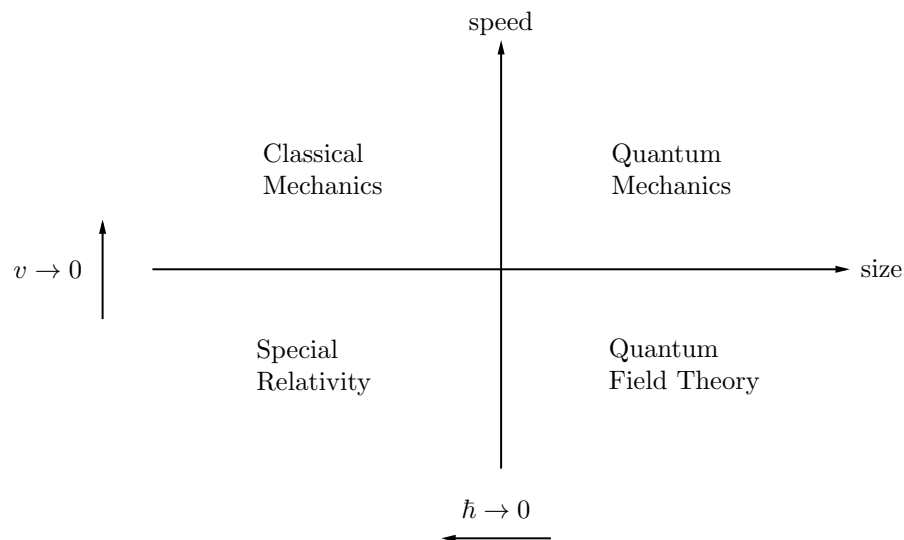


Figure 1.1: Four regimes of mechanics.

Electromagnetism lives in the classical regime, but it is related to other domains as well because

- Electromagnetism is naturally consistent with special relativity.
- As Newtonian (classical) mechanics goes to quantum mechanics for small systems, electromagnetism goes to quantum electrodynamics (QED).

Electromagnetism is one of the four fundamental forces. The other three forces are strong force, weak force, and gravity. Classified by range, electromagnetism is like gravity, both of which are long range forces (range ∞). By strength, electromagnetism is the second strongest force. It is about 100 times weaker than the strong force, which is a short range force that holds nuclei together.

In fact, most forces in our daily life, such as friction, normal force, tension, are governed by electromagnetism. Though electromagnetism and gravity are both long range forces and decay like $\sim r^{-2}$, their strength differs by an order of $\sim 10^{42}$. Then why do gravity and electromagnetism balance in our daily life? First, we live beside a huge mass, the Earth. Second, any naked charge is immediately attracted to opposite charges and becomes neutralized due to the strength of electromagnetism. Hence macroscopically, the net charge of any system is approximately zero.

Electromagnetism is basically the force between charges. Properties of “charges” include:

- Charges comes in two forms, positive and negative. Whether a charge is positive or negative is just a matter of convention, since we only know there are two classes. Opposite classes (charges) attract and like classes (charges) repel. By convention, electrons are negatively charged. Moreover, charges are additive—the total charge of a system is the sum of individual charges.

- Charge is conserved, i.e. charge cannot be created or destroyed. A powerful statement is that charge is *locally* conserved. That is, additional to the fact that charge of the universe is conserved, you cannot create net charges in any isolated system.
- Charge is quantized. They come in discrete lumps called the **elementary charge**, the charge of an electron. However, for most of classical electromagnetism, the quantization is not unimportant.

1.2 Vector Calculus

Vector calculus is the language of electromagnetism: it is useful in electromagnetism, and electromagnetism promoted the development of vector calculus. In this notes, all vectors are bolded: $\mathbf{E}$, $\mathbf{B}$, etc. The component of a vector $\mathbf{A}$ is written with subscripts indicating coordinates, e.g.

$$\mathbf{A} = (A_x, A_y, A_z).$$

1.2.1 Dot Products and Cross Products

The dot product of two vectors $\mathbf{A}$ and $\mathbf{B}$ is defined by

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$$

where θ is the angle between $\mathbf{A}$ and $\mathbf{B}$. Dot products are commutative and distributive. The cross product of $\mathbf{A}$ and $\mathbf{B}$ is defined by

$$\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a unit normal vector to both $\mathbf{A}$ and $\mathbf{B}$. Cross products are distributive and *anti*-commutative, but not associative.

Scalar Triple Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}. \quad (1.1)$$

Vector Triple Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}). \quad (1.2)$$

1.2.2 Gradient, Divergence, and Curl

The del (or nabla) operator is defined as (in Cartesian coordinates):

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}.$$

It is an operator, not a vector that can be multiplied. Many vector identities does not work for ∇ . It can act on a scalar field or a vector field in three ways:

- The **gradient**:

$$\nabla \phi = \hat{\mathbf{x}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial \phi}{\partial z}.$$

The gradient turns a scalar field ϕ into a vector field $\nabla\phi$. It has a close relation with the differential of a scalar field,

$$d\phi = \left(\hat{\mathbf{x}} \frac{\partial\phi}{\partial x} + \hat{\mathbf{y}} \frac{\partial\phi}{\partial y} + \hat{\mathbf{z}} \frac{\partial\phi}{\partial z} \right) \cdot (\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz) = (\nabla\phi) \cdot d\boldsymbol{\ell},$$

where $d\boldsymbol{\ell}$ is the line element in Cartesian coordinates.

- The **divergence**:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.$$

The divergence turns a vector field $\mathbf{A}$ into a scalar field $\nabla \cdot \mathbf{A}$. It is a measure of how much $\mathbf{A}$ spreads out (diverges) from a point. If a vector field has a nonzero divergence at some point, there will be a “source”, or a “sink”. We will see this in electrostatics.

- The **curl**:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ A_x & A_y & A_z \end{vmatrix} = \hat{\mathbf{x}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{\mathbf{y}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{\mathbf{z}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right).$$

The curl, as its name suggests, measures how much a vector field curls around some point. We will see this in magnetostatics.

1.2.3 Product Rules

∇ is a linear differential operator: for a scalar k , scalar fields ϕ , ψ , and vector fields $\mathbf{A}$ and $\mathbf{B}$

$$\nabla(\phi + \psi) = \nabla\phi + \nabla\psi, \quad \nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}, \quad \nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}.$$

$$\nabla(k\phi) = k\nabla\phi, \quad \nabla \cdot (k\mathbf{A}) = k\nabla \cdot \mathbf{A}, \quad \nabla \times (k\mathbf{A}) = k\nabla \times \mathbf{A}.$$

There are six product rules for ∇ :

- Two for gradients,

$$\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi, \tag{1.3}$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}, \tag{1.4}$$

- two for divergences,

$$\nabla \cdot (\phi\mathbf{A}) = \phi(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla\phi), \tag{1.5}$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \tag{1.6}$$

- and two for curls,

$$\nabla \times (\phi\mathbf{A}) = \phi(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla\phi) \tag{1.7}$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}). \tag{1.8}$$

1.2.4 Second Derivatives

We can apply ∇ twice to a scalar field or a vector field by using combinations of gradient, divergence, and curl. There are five ways to construct second derivatives:

- The divergence of a gradient:

$$\nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}.$$

This can be written as $\nabla^2 \phi$. The operator ∇^2 is called the **Laplacian**.

- The curl of a gradient is always zero:

$$\nabla \times (\nabla \phi) = 0.$$

- The gradient of the divergence $\nabla(\nabla \cdot \mathbf{A})$ is seldomly useful.

- The divergence of a curl is always zero:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

- The curl of a curl:

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

The second term is a laplacian of a vector, which just takes the laplacian of each components of a vector.

1.2.5 Integral Theorems

There are three integral theorems, each related to one of the gradient, divergence, and curl:

- The **fundamental theorem of line integrals**:

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla \phi) \cdot d\boldsymbol{\ell} = \phi(\mathbf{b}) - \phi(\mathbf{a}) \quad (1.9)$$

There are two corollary to this: 1. $\int_{\mathbf{a}}^{\mathbf{b}} (\nabla \phi) \cdot d\boldsymbol{\ell}$ is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$. 2. $\oint (\nabla \phi) \cdot d\boldsymbol{\ell} = 0$.

- The **divergence theorem**:

$$\int_V (\nabla \cdot \mathbf{A}) dV = \oint_{\partial V} \mathbf{A} \cdot d\mathbf{a}, \quad (1.10)$$

where $d\mathbf{a}$ is the vector surface element (with direction normal to the surface) and ∂V is the boundary of V . Geometrically, it is saying that if there exists a source in space, there will be some flux flowing out a surface enclosing that source. The divergence theorem is particularly useful in electrostatics.

- **Stokes' theorem**:

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_{\partial S} \mathbf{A} \cdot d\boldsymbol{\ell}, \quad (1.11)$$

where ∂S is the boundary of S . Geometrically, Stokes' theorem says that if there are some curl of vector field $\mathbf{B}$ on a surface, you can find its circulation on the boundary of the surface. We will encounter this very often in magnetostatics.

Note that all these integral theorems turn some kind of derivative a function to the boundary value of that function: line integral of gradient $\leftrightarrow$ value at endpoints; surface integral of curl $\leftrightarrow$ value on boundary loops; volume integral of divergence $\leftrightarrow$ value on boundary surfaces.

1.2.6 Curvilinear Coordinates

Here are some formulas for integral elements, gradients, divergences, curls, and Laplacians in terms of spherical coordinates and cylindrical coordinates:

Spherical Coordinates

- Coordinates: (r, θ, ϕ) , where $r \in [0, \infty)$ is the distance from the origin; $\theta \in [0, \pi]$ is the angle down from the z -axis and is called the **polar angle**; $\phi \in [0, 2\pi]$ is the angle around from the x -axis in counterclockwise direction and is called the **azimuthal angle**.

- Relation to Cartesian coordinates:

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

- Unit vectors for $\mathbf{A} = A_r \hat{\mathbf{r}} + A_\theta \hat{\boldsymbol{\theta}} + A_\phi \hat{\boldsymbol{\phi}}$:

$$\begin{cases} \hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}, \\ \hat{\boldsymbol{\theta}} = \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}, \\ \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}. \end{cases}$$

- Line element and volume element:

$$d\boldsymbol{\ell} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}, \quad dV = r^2 \sin \theta dr d\theta d\phi.$$

- The gradient, divergence, curl, and Laplacian:

$$\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\boldsymbol{\phi}}.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}.$$

$$\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial}{\partial \phi} A_\theta \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\boldsymbol{\theta}} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}.$$

$$\nabla^2 T = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2}.$$

Cylindrical Coordinates

- Coordinates: (r, ϕ, z) , where r is the distance from the z -axis, and ϕ is still the azimuthal angle.

- Relation to Cartesian coordinates:

$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z.$$

- Unit vectors for $\mathbf{A} = A_r \hat{\mathbf{r}} + A_\phi \hat{\boldsymbol{\phi}} + A_z \hat{\mathbf{z}}$:

$$\begin{cases} \hat{\mathbf{r}} = \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}, \\ \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}, \\ \hat{\mathbf{z}} = \hat{\mathbf{z}}. \end{cases}$$

- Line element and volume element:

$$d\boldsymbol{\ell} = dr \hat{\mathbf{r}} + r d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}, \quad dV = r dr d\phi dz.$$

- The gradient, divergence, curl, and Laplacian:

$$\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}.$$

$$\begin{aligned}\nabla \cdot \mathbf{A} &= \frac{1}{r} \frac{\partial}{\partial r}(rA_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}.\end{aligned}$$

$$\nabla \times \mathbf{A} = \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{\mathbf{r}} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \hat{\phi} + \frac{1}{r} \left[\frac{\partial}{\partial r}(rA_\phi) - \frac{\partial A_r}{\partial \phi} \right] \hat{\mathbf{z}}.$$

$$\nabla^2 T = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}.$$

1.2.7 Useful Formulas

Griffiths Problem 1.36

$$\begin{aligned}\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} &= \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} + \oint_{\partial S} f \mathbf{A} \cdot d\boldsymbol{\ell}.\end{aligned}$$

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) dV = \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) dV + \oint_{\partial V} (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a}.$$

Scaling of a Delta function

$$\delta(kx) = \frac{1}{|k|} \delta(x).$$

Griffiths Problem 1.61 Volume integral of a gradient:

$$\int_V (\nabla \phi) dV = \oint_{\partial V} \phi d\mathbf{a}.$$

Volume integral of a curl:

$$\int_V (\nabla \times \mathbf{A}) dV = - \oint_{\partial V} \mathbf{A} \times d\mathbf{a}.$$

Green's identity:

$$\int_V [\phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi)] dV = \oint_{\partial V} (\phi \nabla \psi) \cdot d\mathbf{a}.$$

Green's second identity:

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV = \oint_{\partial V} (\phi \nabla \psi - \psi \nabla \phi) \cdot d\mathbf{a}.$$

Surface integral of a gradient:

$$\int_S \nabla \phi \times d\mathbf{a} = - \oint_{\partial S} \phi d\boldsymbol{\ell}.$$

1.2.8 One More Useful Thing

We are going to use a lot of $1/r$ and $1/r^2$'s. It is useful to remember that

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) = 4\pi \delta^3(\mathbf{z}), \quad \nabla \times \left(\frac{\hat{\mathbf{z}}}{z^2} \right) = 0, \quad \text{and} \quad \nabla \left(\frac{1}{z} \right) = -\frac{\hat{\mathbf{z}}}{z^2},$$

where $\mathbf{z} = \mathbf{r} - \mathbf{r}'$, $z = |\mathbf{r} - \mathbf{r}'|$ and ∇ is with respect to $\mathbf{r}$ while $\mathbf{r}'$ is held constant.

2 ELECTROSTATICS

2.1 The Electric Field

The fundamental problem of electrodynamics is: what is the force on a charge Q (a **test charge**) due to other charge distributions, $q_1, q_2, q_3, \dots$, given their positions? Additionally, we want to know the trajectory of Q subject to other charges. Fortunately, electromagnetism obeys the **principle of superposition**: the interaction between any two charges is completely unaffected by the presence of others. Suppose we know the force on charge Q due to q_1 alone is $\mathbf{F}_1$, and the force due to q_2 alone is $\mathbf{F}_2$. The net force on Q due to both charges is just $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2$. To simplify our problem at the first stage, in **electrostatics**, all source charges are stationary and only the test charge is able to move around.

2.1.1 Coulomb's Law

The force on a test charge Q due to a single point charge q at rest a distance z away is given by **Coulomb's law**,

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{Qq}{z^2} \hat{\mathbf{z}}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ (in SI units) is called the **permittivity of free space**. If q is at position $\mathbf{r}'$ and Q is at position $\mathbf{r}$, their separation vector is

$$\mathbf{z} = \mathbf{r} - \mathbf{r}', \quad z = |\mathbf{r} - \mathbf{r}'|, \quad \hat{\mathbf{z}} = \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|},$$

where z is the magnitude and $\hat{\mathbf{z}}$ is the direction of the separation vector.

2.1.2 The Electric Field

Back to the fundamental problem of electrostatics, suppose we have point charges $q_1, q_2, \dots, q_n$ at distance $z_1, z_2, \dots, z_n$ from Q . By principle of superposition, the total force on Q is

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{Qq_1}{z_1^2} \hat{\mathbf{z}}_1 + \frac{Qq_2}{z_2^2} \hat{\mathbf{z}}_2 + \dots \right) = \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{z_1^2} \hat{\mathbf{z}}_1 + \frac{q_2}{z_2^2} \hat{\mathbf{z}}_2 + \dots \right),$$

or

$$\mathbf{F} = QE \quad (2.2)$$

where

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{z_i^2} \hat{\mathbf{z}}_i. \quad (2.3)$$

The quantity $\mathbf{E}$ is called the **electric field** of the source charges. There are two things to notice. First, the electric field is a position of $\mathbf{r}$ because the separation vectors $\mathbf{z}_i$ depends on the location of the field point. Second, the electric field is independent of the test charge Q , only determined by the configuration of source charges.

If now the source of the field is a continuous charge distribution rather than several point charges, the definition of the electric field

becomes an integral

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{\mathbf{z}}. \quad (2.4)$$

In general, there are three kinds of charge distribution,

- A line charge with charge-per-unit-length, $\lambda(\mathbf{r}')$, and $dq = \lambda d\ell'$,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2} \hat{\mathbf{z}} d\ell'.$$

- A surface charge with charge-per-unit-area, $\sigma(\mathbf{r}')$, and $dq = \sigma da'$,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2} \hat{\mathbf{z}} da'.$$

- A volume charge with charge density, $\rho(\mathbf{r}')$, and $dq = \rho dV'$,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{z}} dV'.$$

(Sometimes we will write dV as $d^3\mathbf{r}$.)

The volume charge density $\rho(\mathbf{r})$ is the most general one. We can write a line charge or a surface charge as a volume charge by using the delta function $\delta(\mathbf{r})$. For example, a line charge $\lambda(z)$ located on the z -axis has a volume charge $\rho(\mathbf{r}) = \lambda(z)\delta(x)\delta(y)$. A surface charge σ on the xy -plane has a volume charge $\rho(\mathbf{r}) = \sigma(x, y)\delta(z)$. A point charge q at $\mathbf{r}$ is just $\rho(\mathbf{r}) = q\delta^3(\mathbf{r})$.

2.2 The Divergence and Curl of Electric Fields

2.2.1 Electric Flux and Gauss's Law

The electric flux Φ_E of an electric field $\mathbf{E}$ through a surface S is defined as

$$\Phi_E \equiv \int_S \mathbf{E} \cdot d\mathbf{a}. \quad (2.5)$$

The electric flux is a measure of how much electric field is passing through S . The area element vector $d\mathbf{a}$ has the direction normal to the area element da . **Gauss's law** states that for a closed surface $S = \partial V$, the electric flux through S measures how much charge is enclosed by it. If S encloses no charge, the electric flux through it is zero. Quantitatively,

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad (2.6)$$

This is **Gauss's law in integral form**. We can naturally write the surface integral as a volume integral using the divergence theorem:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{E}) dV.$$

Meanwhile, the enclosed charge in terms of the charge density ρ is

$$Q_{\text{enc}} = \int_V \rho dV.$$

Then Gauss's law becomes

$$\int_V (\nabla \cdot \mathbf{E}) dV = \int_V \left(\frac{\rho}{\epsilon_0} \right) dV.$$

This is true for any volume, so we arrive at **Gauss's law in differential form**:

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.} \quad (2.7)$$

2.2.2 The Curl of Electric Field

We now know the divergence of $\mathbf{E}$, then what about its curl? For a point charge q at the origin, the electric field is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

Let us first consider the line integral from a point $\mathbf{a}$ to some other point $\mathbf{b}$:

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell}.$$

In spherical coordinates, the line element $d\boldsymbol{\ell} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}$. The line integral then simplifies to

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} = \frac{1}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{q}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right),$$

where r_a is the distance from the charge to $\mathbf{a}$ and r_b is the distance from the charge to $\mathbf{b}$. When $\mathbf{a} = \mathbf{b}$ (in this case $r_a = r_b$), the line integral is around a closed path and we find that

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = 0.$$

By Stokes' theorem,

$$\boxed{\nabla \times \mathbf{E} = 0.} \quad (2.8)$$

We have not specify any choice of coordinates, so this holds for any charge location. Moreover, the principle of superposition allows us to add up all electric fields due to charges to get a total electric field. Since the curl $\nabla \times$ is a linear operator, the curl of the total electric field is the sum of all curls of each electric field, so

$$\nabla \times \mathbf{E} = \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \cdots) = \nabla \times \mathbf{E}_1 + \nabla \times \mathbf{E}_2 + \cdots = 0.$$

Therefore, $\nabla \times \mathbf{E} = 0$ holds for any electric field or any static charge distribution.

2.2.3 Useful Formulas and Results

- The electric field outside a spherically symmetric charge distribution is the same as if all the charges had been concentrated at the center,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$

- The electric field due to an infinite sheet with uniform surface charge σ is

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a unit vector pointing away from the surface. It does not depend on the distance from the sheet.

- The electric field due to an infinite line charge λ is

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}},$$

where r is the distance from the line charge and $\hat{\mathbf{r}}$ is pointing away from the line.

2.3 Electric Potential

2.3.1 Introduction to Potential

In general, it is painful to find the electric field through integrals like (2.4) because they are vector integrals and $\hat{\mathbf{z}}$ is annoying. However, in the last section, we see that the electric field has a nice property that it is curl free. This allows us to reduce a vector problem into a scalar one. Specifically, the line integral of $\mathbf{E}$ from point $\mathbf{a}$ to $\mathbf{b}$ is independent of path. We can thus define a function

$$\phi(\mathbf{r}) \equiv - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell} \quad (2.9)$$

where $\mathcal{O}$ is some arbitrary reference point. This function $\phi(\mathbf{r})$ is called the **electric potential**. This definition indicates that only the potential *difference* is meaningful. The potential difference between two points $\mathbf{a}$ and $\mathbf{b}$ is

$$\phi(\mathbf{b}) - \phi(\mathbf{a}) = - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell}.$$

The fundamental theorem for line integrals says that

$$\phi(\mathbf{b}) - \phi(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla\phi) \cdot d\boldsymbol{\ell}.$$

Hence

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla\phi) \cdot d\boldsymbol{\ell} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell}.$$

This is true for any points $\mathbf{a}$ and $\mathbf{b}$, so

$$\mathbf{E} = -\nabla\phi \quad (2.10)$$

Here are some comments on the electric potential:

- It obeys the principle of superposition,

$$\phi_{\text{total}} = \phi_1 + \phi_2 + \dots$$

- The name “potential” indicates that such quantity has a close connection to potential energy. A positive charge always flows from high potential to low potential, opposite for a negative charge. A surface over which the potential is constant is called an **equipotential**. The units of potential is a **volt**, which is a joule/coulomb.
- As we have mentioned, a scalar function like the potential is easier to manipulate with than a vector function like the electric field. Once we get the potential, taking a gradient $\nabla\phi$ gives us the electric field. You may wonder why a scalar function (ϕ has only one component) contains the same amount of information as a vector function ($\mathbf{E}$ has three components) does. The answer lies in the curl free nature of electric field, $\nabla \times \mathbf{E} = 0$. In other words, electric field is a special vector function that automatically has some constraints on its components.
- The reference point $\mathcal{O}$ is arbitrary. If we change $\mathcal{O}$ to $\mathcal{O}'$, the potential will be off by a constant, but the potential difference is

invariant. After all, a single potential is meaningless. Just like when we talk about the height of a mountain we often refer to the height as “the height above the sea level”. There can still be a natural reference point to set $\phi(\mathcal{O}) = 0$, and this point is the spatial infinity, $\mathcal{O} = \infty$. However, when we have an infinite charge distribution (like an infinite sheet of charge), this potential at infinity blows up and we need to find a new one.

2.3.2 Poisson’s Equation and Laplace’s Equation

A vector function can be determined only if we know both its divergence and curl. For electric field, they are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \times \mathbf{E} = 0.$$

In terms of potential, the divergence of electric field becomes the laplacian of the potential by (2.10),

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.} \tag{2.11}$$

This is called **Poisson’s equation**. In regions where no charge is present, it reduces to **Laplace’s equation**,

$$\nabla^2 \phi = 0.$$

The curl of $\mathbf{E}$ gives no information about ϕ because the curl of a gradient is always zero. Actually this is just the motivation that makes us define the potential. We see that while $\mathbf{E}$ needs two equations to specify, only one equation (Poisson’s equation) is needed to determine ϕ .

2.3.3 The Potential of a Localized Charge Distribution

In general, we are given a charge distribution ρ and we want to find the potential as a function of position. Given that the electric field of a point charge placed at the origin $\mathbf{E} = (1/4\pi\epsilon_0)(1/r^2)\hat{\mathbf{r}}$ and $d\boldsymbol{\ell} = dr\hat{\mathbf{r}} + r d\theta\hat{\boldsymbol{\theta}} + r \sin\theta d\phi\hat{\boldsymbol{\phi}}$ (here ϕ is the azimuthal angle in spherical coordinates),

$$\mathbf{E} \cdot d\boldsymbol{\ell} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr.$$

Setting $\phi(\infty) = 0$, the potential of a point charge q at the origin is

$$\phi(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell} = - \frac{1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

In general, the potential of a point charge q at arbitrary location is

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{z},$$

where z is the distance from q to $\mathbf{r}$. By the principle of superposition, for a continuous charge distribution,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{z}.$$

We can again develop some integrals to obtain the potential for a line charge, a surface charge, or a volume charge:

- A line charge with charge-per-unit-length, $\lambda(\mathbf{r}')$, and $dq = \lambda d\ell'$,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{z} d\ell'.$$

- A surface charge with charge-per-unit-area, $\sigma(\mathbf{r}')$, and $dq = \sigma da'$,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{z} da'.$$

- A volume charge with charge density, $\rho(\mathbf{r}')$, and $dq = \rho dV'$,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} dV'.$$

2.3.4 Boundary Conditions

The electric field is discontinuous across a surface charge σ . Consider a Gaussian surface S enclosing a charge Q over the surface (see Figure 2.1). Note that σ varies over the surface, so our Gaussian surface is infinitesimally small.

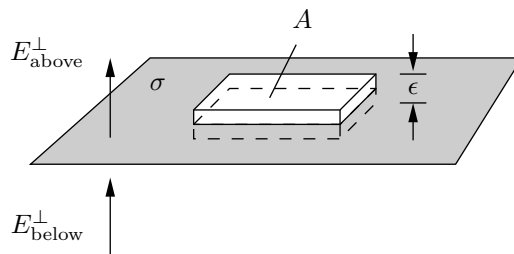


Figure 2.1:

According to Gauss's law,

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{\sigma}{\epsilon_0} A,$$

Only the top and bottom side contribute to the flux as $\epsilon \rightarrow 0$, so

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{\sigma}{\epsilon_0},$$

where $E_{\text{above}}^{\perp}$ is the perpendicular component of $\mathbf{E}$ with respect to the surface, and $E_{\text{below}}^{\perp}$ is the perpendicular component below the surface. The conclusion is that the normal component of $\mathbf{E}$ is discontinuous by an amount σ/ϵ_0 at any boundary. We can also use $\nabla \times \mathbf{E} = 0$ to show that the parallel component of $\mathbf{E}$ to the surface is always continuous. By taking a vertical rectangular loop C with height $\epsilon \rightarrow 0$ and length L ,

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = 0 \implies \mathbf{E}_{\text{above}}^{\parallel} = \mathbf{E}_{\text{below}}^{\parallel}.$$

Hence we can combine the two results into

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (2.12)$$

where $\hat{\mathbf{n}}$ is the unit vector perpendicular to the surface, pointing from “below” to “above”.

The potential is continuous across any boundary because

$$\phi_{\text{above}} - \phi_{\text{below}} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} \rightarrow 0$$

as the path length goes to zero with $\mathbf{E}$ finite. The gradient of ϕ has a discontinuity by (2.12)

$$\nabla\phi_{\text{above}} - \nabla\phi_{\text{below}} = -\frac{\sigma}{\epsilon_0}\hat{\mathbf{n}} \quad \text{or} \quad \frac{\partial\phi_{\text{above}}}{\partial n} - \frac{\partial\phi_{\text{below}}}{\partial n} = -\frac{\sigma}{\epsilon_0} \quad (2.13)$$

where $\partial\phi/\partial n = \nabla\phi \cdot \hat{\mathbf{n}}$ is called the **normal derivative** of ϕ , the rate of change of ϕ in the direction perpendicular to the surface.

2.3.5 Useful Formulas and Results

- The potential outside a spherically symmetric charge distribution is given by

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

The potential inside spherical shell of radius R is constant,

$$\phi(r < R) = \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

2.4 Work and Energy in Electrostatics

2.4.1 The Work It Takes to Move a Charge

Suppose we want to move a test charge Q from $\mathbf{a}$ to $\mathbf{b}$ in an electric field $\mathbf{E}$. The force on the charge by the electric field is $\mathbf{F} = Q\mathbf{E}$, so we must apply at least $\mathbf{F} = -Q\mathbf{E}$ to move it. The work done to move Q from $\mathbf{a}$ to $\mathbf{b}$ is then

$$W = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\boldsymbol{\ell} = -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} = Q[\phi(\mathbf{b}) - \phi(\mathbf{a})].$$

Such work is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$, so we call the electrostatic force *conservative*. Rearranging the equation gives

$$\phi(\mathbf{b}) - \phi(\mathbf{a}) = \frac{W}{Q}.$$

It means the potential difference between points $\mathbf{a}$ and $\mathbf{b}$ is equal to the work per unit charge required to carry a particle from $\mathbf{a}$ to $\mathbf{b}$. If we set the reference point at infinity, the work to bring Q to a point $\mathbf{r}$ from infinity is

$$W = Q[\phi(\mathbf{r}) - \phi(\infty)] = Q\phi(\mathbf{r}).$$

The essence of the potential is the potential energy (the work it takes to create the system) per unit charge.

2.4.2 The Energy of a Point Charge Distribution

Now we want to assemble a collection of charges. How much work should be done? Let us start by bringing two charges together. Moving charge q_1 to $\mathbf{r}_1$ requires no work since there is no electric field present initially. In the presence of q_1 , moving q_2 from infinity to $\mathbf{r}_2$ requires work

$$W_2 = q_2\phi(\mathbf{r}_2) = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right),$$

where $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$ is the distance between q_1 and q_2 . To bring in q_3 to $\mathbf{r}_3$, we need to counter both fields from q_1 and q_2 , and so

$$W_3 = q_3(\phi_1(\mathbf{r}_3) + \phi_2(\mathbf{r}_3)) = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right).$$

Similarly, bringing in charge q_4 requires

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{r_{14}} + \frac{q_2}{r_{24}} + \frac{q_3}{r_{34}} \right).$$

The total work needed to bring the first 4 charges together is

$$W_{\text{total}} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right).$$

You may find the general rule to be

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}} = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}.$$

This sum does not depend on the order to assemble the charges. Pulling out a factor of q_i ,

$$W = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j \neq i} \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_{ij}} \right).$$

The term in the bracket is the potential at $\mathbf{r}_i$ due to all other charges. In conclusion, the work to assemble a configuration of charges, or the energy stored in such a configuration, is

$$W = \frac{1}{2} \sum_{i=1}^n q_i \phi(\mathbf{r}_i) \quad (2.14)$$

2.4.3 The Energy of a Continuous Charge Distribution

For a continuous charge distribution, the sum becomes an integral,

$$W = \frac{1}{2} \int \rho(\mathbf{r}) \phi(\mathbf{r}) dV.$$

It turns out that this work can be rewritten in terms of the electric field. By the differential form of Gauss's law,

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E} \implies W = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) \phi dV.$$

Integrating by parts,

$$W = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla \phi) dV + \oint \phi \mathbf{E} \cdot d\mathbf{a} \right] = \frac{\epsilon_0}{2} \left(\int_V \mathbf{E}^2 dV + \oint_S \phi \mathbf{E} \cdot d\mathbf{a} \right).$$

The volume we are integrating over can be the volume that encloses all charges, but it can extend all the way to infinity because the volume with $\rho = 0$ contributes nothing to the integral. Since generally E^2 exist over all space, as we extend the volume to infinity, the first integral increase, which means the second integral must decrease. In fact, at infinity, the second integral over the surface $S = \partial V$, for a finite charge distribution, vanishes because $E \sim 1/r^2$, $\phi \sim 1/r$, and S grows like r^2 . Hence the integral decays like $1/r$. In conclusion, when integrated over all space, the energy stored in the electric field is

$$\boxed{W = \frac{\epsilon_0}{2} \int \mathbf{E}^2 dV.} \quad [\text{all space}] \quad (2.15)$$

2.5 Conductors

2.5.1 Basic Properties

In a **conductor**, electrons are free to move (they are called **delocalized electrons**). A perfect or ideal conductor can contain an unlimited supply of free charges. Here are some electrostatic properties of ideal conductors:

- $\mathbf{E} = 0$ inside a conductor. Otherwise, the charges are free to move and it is no longer electrostatic. Consider a conductor placed in an external electric field $\mathbf{E}_{\text{ext}}$. It will induce net charges until the electric field due to these **induced charges** precisely cancels $\mathbf{E}_{\text{ext}}$ inside the conductor. Practically, the timescale (called relaxation time τ) for the movement of charges is negligible (e.g. $\tau \sim 10^{-19}$ s for copper).
- The net charge density $\rho = 0$ inside a conductor according to Gauss's law, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. If $\mathbf{E} = 0$, then so is ρ . This causes the next property $\rightarrow$
- Any net charge must reside on the surface.
- A conductor is an equipotential. For any two points $\mathbf{a}$ and $\mathbf{b}$ on or inside the conductor,

$$\phi(\mathbf{b}) - \phi(\mathbf{a}) = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\ell = 0 \implies \phi(\mathbf{a}) = \phi(\mathbf{b}).$$

- The electric field $\mathbf{E}$ is perpendicular to the surface *just outside* the conductor. According the boundary condition of $\mathbf{E}$ of a surface charge (see Section 2.3.4), its parallel component is continuous. Inside the conductor, $\mathbf{E}^{\parallel} = 0$ because there is no field at all, which means the outside parallel component must also vanish. Hence only the perpendicular component survives.

2.5.2 Induced Charges

If a charge $+q$ is brought near a neutral conductor, the negative charges on the conductor is attracted by $+q$ and the positive charges is repelled. Eventually the conductor will be attracted by $+q$ since the negative charges are closer (attracting force is larger). If there is some hollow cavity in the conductor, and some charge $+q$ is inside the cavity, there will be an induced charge around cavity's surface. The net induced charge is exactly $-q$, resulting in a net charge $+q$ on the outer surface of the conductor. Hence even though you cannot see through the conductor, you will know there is a charge $+q$ inside the cavity.

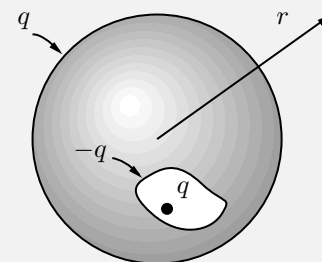
Example 2.1. A spherical conductor with a cavity

There is a spherical conductor with a weirdly shaped cavity inside. There is a charge q within the cavity. What is the electric field outside the sphere?

The electric field outside the sphere is simply

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$

regardless of the geometry of the cavity. The influence of the charge q inside the cavity is shielded by the induced charge $-q$ on the inner surface of the conductor. The leftover $+q$ on the outer surface will distribute uniformly along the sphere. For a more rigorous explanation, we need to rely on the uniqueness theorem discussed in the next chapter.



If there is no charge inside the cavity, then the field inside the cavity vanishes regardless of the charge distribution outside the conductor. We can prove this by taking an arbitrary loop that passes through both the conductor and the cavity. The field inside the conductor is always zero, so the line integral of $\mathbf{E}$ inside the cavity should also be zero to satisfy $\oint \mathbf{E} \cdot d\boldsymbol{\ell}$. This is true for any loop so $\mathbf{E} = 0$ everywhere inside the cavity.

2.5.3 Surface Charge and the Force on a Conductor

The electric field inside a conductor is zero, so by boundary condition (2.12), the field just outside the conductor is

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}.$$

The surface charges will then experience a force due to this field. The force per unit area $\mathbf{f}$ is $\sigma\mathbf{E}$. A problem arises when there is discontinuity in field at the surface charge, then which electric field $\mathbf{E}$ should we use? The answer is the average of the field above and below:

$$\mathbf{f} = \sigma\mathbf{E}_{\text{average}} = \frac{1}{2}\sigma(\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}).$$

To argue this, we can divide the electric field at the location of an infinitesimal patch of charge into two parts: the field due to the patch and the field due to other charges,

$$\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}.$$

The force on the patch is only due to $\mathbf{E}_{\text{other}}$ because the patch cannot exert a force on itself. This is good because $\mathbf{E}_{\text{other}}$ is continuous. The discontinuity results entirely from the patch, which has $\sigma/2\epsilon_0$ on each side, pointing away from the surface. We will then have

$$\mathbf{E}_{\text{above}} = \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}, \quad \mathbf{E}_{\text{below}} = \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}.$$

Therefore,

$$\mathbf{E}_{\text{other}} = \frac{1}{2}(\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}) = \mathbf{E}_{\text{average}}.$$

This statement applies to any surface charge. For a conductor, the average field is $\sigma/2\epsilon_0\hat{\mathbf{n}}$, so the force per unit area is

$$\mathbf{f} = P = \frac{1}{2\epsilon_0}\sigma^2\hat{\mathbf{n}}, \tag{2.16}$$

which is called the **electrostatic pressure** on the surface. In terms of the electric field just outside the surface,

$$P = \frac{\epsilon_0}{2}\mathbf{E}^2. \tag{2.17}$$

2.5.4 Capacitors

Consider two conductors, one with charge $+Q$ and the other $-Q$. We can somewhat calculate the potential between them,

$$\phi = \phi_+ - \phi_- = - \int_{(-)}^{(+)} \mathbf{E} \cdot d\boldsymbol{\ell}.$$

The electric field at any point $\mathbf{r}$ is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r^2} \hat{\mathbf{r}} dV$$

by definition. The problem is we have no knowledge on ρ , but we do know that $\mathbf{E}$ is proportional to ρ (and Q). It turns out that doubling the charge on each conductor does not change the arrangement of ρ but only doubles it. Hence when Q is doubled, both $\mathbf{E}$

and ϕ are doubled. We define the constant of proportionality as the **capacitance** of the arrangement:

$$C \equiv \frac{Q}{\phi}. \quad (2.18)$$

The capacitance depends only on the geometry and the arrangement of the two conductors. The SI units of the capacitance is **farads** (F). 1 F turns out to be an enormous amount of capacitance—practical units are μF , nF , or pF . Here is a list of capacitance of some common geometries of conductors:

- Parallel plates with area A and separation d and $d \ll \sqrt{A}$:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad \phi = \frac{Q}{A\epsilon_0} d, \quad C = \frac{A\epsilon_0}{d}.$$

- Concentric spherical metal shells with radii a and b ($a < b$):

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}, \quad \phi = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right), \quad C = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

- Capacitance per unit length of two coaxial cylindrical conductor of radii a and b ($a < b$):

$$C/\ell = \frac{2\pi\epsilon_0}{\ln(b/a)}.$$

These formulas work for regions between the plates/conductors. Outside the plates, the electric field vanishes and the potential is constant.

A capacitor can store energy when you charge it up by moving electrons from the positive plate to the negative one. Consider an intermediate stage where there are charge q on the positive plate and $-q$ on the other. The potential difference between is $\phi = q/C$. The work necessary to move a charge dq through this potential difference is

$$dW = \left(\frac{q}{C} \right) dq.$$

The total work to charge up a capacitor is then

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} C \phi^2 = \frac{1}{2} Q \phi,$$

where ϕ is the final potential of the capacitor. We can also take the differential of the work to get the force on the capacitor:

$$dW = \frac{1}{2} Q^2 d \left[\frac{1}{C(\mathbf{r})} \right] = \frac{1}{2} Q^2 \nabla \left[\frac{1}{C(\mathbf{r})} \right] \cdot d\mathbf{r} = \mathbf{F} \cdot d\mathbf{r},$$

which means

$$\mathbf{F} = \frac{1}{2} Q^2 \nabla \left[\frac{1}{C(\mathbf{r})} \right].$$

This result can be seen as the force needed to assemble the two conductors to form a capacitor. Alternatively, it can be interpreted as the force needed to keep the two conductors at rest, opposing their attraction.

2.5.5 Useful Formulas and Results

- **Griffiths Problem 2.57:** The surface charge density of an ellipsoid conductor can be determined explicitly if you know the total charge Q on it:

$$\sigma = \frac{Q}{4\pi abc} \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right)^{-1/2}$$

for an ellipsoid that obeys the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

- **Griffiths Problem 2.59:** Suppose a conductor carrying a net charge Q , when placed in an external electric field $\mathbf{E}_e$, experiences a force $\mathbf{F}$. If the external field is uniform and is now reversed ($\mathbf{E}_e \rightarrow -\mathbf{E}_e$), the force also reverses ($\mathbf{F} \rightarrow -\mathbf{F}$).

3 POTENTIALS

3.1 Laplace's Equation

In previous sections we already mentioned that calculating the potential ϕ is generally easier than calculating the electric field $\mathbf{E}$, unless there are some symmetries that allow us to use Gauss's law. However, it is still difficult to solve an integral even it is a scalar one,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} dV.$$

Aside from the integral, ρ is also unknown in most problems involving conductors. We may only know the total charge, or the potential of the conductor. If we want the potential over whole space, it is more convenient to work with the differential form (Poisson's equation):

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.$$

Moreover, we are usually interested in ϕ where no charge is present ($\rho = 0$). Poisson's equation then reduces to Laplace's equation,

$$\nabla^2 \phi = 0.$$

Subject to some boundary conditions, we may find the solution to Laplace's equation (called **harmonic functions**).

3.1.1 Laplace's Equation in Three Dimensions

Suppose we have the solution $\phi(\mathbf{r})$ for Laplace's equation. It has two nice properties:

Theorem 3.1.

The value of ϕ at point $\mathbf{r}$ is the average value of ϕ over a spherical surface of radius R centered at $\mathbf{r}$:

$$\phi(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} \phi da.$$

Proof. Consider a point charge q outside the sphere of radius R . Choose a coordinate so that the center of the sphere is at the origin and q is on the z -axis. The potential on the surface is

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad \text{where} \quad r^2 = z^2 + R^2 + 2zR \cos \theta$$

in spherical coordinates. The average potential over the surface is

$$\begin{aligned} \phi_{\text{average}} &= \frac{1}{4\pi R^2} \frac{q}{4\pi\epsilon_0} \int_0^{2\pi} d\phi \int_0^\pi d\theta [z^2 + R^2 - 2zR \cos \theta]^{-1/2} R^2 \sin \theta \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{2zR} \left[\sqrt{z^2 + R^2 - 2zR \cos \theta} \right]_0^\pi \\ &= \frac{q}{4\pi\epsilon_0} \frac{1}{2zR} [|z + R| - |z - R|] = \frac{1}{4\pi\epsilon_0} \frac{q}{z}. \end{aligned}$$

This is exactly the potential due to q at the center of the sphere. By principle of superposition, this works for any charge distribution outside the sphere. $\square$

As a consequence, ϕ can have no local maxima or minima. All extreme values of ϕ must occur at the boundaries. There are more similar properties in electrostatics. Here is a list of them:

- The average ϕ over a spherical surface of radius R due to a total charge Q *inside* the sphere is

$$\phi_{\text{average}} = \frac{Q_{\text{enc}}}{4\pi\epsilon_0 R}.$$

- **Earnshaw's theorem:** A charged particle cannot be held in a stable equilibrium by electrostatic forces alone.
- The average electric field (the *vector* average, not the flux) $\mathbf{E}_{\text{average}}$ due to charges inside the sphere is zero.
- The (vector) average electric field $\mathbf{E}_{\text{average}}$ over a spherical surface, due to charges outside the sphere, is the same as the field at the center.
- The (vector) average electric field over a spherical *volume* of radius R , due to charges outside the sphere, is the same as the field at the center.
- The (vector) average electric field over the spherical volume, due to charges inside the sphere, equals

$$\mathbf{E}_{\text{average}} = -\frac{1}{4\pi\epsilon_0} \frac{\mathbf{p}}{R^3},$$

where $\mathbf{p}$ is the total dipole moment (which will be discussed later in this chapter) of all the interior charges.

3.1.2 Boundary Conditions and the First Uniqueness Theorem

To determine ϕ that satisfies Laplace's equation, we also need boundary conditions. For three dimensional Laplace's equation, knowing ϕ_{boundary} uniquely determines ϕ . This is one of the **uniqueness theorem** in electrostatics. There are many other uniqueness theorems similar to this one, but we will present and proof two most useful ones:

Theorem 3.2. First uniqueness theorem

The solution to Laplace's equation in some volume V is uniquely determined if ϕ is specified on the boundary surface $S = \partial V$.

Proof. Suppose there are two solutions to Laplace's equation subject to the same boundary condition:

$$\nabla^2\phi_1 = 0 \quad \text{and} \quad \nabla^2\phi_2 = 0.$$

Let their difference be $\phi_3 \equiv \phi_1 - \phi_2$ and ϕ_3 obviously obeys Laplace's equation as well:

$$\nabla^2\phi_3 = \nabla^2\phi_1 - \nabla^2\phi_2 = 0.$$

The boundary condition is that $\phi_3 = 0$ at the boundary because ϕ_1 and ϕ_2 have the same boundary condition. By the second property of Laplace's equation (all extrema occur on the boundaries), the maximum and minimum of ϕ_3 are both zero. Hence $\phi_3 = 0$

everywhere, or equivalently,

$$\phi_1 = \phi_2.$$

□

The uniqueness theorem is more powerful than it seems. As long as you come up with a solution to Laplace's equation, and it has the correct value on the boundaries, it is *the* solution.

What if there are some charge inside the region we are interested in (in case ϕ obeys Poisson's equation)?

$$\nabla^2 \phi_1 = -\frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla^2 \phi_2 = -\frac{\rho}{\epsilon_0}.$$

We will also arrive at

$$\nabla^2 \phi_3 = \nabla^2 \phi_1 - \nabla^2 \phi_2 = -\frac{\rho}{\epsilon_0} + \frac{\rho}{\epsilon_0} = 0.$$

ϕ_3 satisfies Laplace's equation and has zero values on all boundaries, so $\phi_3 = 0$ and $\phi_1 = \phi_2$.

3.1.3 Conductors and the Second Uniqueness Theorem

In some cases, we will know the potential on conductors (such as when it is connected to a battery with fixed potential, or when it is grounded as $\phi = 0$). Sometimes, however, we do not know the potential on conductors, but instead the charges on the conducting surfaces. When Q_a and Q_b are deposited on two conductors, they move around and rearrange themselves so we do not know the charge density on conductors either. Suppose there are some charge ρ in the region between conductors, here is the second uniqueness theorem.

Theorem 3.3. Second uniqueness theorem

In a volume V outside conductors and containing a specified charge density ρ , the electric field is uniquely determined if the total charge on each conductor is given.

Proof. Suppose there are two fields satisfying the conditions of the problem. By Gauss's law,

$$\nabla \cdot \mathbf{E}_1 = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \cdot \mathbf{E}_2 = \frac{\rho}{\epsilon_0}.$$

They both obey Gauss's law in integral form for a Gaussian surface S_i enclosing each conductor with charge Q_i :

$$\oint_{S_i} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{Q_i}{\epsilon_0} \quad \text{and} \quad \oint_{S_i} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{Q_i}{\epsilon_0}.$$

For the boundary S_{total} enclosing all charges,

$$\oint_{S_{\text{total}}} \mathbf{E}_1 \cdot d\mathbf{a} = \frac{Q_{\text{total}}}{\epsilon_0} \quad \text{and} \quad \oint_{S_{\text{total}}} \mathbf{E}_2 \cdot d\mathbf{a} = \frac{Q_{\text{total}}}{\epsilon_0}.$$

Let $\mathbf{E}_3 \equiv \mathbf{E}_1 - \mathbf{E}_2$ so that

$$\nabla \cdot \mathbf{E}_3 = 0 \quad \text{and} \quad \oint_{S_i} \mathbf{E}_3 \cdot d\mathbf{a} = 0$$

for each boundary surface S_i . A conductor has a property that its surface is an equipotential. Thus, ϕ_3^i is a constant (not necessarily

the same constant) over each conducting surface S_i . Using the product rule,

$$\nabla \cdot (\phi_3 \mathbf{E}_3) = \phi_3 (\nabla \cdot \mathbf{E}_3) + \mathbf{E}_3 \cdot (\nabla \phi_3) = -|\mathbf{E}_3|^2.$$

Integrating over V on both sides and invoke the divergence theorem,

$$\int_V \nabla \cdot (\phi_3 \mathbf{E}_3) dV = \oint_{\partial V} \phi_3 \mathbf{E}_3 \cdot d\mathbf{a} = - \int_V |\mathbf{E}_3|^2 dV.$$

Note that ∂V covers all boundaries, the conducting surface and the outer surface at infinity. We can integrate them separately,

$$\oint_{\partial V} \phi_3 \mathbf{E}_3 \cdot d\mathbf{a} = \sum_i \oint_{S_i} \phi_3^i \mathbf{E}_3 \cdot d\mathbf{a} = \sum_i \phi_3^i \oint_{S_i} \mathbf{E}_3 \cdot d\mathbf{a} = \sum_i \phi_3^i \cdot (0) = 0.$$

We can pull out ϕ_3^i outside of the integral because each conducting surface is an equipotential. Hence

$$\int_V |\mathbf{E}_3|^2 dV = 0.$$

The integrand is non-negative, so $\mathbf{E}_3 = 0$ everywhere, or equivalently, $\mathbf{E}_1 = \mathbf{E}_2$. □

3.1.4 The Third Uniqueness Theorem

Theorem 3.4. Third Uniqueness Theorem

The electric field is uniquely determined when the charge density ρ is given in some volume V and *either* ϕ or its normal derivative $\partial\phi/\partial n$ is specified each boundary surface.

Proof. To be written. □

3.2 The Method of Images

3.2.1 The Classic Image Problem

Consider a charge q at a distance d above an infinite grounded conducting plane. What is the potential ϕ in the region above the plane due to both q and the induced charges on the plane?

Let us set up the coordinates such that the conducting plane lies in the xy -plane and q is at $(0, 0, d)$. Mathematically, the problem asks to solve Poisson's equation in the region $z > 0$, subject to the boundary conditions

$$\phi(z = 0) = 0 \quad \text{and} \quad \phi(\infty) = 0.$$

The corollary of the first uniqueness theorem works here.

Solution: we will use a nice way, called the **method of images**, to find the potential. Consider a different situation where there is

no grounded plane, but instead there is another charge $-q$ at $(0, 0, -d)$. This configuration produces a potential of

$$\phi(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]. \quad (3.1)$$

It satisfies

$$\phi(z=0) = 0 \quad \text{and} \quad \phi(\infty) = 0,$$

and the only charge in the region $z > 0$ is the point charge q at $(0, 0, d)$. We note that these are exactly the same conditions of the original problem. Thus the potential (3.1) solves the Poisson's equation above the plane, and it satisfies all boundary conditions. It is *the* solution for $z \geq 0$.

3.2.2 Induced Surface Charge

Now there is ϕ , we can use the jump condition (2.13) across a surface to derive the charge density on the plane:

$$\sigma = -\epsilon_0 \left. \frac{\partial \phi}{\partial z} \right|_{z=0}.$$

From (3.1),

$$\frac{\partial \phi}{\partial z} = \frac{1}{4\pi\epsilon_0} \left\{ \frac{-q(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{q(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right\}.$$

Thus,

$$\sigma = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}} = \frac{-qd}{2\pi(r^2 + d^2)^{3/2}}, \quad (3.2)$$

where $r^2 = x^2 + y^2$. The induced total charge on the surface is

$$Q = \int \sigma \, da = \int_0^{2\pi} d\phi \int_0^\infty r \, dr \frac{-qd}{2\pi(r^2 + d^2)^{3/2}} = \left. \frac{qd}{\sqrt{r^2 + d^2}} \right|_0^\infty = -q$$

as expected.

3.2.3 Force and Energy

Since the potential in $z \geq 0$ is the same as the potential produced by $+q$ and $-q$ mentioned in the method of images, then so is the electric field. Hence the force on q is

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}.$$

The energy is not the same. If there were two point charges q and $-q$, the energy stored is

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}.$$

However, the energy stored in a single charge and conducting plane is half of it. We can calculate it explicitly by considering the work on q to bring it from infinity:

$$W = \int_\infty^d \mathbf{F} \cdot d\boldsymbol{\ell} = \frac{1}{4\pi\epsilon_0} \int_\infty^d \frac{q^2}{4z^2} dz = \frac{1}{4\pi\epsilon_0} \left(-\frac{q^2}{4z} \right) \Big|_\infty^d = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.$$

Inducing charges does not require work since they are at equipotential $\phi = 0$. If we consider the energy stored in electric fields, only the upper half space $z \geq 0$ has a field (since $\mathbf{E} = 0$ in conductors), as oppose to electric field produced by actual q and $-q$, where the electric field exist over whole space. Therefore, the energy stored is halved.

3.3 Separation of Variables

The method of images is an elegant method, but it works only in special cases. We should look for methods to solve for Laplace's equation directly. In this section, we will introduce the method of **separation of variables**, which is a useful method to solve for partial differential equations. In electrostatics, separation of variables works in scenarios where the potential ϕ or the charge density σ is specified on the boundaries.

3.3.1 General Ideas

The idea is simple: we want solutions that are separable which look like

$$\phi(x_1, x_2, x_3) = \phi_1(x_1)\phi_2(x_2)\phi_3(x_3)$$

where x_i is some general coordinates. We can pick x_i according to the geometry of the problem. For example, if the problem contains some rectangular plane, we can work with Cartesian coordinates; if the problem involves spheres, we should use spherical coordinates.

The set of all solutions to Laplace's equation is a vector space because Laplace's equation is linear: for

$$\nabla^2\phi_1 = 0 \quad \text{and} \quad \nabla^2\phi_2 = 0 \quad \implies \quad \nabla^2(\alpha\phi_1 + \beta\phi_2) = 0.$$

Given a vector space, the **basis vectors** are a set of vectors $\{v_i(\mathbf{r})\}$ for $i = 1, 2, 3, \dots$ such that any vector $v(\mathbf{r})$ can be written as

$$v(\mathbf{r}) = \sum_i \alpha_i v_i(\mathbf{r})$$

for some α_i . This is called the **completeness** of the basis. We can also define an **inner product** (a linear map from two vectors to a scalar) for our vectors. It turns out that a good inner product for solutions to Laplace's equation is

$$\langle A(\mathbf{r}) | B(\mathbf{r}) \rangle = \int d^3\mathbf{r} A^*(\mathbf{r}) B(\mathbf{r}).$$

Given an inner product, we can always choose a basis $\{v_i(\mathbf{r})\}$, $i = 1, 2, 3, \dots$ such that the inner product of them satisfy

$$\langle v_i | v_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Then $\{v_i(\mathbf{r})\}$ is called an **orthonormal basis**. Given a vector $v(\mathbf{r}) = \sum_i \alpha_i v_i(\mathbf{r})$, we can solve for its "components" α_i in the basis using

$$\langle v_j | v \rangle = \left\langle v_j \left| \sum_i \alpha_i v_i \right. \right\rangle = \sum_i \alpha_i \langle v_j | v_i \rangle = \sum_i \alpha_i \delta_{ij} = \alpha_j.$$

I.e. $\alpha_j = \langle v_j | v \rangle$. In general, if we find a basis, we can easily prove whether it is a orthonormal basis just by taking inner products. The proof of completeness is much harder and we should rely on mathematicians.

3.3.2 Cartesian Coordinates

In Cartesian coordinates, the separable solutions looks like

$$\phi(\mathbf{r}) = X(x)Y(y)Z(z),$$

where $X(x)$ is a function only of x , likewise for $Y(y)$ and $Z(z)$. The Laplace's equation in Cartesian coordinates is

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0.$$

Substitute $\phi(\mathbf{r}) = X(x)Y(y)Z(z)$,

$$\left(\frac{\partial^2 X}{\partial x^2}\right)YZ + \left(\frac{\partial^2 Y}{\partial y^2}\right)XZ + \left(\frac{\partial^2 Z}{\partial z^2}\right)XY = 0.$$

Dividing XYZ on both sides,

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = 0.$$

This ensures that each of the three terms is a constant. Why is that? We can rewrite the equation as

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} - \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2}.$$

The LHS depends only on x , while the RHS depends only on y and z . Varying x does nothing to the RHS, so it should do nothing to the LHS either, which means both sides are constant. We can also prove this for y and z using similar arguments. With these facts, let us set

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = c_1, \quad \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = c_2, \quad \text{and} \quad \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = c_3,$$

with $c_1 + c_2 + c_3 = 0$ to satisfy Laplace's equation. Now let us get some solutions. Consider the x equation,

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = c_1 \quad \Rightarrow \quad \frac{\partial^2 X}{\partial x^2} = c_1 X.$$

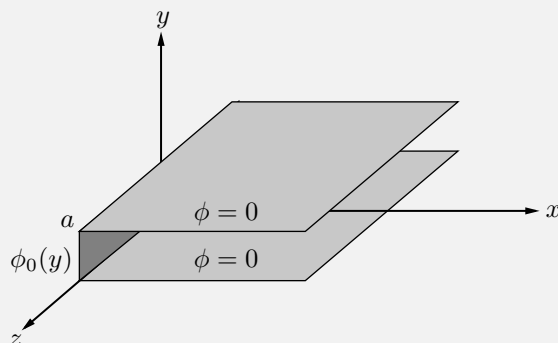
This equation has general solution

$$X(x) = \begin{cases} Ae^{\sqrt{c_1}x} + Be^{-\sqrt{c_1}x} & \text{if } c_1 > 0, \\ Ae^{i\sqrt{-c_1}x} + Be^{-i\sqrt{-c_1}x} = A' \cos(\sqrt{-c_1}x) + B' \sin(\sqrt{-c_1}x) & \text{if } c_1 < 0, \\ A + Bx & \text{if } c_1 = 0. \end{cases}$$

We can do the same for y and z . The coefficients A and B is usually determined by boundary conditions. In fact, the most general solution should consist of different c_1 's, c_2 's, and c_3 's, and there are A and B for each c .

Example 3.1. Two grounded plates

Two infinite grounded metal plates lie parallel to the xz plane, one at $y = 0$, the other at $y = a$. The left end, at $x = 0$, is closed off with an infinite strip insulated from the two plates, and maintained at a specific potential $\phi_0(y)$. We are to find the potential between the plates.



Let us first list out some conditions we know: 1. This configuration is independent of z , so we are going to solve a two-dimensional Laplace's equation,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

2. The boundary conditions are

$$\begin{cases} \phi(x, 0) = 0, \\ \phi(x, a) = 0, \\ \phi(0, y) = \phi_0(y), \\ \phi(\infty, y) = 0. \end{cases}$$

3. The solution is unique because the potential on all boundaries are specified.

As previously mentioned, consider the separable solution

$$\phi(x, y) = X(x)Y(y),$$

and we will arrive at

$$\frac{1}{X} \frac{d^2 X}{dx^2} = c_1, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = c_2, \quad \text{and} \quad c_1 + c_2 = 0.$$

It is convenient to set the constant $c_1 = k^2$

$$\frac{d^2 X}{dx^2} = k^2 X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y.$$

These equations are easy to solve, the solution to the x equation is

$$X(x) = Ae^{kx} + Be^{-kx}.$$

Let us now use some boundary conditions: first, at infinity $\phi(\infty, y) = 0$. This set $A = 0$ in $X(x)$ because otherwise it blows up at infinity. Thus, $X(x) = Be^{-kx}$. The y equation has solutions of the form

$$Y(y) = C \sin ky + D \cos ky.$$

You may find that it does no harm to let $d^2 Y/dy^2 = k^2 Y$ and let the x equation take the negative constant. However, the boundary condition $\phi(x, 0) = \phi(x, a) = 0$ rules out this possibility. There is no real exponential of y that satisfies such boundary condition, so $Y(y)$ has to be sinusoidal. In fact, this boundary condition tells us that

$$D = 0, \quad \text{and} \quad k = \frac{n\pi}{a}, \quad n = 1, 2, 3 \dots$$

Note that n can also be zero or negative integers, but $n = 0$ gives a trivial solution where $\phi = 0$ everywhere, and n negative is not linearly independent from positive n results. The final form of the separable solution would be

$$\phi(x, y) = Ce^{-kx} \sin ky, \quad k = \frac{n\pi}{a}.$$

The general solution to Laplace's equation in this problem is then

$$\phi_{\text{general}}(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin\left(\frac{n\pi y}{a}\right).$$

There is one boundary condition not used, $\phi(0, y) = \phi_0(y)$. Setting $x = 0$ in the general solution gives

$$\phi(0, y) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi y}{a}\right) = \phi_0(y).$$

Fortunately, the sine functions automatically forms a complete and orthogonal (not necessarily orthonormal) basis as desired, and we can use Fourier (sine) series to calculate the coefficients. The coefficients are given by

$$C_n = \frac{2}{a} \int_0^a \phi_0(y) \sin\left(\frac{n\pi y}{a}\right) dy$$

for any specified potential $\phi_0(y)$. For example, consider the constant potential $\phi_0(y) = \phi_0$. The coefficients are

$$C_n = \frac{2\phi_0}{a} \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy = \frac{2\phi_0}{n\pi} (1 - \cos n\pi) = \begin{cases} 0 & n \text{ even} \\ \frac{4\phi_0}{n\pi} & n \text{ odd.} \end{cases}$$

The potential between the plates is

$$\phi(x, y) = \frac{4\phi_0}{\pi} \sum_{\text{odd } n} \frac{1}{n} e^{-n\pi x/a} \sin\left(\frac{n\pi y}{a}\right).$$

Example 3.2. Potential due to a surface charge

Consider a configuration shown on the right (infinite in z). In last example we literally solved Laplace's equation in the upper half space of this configuration. The potential takes the form

$$\phi(x, y > 0) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right) e^{-n\pi y/a}.$$

The lower half space should have similar form, but with the sign in the exponent flipped to satisfy $\phi(\infty) = 0$,

$$\phi(x, y < 0) = \sum_{n=1}^{\infty} \beta_n \sin\left(\frac{n\pi x}{a}\right) e^{n\pi y/a}.$$

There are additional boundary conditions for $\phi(x, y)$: continuity in ϕ and discontinuity in $\nabla\phi$ at the surface charge. The continuity condition requires $\phi(x, y \rightarrow 0^+) = \phi(x, y \rightarrow 0^-)$, with

$$\phi(x, y \rightarrow 0^+) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right), \quad \phi(x, y \rightarrow 0^-) = \sum_{n=1}^{\infty} \beta_n \sin\left(\frac{n\pi x}{a}\right).$$

This gives $\alpha_n = \beta_n$, or equivalently,

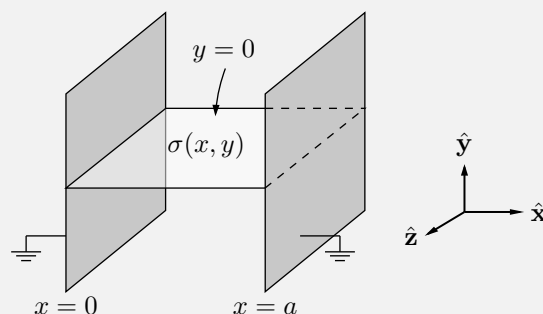
$$\phi(x, y) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right) e^{-n\pi|y|/a}.$$

The jump condition (2.13) says

$$\nabla_{\text{above}}\phi - \nabla_{\text{below}}\phi = -\frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}.$$

Here $\hat{\mathbf{n}}$ is just $\hat{\mathbf{y}}$, so

$$\left. \frac{\partial\phi}{\partial y} \right|_{0^+} - \left. \frac{\partial\phi}{\partial y} \right|_{0^-} = -\frac{\sigma}{\epsilon_0}.$$



We have

$$\frac{\partial \phi}{\partial y} = \begin{cases} \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right) \left(-\frac{n\pi}{a}\right) e^{-n\pi y/a} & y > 0, \\ \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right) \left(\frac{n\pi}{a}\right) e^{n\pi y/a} & y < 0. \end{cases}$$

Evaluating the derivatives at $y = 0$, we will find that

$$\sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{n\pi x}{a}\right) \left(\frac{2n\pi}{a}\right) = \frac{\sigma(x)}{\epsilon_0},$$

which means

$$\alpha_n = \frac{1}{n\pi\epsilon_0} \int_0^a \sigma(x) \sin\left(\frac{n\pi x}{a}\right) dx.$$

What if there is an infinite line charge λ at $(x, y) = (d, 0)$ instead of a surface charge? We can just write it as a surface charge with density

$$\sigma(x) = \lambda \delta(x - d).$$

Then the coefficients will be

$$\alpha_n = \frac{1}{n\pi\epsilon_0} \int_0^a \lambda \delta(x - d) \sin\left(\frac{n\pi x}{a}\right) dx = \frac{\lambda}{n\pi\epsilon_0} \sin\left(\frac{n\pi d}{a}\right).$$

Therefore, the potential is given by

$$\phi(x, y) = \sum_{n=1}^{\infty} \frac{\lambda}{n\pi\epsilon_0} \sin\left(\frac{n\pi d}{a}\right) \sin\left(\frac{n\pi x}{a}\right) e^{-n\pi|y|/a}.$$

3.3.3 Spherical Coordinates

Spherical coordinates help when there are spherical systems, such as determining the potential on a spherical conducting shell. In spherical coordinates, Laplace's equation looks completely different (because the Laplacian operator varies with choices of coordinates),

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0.$$

(Here we use V to denote the potential because ϕ so that it does not confuse with the azimuthal angle ϕ .) Now assume that the problem has azimuthal symmetry so V is independent of ϕ , which will simplify our calculation a lot. Laplace's equation then becomes

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0.$$

As usual, consider the separable solution

$$V(r, \theta) = R(r)\Theta(\theta).$$

Putting this into Laplace's equation,

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0.$$

The first term depends only on r , while the second term only on θ , so each must be a constant. Let the separation constant be $\ell(\ell + 1)$,

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell + 1), \quad \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\ell(\ell + 1).$$

First, the polar equation,

$$\frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -[\ell(\ell + 1) \sin \theta] \Theta.$$

The solutions to this equation are **Legendre polynomials**, P_ℓ , with argument $\cos \theta$,

$$\Theta(\theta) = P_\ell(\cos \theta).$$

A simple way to calculate Legendre polynomials is by the **Rodrigues formula**:

$$P_\ell(x) \equiv \frac{1}{2^\ell \ell!} \left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell.$$

The normalization condition usually sets $P_\ell(1) = 1$. Here are some low order Legendre polynomials:

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= (3x^2 - 1)/2 \\ P_3(x) &= (5x^3 - 3x)/2 \\ P_4(x) &= (35x^4 - 30x^2 + 3)/8 \\ P_5(x) &= (63x^5 - 70x^3 + 15x)/8 \\ &\dots \end{aligned}$$

Note that P_ℓ is even when ℓ is even, and P_ℓ is odd when ℓ is odd. Legendre polynomials are well-studied polynomials in mathematics. They form a complete and orthogonal set when integrated from -1 to 1 ,

$$\int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = \int_0^\pi P_\ell(\cos \theta) P_{\ell'}(\cos \theta) \sin \theta d\theta = \frac{2}{2\ell + 1} \delta_{\ell\ell'}.$$

You may wonder where is the other solution for a specified ℓ since we are dealing with a second order differential equation. The answer is that other solution blows up at $\theta = 0$ and $\theta = \pi$, so they are not good *physical* solutions. For example, for $\ell = 0$, the second solution is $\Theta(\theta) = \ln[\tan(\theta/2)]$. (If the problem excludes the z -axis, the second solution should also be considered.) For if we do not have azimuthal symmetry, the solutions to angular equations will be **spherical harmonics**, Y_ℓ^m , which contains a more complicated polynomial called **associated Legendre polynomials**, P_ℓ^m , with two quantum numbers.

The radial equation is

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \ell(\ell + 1)R.$$

It has solution of the form

$$R(r) = Ar^\ell + \frac{B}{r^{\ell+1}},$$

so the general solution to Laplace's equation in spherical coordinates is

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta). \quad (3.3)$$

Example 3.3. Potential inside and outside a sphere

The potential $V_0(\theta)$ is specified on the surface of a sphere of radius R .

First let us find the potential inside the sphere. By analyzing (3.3), we find that $B_\ell = 0$ for all ℓ or otherwise the potential

blows up at the origin,

$$V(r, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta).$$

Maching the boundary condition at $r = R$ to $V_0(\theta)$, we have

$$V(R, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} P_{\ell}(\cos \theta) = V_0(\theta).$$

Knowing that Legendre polynomials form a complete and orthogonal set,

$$\int_{-1}^1 P_{\ell}(x) P_{\ell'}(x) dx = \int_0^{\pi} P_{\ell}(\cos \theta) P_{\ell'}(\cos \theta) \sin \theta d\theta = \frac{2}{2\ell + 1} \delta_{\ell\ell'},$$

we can easily find all coefficients A_{ℓ} by

$$A_{\ell} R^{\ell} \frac{2}{2\ell + 1} = \int_0^{\pi} V_0(\theta) P_{\ell}(\cos \theta) \sin \theta d\theta \implies A_{\ell} = \frac{2\ell + 1}{2R^{\ell}} \int_0^{\pi} V_0(\theta) P_{\ell}(\cos \theta) \sin \theta d\theta.$$

For potential outside the sphere, $A_{\ell} = 0$ for all ℓ in (3.3) or otherwise the potential blows up at infinity,

$$V(r, \theta) = \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{r^{\ell+1}} P_{\ell}(\cos \theta).$$

At the boundary,

$$V(R, \theta) = \sum_{\ell=0}^{\infty} \frac{B_{\ell}}{R^{\ell+1}} P_{\ell}(\cos \theta) = V_0(\theta).$$

Using similar methods, we will arrive at

$$B_{\ell} = \frac{2\ell + 1}{2} R^{\ell+1} \int_0^{\pi} V_0(\theta) P_{\ell}(\cos \theta) \sin \theta d\theta.$$

Example 3.4. Neutral conducting sphere inside a uniform electric field

An uncharge metal sphere of radius R is placed in an otherwise uniform electric field $\mathbf{E} = E_0 \hat{\mathbf{z}}$. There will be induced charges on the sphere such that the total electric field vanishes inside the sphere, and they distort the field just outside the sphere (see Figure 3.1). We are to find the potential in the region outside the sphere.

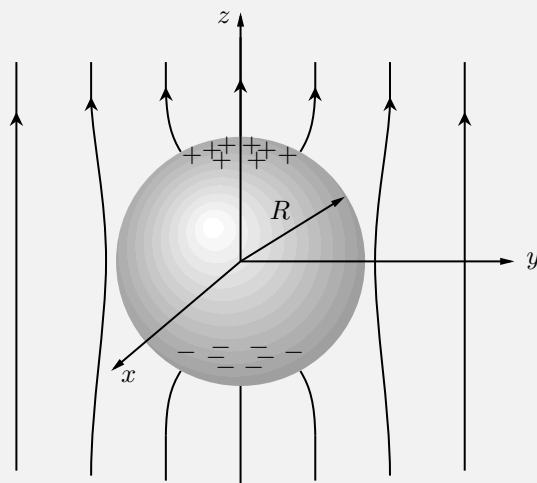


Figure 3.1: Induced charges on the conducting sphere and the distorted electric field lines.

We may set the conducting sphere to be $V = 0$. Due to the uniform electric field, V is not zero at large z . Instead, it should be

$$V \rightarrow -E_0 z + C,$$

far from the sphere. Furthermore, $C = 0$ because the whole xy -plane has $V = 0$ by symmetric distribution of induced charges. Thus the final boundary conditions are

$$V(R, \theta) = 0 \quad \text{and} \quad V \rightarrow -E_0 r \cos \theta \quad \text{for } r \gg R.$$

Using (3.3) and the first boundary condition,

$$A_\ell R^\ell + \frac{B_\ell}{R^{\ell+1}} = 0 \quad \implies \quad B_\ell = -A_\ell R^{2\ell+1},$$

so

$$V(r, \theta) = \sum_{\ell=0}^{\infty} A_\ell \left(r^\ell - \frac{R^{2\ell+1}}{r^{\ell+1}} \right) P_\ell(\cos \theta).$$

At $r \gg R$, the second term in the bracket is negligible, so the asymptotic boundary condition requires

$$\sum_{\ell=0}^{\infty} A_\ell r^\ell P_\ell(\cos \theta) = -E_0 r \cos \theta.$$

Matching the terms on both sides, the only nonzero term has $\ell = 1$. Knowing $P_\ell(\cos \theta) = \cos \theta$, the above equation immediately yields

$$A_1 = -E_0 \quad \text{and so} \quad V(r, \theta) = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos \theta.$$

By inspection, we see that the first term is exactly due to the external field, so the second term must be the field due to the induced charges. Additionally, the induced surface charge can be calculated using (2.13):

$$\begin{aligned} \sigma(\theta) &= -\epsilon_0 \left(\left. \frac{\partial V}{\partial r} \right|_{R^+} - \left. \frac{\partial V}{\partial r} \right|_{R^-} \right) \\ &= -\epsilon_0 \left. \frac{\partial V}{\partial r} \right|_{r=R} = \epsilon_0 E_0 \left(1 + 2 \frac{R^3}{r^3} \right) \cos \theta \Big|_{r=R} = 3\epsilon_0 E_0 \cos \theta. \end{aligned}$$

Note that even though the potential due to a point charge decays like $1/r$, the one due to induced charges in this problem decays like $1/r^2$. Such is an attribute of the potential due to separate opposite charges with net charge zero, which will be relevant to the multipole expansion discussed in the following section.

3.4 Multipole Expansion

3.4.1 Approximate Potentials at Large Distances

It is already known that the potential of some charge distribution will decay like a point charge, but we are interested in more detailed behavior of it at large distance. The potential at $\mathbf{r}$ of any localized charge distribution is given by

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} dV',$$

where $z = |\mathbf{r} - \mathbf{r}'|$ is the distance between $\mathbf{r}$ and $\mathbf{r}'$. Let α be the angle between $\mathbf{r}$ and $\mathbf{r}'$. Then by the law of cosines,

$$z^2 = r^2 + (r')^2 - 2rr' \cos \alpha = r^2 \left[1 + \left(\frac{r'}{r} \right)^2 - 2 \left(\frac{r'}{r} \right) \cos \alpha \right].$$

We can write z as

$$z = r\sqrt{1 + \epsilon}, \quad \text{with} \quad \epsilon \equiv \left(\frac{r'}{r} \right) \left(\frac{r'}{r} - 2 \cos \alpha \right).$$

With $\epsilon \ll 1$,

$$\begin{aligned} \frac{1}{z} &= \frac{1}{r} (1 + \epsilon)^{-1/2} = \frac{1}{r} \left(1 - \frac{1}{2} \epsilon + \frac{3}{8} \epsilon^2 - \frac{5}{16} \epsilon^3 + \dots \right) \\ &= \frac{1}{2} \left[1 - \frac{1}{2} \left(\frac{r'}{r} \right) \left(\frac{r'}{r} - 2 \cos \alpha \right) + \frac{3}{8} \left(\frac{r'}{r} \right)^2 \left(\frac{r'}{r} - 2 \cos \alpha \right)^2 - \frac{5}{16} \left(\frac{r'}{r} \right)^3 \left(\frac{r'}{r} - 2 \cos \alpha \right)^2 + \dots \right] \\ &= \frac{1}{r} \left[1 + \left(\frac{r'}{r} \right) (\cos \alpha) + \left(\frac{r'}{r} \right)^2 \left(\frac{3 \cos^2 \alpha - 1}{2} \right) + \left(\frac{r'}{r} \right)^3 \left(\frac{5 \cos^3 \alpha - 3 \cos \alpha}{2} \right) + \dots \right] \end{aligned}$$

where the last step is collecting like powers of (r'/r) . The coefficients of (r'/r) are Legendre polynomials, so we can write compactly

$$\frac{1}{z} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r} \right)^n P_n(\cos \alpha).$$

Putting this back to the integral and move out the r 's,

$$\boxed{\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') dV'.} \quad (3.4)$$

This equation is *exact*. It is called the **multipole expansion** of ϕ in powers of $1/r$. The $(n = 0)$ term is called the **monopole**; the $(n = 1)$ term is the **dipole**; higher terms include quadrupole, octopole, etc. Note that because of the angle α , the multipole expansion depends on the choice of coordinates. If your $\mathbf{r}$ is along the z -axis, then α is just the polar angle.

3.4.2 The Monopole and Dipole Terms

The monopole term (first approximation term) is simple:

$$\phi_{\text{mon}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r},$$

where $Q = \int \rho dV$ is the total charge and r is the distance from the origin. For a point charge, the monopole term is the exact potential because all higher terms vanish. The dipole term is

$$\phi_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int r' \cos \alpha \rho(\mathbf{r}') dV' = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \hat{\mathbf{r}} \cdot \int \mathbf{r}' \rho(\mathbf{r}') dV',$$

where in the last equality we use $r' \cos \alpha = \hat{\mathbf{r}} \cdot \mathbf{r}'$. The integral does not depend on $\mathbf{r}$, so it is an intrinsic property of the configuration.

It is defined as the **dipole moment**:

$$\boxed{\mathbf{p} \equiv \int \mathbf{r}' \rho(\mathbf{r}') dV'} \quad (3.5)$$

and so

$$\phi_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}. \quad (3.6)$$

In Einstein summation notation, these are

$$p_i = \int d^3\mathbf{r}' \rho(\mathbf{r}') r'_i \quad \text{and} \quad \phi_{\text{dip}}(r^i) = \frac{1}{4\pi\epsilon_0} \frac{p_i \hat{r}^i}{r^2}.$$

We can also work out the quadrupole moment and its associated potential term:

$$Q_{ij} = \int d^3\mathbf{r}' \rho(\mathbf{r}') r'^2 \left(\hat{r}'_i \hat{r}'_j - \frac{1}{3} \delta_{ij} \right) \quad \text{and} \quad \phi_{\text{quad}} = \frac{1}{4\pi\epsilon_0} \frac{3}{2} \frac{Q_{ij} (\hat{r}^i \hat{r}^j - \frac{1}{3} \delta^{ij})}{r^3}.$$

A **physical dipole** has no net charge:

$$\mathbf{p} = q\mathbf{r}'_+ - q\mathbf{r}'_- = q(\mathbf{r}'_+ - \mathbf{r}'_-) = q\mathbf{d},$$

where $\mathbf{d}$ is the separation from negative to positive charge. This is not a pure dipole though, as there will be higher order terms. To construct a **perfect** (point) **dipole**, you should make $d \rightarrow 0$ and $q \rightarrow \infty$ so that $qd = p$ stays fixed. The electric field of a perfect dipole is given by

$$\mathbf{E}_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}].$$

We also see that this dipole moment does not depend on the choice of reference point. In fact, a dipole moment is invariant under a change of reference point when the net charge of the configuration is zero. Suppose we move the origin by $\mathbf{a}$ and calculate the dipole with respect to the new origin $\bar{O}$:

$$\bar{\mathbf{p}} = \int \bar{\mathbf{r}}' \bar{\rho}(\bar{\mathbf{r}}') dV' = \int (\mathbf{r}' - \mathbf{a}) \rho(\mathbf{r}') dV' = \int \mathbf{r}' \rho(\mathbf{r}') dV' - \mathbf{a} \int \rho(\mathbf{r}') dV' = \mathbf{p} - Q\mathbf{a}.$$

If $Q = 0$, then $\bar{\mathbf{p}} = \mathbf{p}$.

Dipole moments are additive because they are vectors. For example, consider a configuration with four charges at the corners of a square. Each charge has neighbor charges with opposite signs (but same magnitude). The dipole moment is

$$\mathbf{p} \begin{bmatrix} + & - \\ - & + \end{bmatrix} = \mathbf{p} \begin{bmatrix} + \\ - \end{bmatrix} + \mathbf{p} \begin{bmatrix} - \\ + \end{bmatrix} = \mathbf{p} \begin{bmatrix} + & - \end{bmatrix} + \mathbf{p} \begin{bmatrix} - & + \end{bmatrix} = 0,$$

however you add the dipoles. This is a quadrupole in which both monopole and dipole terms vanish.

3.4.3 Dipoles in an Electric Field

Consider a uniform electric field $\mathbf{E}$ and two opposite charges held by some bound force (Figure 3.2). The origin is at the midpoint of the two charges. Charges $+q$ and $-q$ experience opposite forces from the electric field, $\mathbf{F}_+ = q\mathbf{E}$ and $\mathbf{F}_- = -q\mathbf{E}$, respectively. There is no net force, but it is clear on the figure that there will be a torque on the system:

$$\begin{aligned} \boldsymbol{\tau} &= (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) \\ &= [(\mathbf{d}/2) \times (q\mathbf{E})] + [(-\mathbf{d}/2) \times (-q\mathbf{E})] = q\mathbf{d} \times \mathbf{E}. \end{aligned}$$

Therefore, the dipole experiences a torque

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E} \tag{3.7}$$

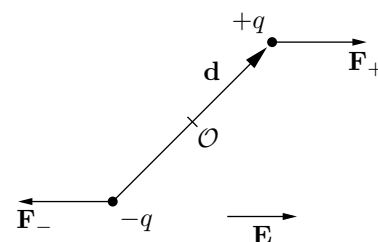


Figure 3.2: A dipole in a constant electric field, experiencing a torque.

We can also calculate the interaction energy in the dipole: let $\phi(\mathcal{O}) = 0$, then

$$U = q\phi(\mathbf{d}/2) - q\phi(-\mathbf{d}/2) = q[-\mathbf{E} \cdot (\mathbf{d}/2)] - q[-\mathbf{E} \cdot (-\mathbf{d}/2)] = -q\mathbf{d} \cdot \mathbf{E} = -\mathbf{p} \cdot \mathbf{E}.$$

It is saying that when $\mathbf{p}$ is aligned with $\mathbf{E}$, the system has minimum energy (a stable equilibrium). When they are anti-parallel, the system has the maximum energy (an unstable equilibrium).

When the electric field is not uniform, there will be a net force on the dipole in addition to the torque. Assuming the dipole is short so that the change in electric field small, the force is

$$\mathbf{F} = \mathbf{F}_+ + \mathbf{F}_- = q(\mathbf{E}_+ - \mathbf{E}_-) = q(\delta\mathbf{E}).$$

In the x -direction, $\delta E_x = (\nabla E_x) \cdot \mathbf{d}$, and similar for E_y and E_z . It is convenient to write the change as

$$\delta\mathbf{E} = (\mathbf{d} \cdot \nabla)\mathbf{E} \implies \mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}.$$

3.4.4 Comments on the Choice of Origin for Multipole Expansion

There is a useful idea called the **elementary moment**, which is a charge distribution that is infinitesimally small, but has a non-zero value of one of the multipole moments and zero for all other moments when its location is at the origin. Here is some example: when the charge distribution is small but has a charge q , then the charge density is a delta function $\rho(\mathbf{r}) = q\delta^3(\mathbf{r} - \mathbf{r}_q)$. We will get moments

$$\begin{aligned} q_{\text{total}} &= \int d^3\mathbf{r} q\delta^3(\mathbf{r} - \mathbf{r}_q) = q, \\ p_i &= \int d^3\mathbf{r} q\delta^3(\mathbf{r} - \mathbf{r}_q) r\hat{r}_i = qr_q\hat{r}_{q,i}, \\ Q_{ij} &= \int d^3\mathbf{r} q\delta^3(\mathbf{r} - \mathbf{r}_q) r^2 \left(\hat{r}_i\hat{r}_j - \frac{1}{3}\delta_{ij} \right) = qr_q^2 \left(\hat{r}_{q,i}\hat{r}_{q,j} - \frac{1}{3}\delta_{ij} \right), \dots \end{aligned}$$

If we choose $\mathbf{r}_q$ to be the origin, all higher terms vanish (because they have multiplicative factors of r_q) except the monopole.

An elementary dipole has two equal charges separated by $d\hat{\mathbf{p}}$, with midpoint located at $\mathbf{r}_p$. It has charge density

$$\rho(\mathbf{r}) = q\delta^3(\mathbf{r} - \mathbf{r}_+) - q\delta^3(\mathbf{r} - \mathbf{r}_-) \quad \text{with} \quad \mathbf{r}_+ = \mathbf{r}_p + \frac{d}{2}\hat{\mathbf{p}}, \quad \mathbf{r}_- = \mathbf{r}_p - \frac{d}{2}\hat{\mathbf{p}}.$$

Evidently, $q_{\text{total}} = 0$, the monopole vanishes. The dipole moment is

$$p_i = \int d^3\mathbf{r} [q\delta^3(\mathbf{r} - \mathbf{r}_+) - q\delta^3(\mathbf{r} - \mathbf{r}_-)] r\hat{r}_i = qd(\hat{\mathbf{p}})_i.$$

The quadrupole moment is a bit complicated:

$$\begin{aligned}
Q_{ij} &= \int d^3\mathbf{r} [q\delta^3(\mathbf{r} - \mathbf{r}_+) - q\delta^3(\mathbf{r} - \mathbf{r}_-)] r^2 \left(\hat{r}_i \hat{r}_j - \frac{1}{3} \delta_{ij} \right) \\
&= q \left[r_+^2 \left(\hat{r}_{+,i} \hat{r}_{+,j} - \frac{1}{3} \delta_{ij} \right) - r_-^2 \left(\hat{r}_{-,i} \hat{r}_{-,j} - \frac{1}{3} \delta_{ij} \right) \right] \\
&= q \left\{ \left[(\mathbf{r}_+)_i (\mathbf{r}_+)_j - \frac{1}{3} r_+^2 \delta_{ij} \right] - \left[(\mathbf{r}_-)_i (\mathbf{r}_-)_j - \frac{1}{3} r_-^2 \delta_{ij} \right] \right\} \\
&= q \left\{ [(\mathbf{r}_+)_i (\mathbf{r}_+)_j - (\mathbf{r}_-)_i (\mathbf{r}_-)_j] - \frac{1}{3} (r_+^2 - r_-^2) \delta_{ij} \right\} \\
&= q \left\{ \left[\left(\mathbf{r}_p + \frac{d}{2} \hat{\mathbf{p}} \right)_i \left(\mathbf{r}_p + \frac{d}{2} \hat{\mathbf{p}} \right)_j - \left(\mathbf{r}_p - \frac{d}{2} \hat{\mathbf{p}} \right)_i \left(\mathbf{r}_p - \frac{d}{2} \hat{\mathbf{p}} \right)_j \right] - \frac{1}{3} \left(\left| \mathbf{r}_p + \frac{d}{2} \hat{\mathbf{p}} \right|^2 - \left| \mathbf{r}_p - \frac{d}{2} \hat{\mathbf{p}} \right|^2 \right) \delta_{ij} \right\} \\
&= q \left\{ d [(\mathbf{r}_p)_i (\hat{\mathbf{p}})_j + (\hat{\mathbf{p}})_i (\mathbf{r}_p)_j] - \frac{1}{3} \left[\left(r_p^2 + \frac{d^2}{4} + d \hat{\mathbf{r}}_p \cdot \hat{\mathbf{p}} \right) - \left(r_p^2 + \frac{d^2}{4} - d \hat{\mathbf{r}}_p \cdot \hat{\mathbf{p}} \right) \right] \delta_{ij} \right\} \\
&= qd \left[(\mathbf{r}_p)_i (\hat{\mathbf{p}})_j + (\hat{\mathbf{p}})_i (\mathbf{r}_p)_j - \frac{2}{3} \hat{\mathbf{r}}_p \cdot \hat{\mathbf{p}} \delta_{ij} \right].
\end{aligned}$$

If we choose $\mathbf{r}_p$ as the origin, only the dipole term survives and higher terms vanish since they have multiplicative factors of r_p . Hence the elementary dipole is a pure dipole. We can get other elementary multipoles using the same method. The lowest non-zero multipole moment is coordinates invariant, and higher terms vanish if we choose the location of the elementary multipole as the origin.

The idea of multipole expansion and elementary moments are useful when it is hard to track charge density $\rho(\mathbf{r})$ of individual elements. For example, the elementary dipole is an excellent approximation for polar molecules like water. Tracking the electron cloud density and calculate the potential is nearly impossible, especially when there are moles of them.

4 ELECTRIC FIELDS IN MATTER

This chapter is more about insulators, or **dielectrics**, which is an opposite of conductors. Most charges in dielectrics do not move freely.

4.1 Polarization

4.1.1 Induced Dipoles

The dipole we discussed in the last section is a **permanent dipole**—it exists before an external electric field is applied. There are many natural permanent dipoles: water molecule (H_2O) is a typical example, where electron clouds are attracted from the two hydrogen atom to the oxygen atom. **Induced dipoles** are dipoles induced by external electric fields. For example, a neutral atom (with an electron cloud surrounding a positive nucleus) can get **polarized** by an electric field $\mathbf{E}$. That is, the electron cloud and the nucleus are pulled apart by the external field, reaching an equilibrium as a dipole with the dipole moment $\mathbf{p}$ parallel to $\mathbf{E}$. In general, the induced dipole moment is approximately proportional to the field,

$$\mathbf{p} = \alpha \mathbf{E}, \quad (4.1)$$

where the constant of proportionality α is called the **atomic polarizability**.

Example 4.1. Atomic polarizability of an atom

A simple model of an atom contains a point nucleus $+q$ surrounded by a uniformly charged spherical cloud $-q$ of radius a (Figure 4.1 left).

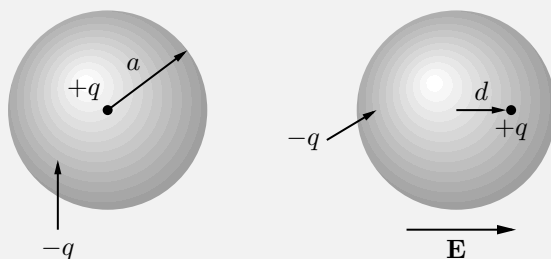


Figure 4.1: An atomic model without (left) and with (right) an external electric field.

Now apply an external electric field. It can be approximated that the nucleus has shifted a distance d to the right relative to the electron cloud (which remains its spherical shape). The electron cloud produces an electric field such that it balances out the external electric field at the nucleus (Figure 4.1 right): $E = E_e$. The field at radius d inside a uniformly charged sphere is

given by

$$E_e = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}.$$

At equilibrium,

$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} \implies p = qd = (4\pi\epsilon_0 a^3)E.$$

The polarizability is $\alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 V$, where V is the volume of the atom. We see that the magnitude of the dipole does not depend on the direction in which the electric field applies. You may encounter some charge distribution that does not have a spherical symmetry, such as a carbon dioxide (CO_2) with linear symmetry. In this case, the polarizability becomes a **polarizability tensor** α_{ij} such that

$$\begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}.$$

4.1.2 Polarization

For a volume of atoms or molecules, the dipole moment of each individual elements adds up because they all tend to get aligned with the electric field. It is convenient to define the **polarization**:

$$\mathbf{P} \equiv \frac{d\mathbf{p}}{dV} \implies \mathbf{p}_{\text{total}} = \int_V \mathbf{P}(\mathbf{r}) dV. \quad (4.2)$$

In later sections, we will discuss the field responsible for $\mathbf{P}$, together with fields induced by $\mathbf{P}$.

4.2 The Field of a Polarized Object

4.2.1 Bound Charges

Given the polarization $\mathbf{P}$ of an object, what is the field caused by it? Again, it is easier to work with potential rather than electric field itself. The potential for a single dipole is

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{z}}}{z^2},$$

where $\mathbf{z} = \mathbf{r} - \mathbf{r}'$ is the displacement from the dipole to $\mathbf{r}$. We can write this in terms of polarization,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{z^2} dV'.$$

What is the physical interpretation of this? Here is a useful relation:

$$\nabla' \left(\frac{1}{z} \right) = \nabla' \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) = \frac{\hat{\mathbf{z}}}{z^2}.$$

where ∇' is the gradient with respect to source coordinates $\mathbf{r}'$, not $\mathbf{r}$. Then

$$\begin{aligned}\phi(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \int_V \mathbf{P} \cdot \nabla' \left(\frac{1}{z} \right) dV' \\ &= \frac{1}{4\pi\epsilon_0} \left[\int_V \nabla' \cdot \left(\frac{\mathbf{P}}{z} \right) dV' - \int_V \frac{1}{z} (\nabla' \cdot \mathbf{P}) dV' \right] \\ &= \frac{1}{4\pi\epsilon_0} \oint_{\partial V} \frac{\mathbf{P}}{z} \cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{z} (\nabla' \cdot \mathbf{P}) dV'.\end{aligned}$$

where in the second equality we used integration by parts, and in the third equality the divergence theorem. Observing that they both have a factor of $1/z$, they look like some kind of potential. The first term is like the potential of a surface charge

$$\boxed{\sigma_b \equiv \mathbf{P} \cdot \hat{\mathbf{n}}}. \quad (4.3)$$

The second term is like the potential of a volume charge

$$\boxed{\rho_b \equiv -\nabla \cdot \mathbf{P}} \quad (4.4)$$

and so

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \oint_{\partial V} \frac{\sigma_b}{z} da' + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{z} dV'.$$

We can interpret potential of a polarized object to be the sum of potential due to these surface charges and volume charges. These charges are called **bound charges**, i.e., they are not free to move like charges in a conductor. Since the potential is caused by bound charges, then so is the electric field. We can solve for it using any methods just by seeing bound charges as actual charges.

Example 4.2. A uniformly polarized sphere

Find the potential produced by a uniformly polarized sphere of radius R with polarization $\mathbf{P} = P\hat{\mathbf{z}}$. We have

$$\rho_b = 0 \quad \text{and} \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P \cos \theta,$$

where θ is the polar angle. Example 3.4 already calculated the potential due to this surface charge:

$$\phi(\mathbf{r}) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta, & r \leq R, \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta, & r \geq R. \end{cases}$$

Using $r \cos \theta = z$, the electric inside the sphere is uniform:

$$\mathbf{E} = -\nabla\phi = -\frac{P}{3\epsilon_0} \hat{\mathbf{z}} = -\frac{1}{3\epsilon_0} \mathbf{P}, \quad r < R.$$

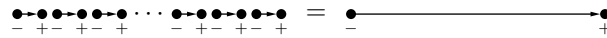
The potential outside the sphere is

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} \quad \text{with} \quad \mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}, \quad r \geq R.$$

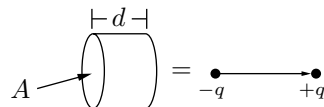
This is exactly the potential due to an elementary dipole at the origin.

4.2.2 Physical Interpretation of Bound Charges

Even though we define the bound charges by just manipulating integrals, it has physical meanings. First, the surface bound charges: consider a uniform polarization $\mathbf{P}$. Choose a long string of dipoles:



All intermediate charges cancel out, the net charges appear at the boundaries. Consider then a tube with uniform $\mathbf{P}$,



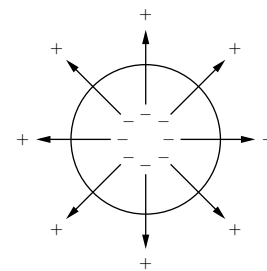
The bound charges on the ends are $+q$ and $-q$. By the definition of polarization, the dipole moment of this tube is $p = P(Ad) = qd$. Thus the surface charge density on one end is $\sigma_b = q/A = P$. If the ends have some angle θ with respect to vertical, the total charge says the same, but the area of the end becomes $A/\cos\theta$. This gives the surface charge density

$$\sigma_b = \frac{q}{A/\cos\theta} = P \cos\theta = \mathbf{P} \cdot \hat{\mathbf{n}}.$$

Volume bound charges comes from nonuniform polarization.

A diverging $\mathbf{P}$ results in an accumulation of negative charge (shown on the right). An equivalent amount of positive charge is pushed out of the volume: it has surface charge density $\mathbf{P} \cdot \hat{\mathbf{n}}$. Hence the charge inside the volume is

$$\int_V \rho_b dV = - \oint_{\partial V} \mathbf{P} \cdot d\mathbf{a} = - \int_V (\nabla \cdot \mathbf{P}) dV.$$



This works for any volume, so

$$\rho_b = -\nabla \cdot \mathbf{P}.$$

4.3 The Electric Displacement

4.3.1 Gauss's Law in the Presence of Dielectrics

In the last section we see that the field due to a polarized object is just the field due to bound charges. Now we should take into account all other fields from **free charges** ρ_f , any charge that is not a result of polarization. There may be free charges (like ions) in dielectrics, so

$$\rho = \rho_b + \rho_f.$$

By Gauss's law,

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f \implies \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f.$$

We call the term in the parenthesis **electric displacement** $\mathbf{D}$:

$$\boxed{\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}.} \tag{4.5}$$

Gauss's law in terms of $\mathbf{D}$ is

$$\nabla \cdot \mathbf{D} = \rho_f \iff \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}},$$

where $Q_{f_{\text{enc}}}$ is the total free charge enclosed in the volume. This is particularly useful because most of the time we can only control free charges. The free charges will polarize the dielectrics and then there will be induced bound charges. We can directly calculate $\mathbf{D}$ using Gauss's law when there is symmetry.

4.3.2 A Note on Electric Displacement

The electric displacement looks a lot like the electric field. You may think the free charge is “sourcing” the electric displacement. You may also try to use other methods to calculate $\mathbf{D}$ just like you did for $\mathbf{E}$ (with an additional factor ϵ_0). This is a dangerous idea, for there is no “Coulomb's law” for $\mathbf{D}$:

$$\mathbf{D} \neq \frac{1}{4\pi} \int \frac{\rho_f(\mathbf{r}')}{r^2} \hat{\mathbf{r}} dV'.$$

The key reason is you need both the divergence and the curl to specify a vector field, but we know only the divergence of $\mathbf{D}$. Recall that the electric field is a special vector field since there is an underlying electrostatic constraint on it: $\nabla \times \mathbf{E} = 0$. We can of course calculate the curl of $\mathbf{D}$:

$$\nabla \times \mathbf{D} = \epsilon_0(\nabla \times \mathbf{E}) + \nabla \times \mathbf{P} = \nabla \times \mathbf{P},$$

which, in general, does not vanish. As a result, there is no “potential” for $\mathbf{D}$.

4.3.3 Boundary Conditions

By the same method discussed in Section 2.3.4, $\mathbf{D}$ has some boundary conditions:

- Gauss's law in terms of $\mathbf{D}$ gives a discontinuous perpendicular component:

$$D_{\text{above}}^\perp - D_{\text{below}}^\perp = \sigma_f.$$

- $\nabla \times \mathbf{D} = \nabla \times \mathbf{P}$ gives a discontinuous parallel component:

$$\mathbf{D}_{\text{above}}^\parallel - \mathbf{D}_{\text{below}}^\parallel = \mathbf{P}_{\text{above}}^\parallel - \mathbf{P}_{\text{below}}^\parallel.$$

4.4 Linear Dielectrics

4.4.1 Susceptibility, Permittivity, Dielectric Constant

In many substances, the polarization is proportional to the electric field:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \tag{4.6}$$

where the dimensionless constant of proportionality χ_e is called the **electric susceptibility** of the material. Materials with this property are called **linear dielectrics**. Note that $\mathbf{E}$ in (4.6) is the *total* electric field—the sum of external field and the field due to polarization itself. It is more convenient to work with $\mathbf{D}$ because it is related to free charges that we can control. $\mathbf{D}$ can be written

as

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0(1 + \chi_e) \mathbf{E}, \quad \text{or} \quad \mathbf{D} = \epsilon \mathbf{E},$$

where

$$\epsilon \equiv \epsilon_0(1 + \chi_e) \quad (4.7)$$

The new ϵ is called the **permittivity** of the material. In vacuum, where there is no polarization, ϵ_0 is called the **permittivity of free space**. We also call $\epsilon_r = 1 + \chi_e = \epsilon/\epsilon_0$ the **relative permittivity**, or **dielectric constant**.

In a space filled entirely of homogeneous (susceptibility independent of position) linear dielectric, the curl of $\mathbf{P}$ vanishes: $\nabla \times \mathbf{P} = \epsilon_0 \chi_e \nabla \times \mathbf{E}$. Then in this case,

$$\nabla \cdot \mathbf{D} = \rho_f \quad \text{and} \quad \nabla \times \mathbf{D} = 0,$$

so $\mathbf{D}$ can be found as though there were no dielectric,

$$\mathbf{D} = \epsilon_0 \mathbf{E}_{\text{vac}},$$

where $\mathbf{E}_{\text{vac}}$ is the field produced by the same free charge in the absence of dielectric. Thus, the actual $\mathbf{E}$ obeys

$$\mathbf{E} = \frac{1}{\epsilon} \mathbf{D} = \frac{1}{\epsilon_r} \mathbf{E}_{\text{vac}}.$$

In conclusion, for a charge distribution immersed in a large homogeneous linear dielectric, we can simply get the electric field by replacing ϵ_0 with ϵ in the known electric field produced by free charges in vacuum.

Example 4.3. Parallel-plate capacitor with dielectric material

Consider a parallel-plate capacitor, with charge Q and $-Q$ on plates, filled with insulating material of dielectric constant ϵ_r . What effect does this have on its capacitance?

We already know the electric field when no dielectric is present: it is between the plate, with

$$\mathbf{E}_{\text{vac}} = \frac{\sigma_f}{\epsilon_0} \hat{\mathbf{n}} \implies \phi_{\text{vac}} = \frac{Q}{A\epsilon_0} d.$$

Using the conclusion above, we replace ϵ_0 with ϵ when there is dielectric,

$$\mathbf{E} = \frac{\sigma_f}{\epsilon} \hat{\mathbf{n}} \implies \phi = \frac{Q}{A\epsilon} d.$$

The capacitance is defined by $C = Q/\phi$, so

$$C = \epsilon_r C_{\text{vac}} = \frac{A\epsilon}{d}.$$

You can also check by explicit calculation starting from $\mathbf{D}$. Assume (and in fact) the polarization of the dielectric is constant. We can guess the electric displacement is along the z -direction. By the jump condition of $D^\perp$:

$$D_{\text{above}} - D_{\text{below}} = \sigma_f.$$

D_{above} is zero above the plate because $\mathbf{E} = 0$ and $\mathbf{P} = 0$ above the plate. Thus, $D_{\text{below}} = -\sigma_f$ and $E_{\text{below}} = -\sigma_f/\epsilon$. You get the same answer.

4.4.2 Boundary Value Problems with Linear Dielectrics

In a homogeneous isotropic linear dielectric, the bound charge density is proportional to the free charge density:

$$\rho_b = -\nabla \cdot \mathbf{P} = -\nabla \cdot \left(\epsilon_0 \frac{\chi_e}{\epsilon} \mathbf{D} \right) = - \left(\frac{\chi_e}{1 + \chi_e} \right) \rho_f.$$

If no free charge is inside the dielectric, then $\rho = 0$, i.e. all net charge must be on the surface. Then the potential inside a dielectric obeys Laplace's equation. We can use methods learned in Chapter 3 to determine the potential. It is convenient to write the boundary condition of displacement field as

$$\epsilon_{\text{above}} E_{\text{above}}^{\perp} - \epsilon_{\text{below}} E_{\text{below}}^{\perp} = \sigma_f,$$

or

$$\epsilon_{\text{above}} \frac{\partial \phi_{\text{above}}}{\partial n} - \epsilon_{\text{below}} \frac{\partial \phi_{\text{below}}}{\partial n} = -\sigma_f.$$

4.4.3 Energy in Dielectric Systems

Recall that the energy stored in the electric field is given by (2.15),

$$W = \frac{\epsilon_0}{2} \int \mathbf{E}^2 dV.$$

In the presence of linear dielectric, the energy in the system is

$$W = \frac{\epsilon_0}{2} \int \epsilon_r \mathbf{E}^2 dV = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dV. \quad (4.8)$$

Proof. Assume the dielectric is fixed in position. Now suppose there is already some ρ_f present, and we bring in a small free charge $\delta\rho_f$. The work done on building up this small free charge is

$$\delta W = \int (\delta\rho_f) \phi dV = \int [\nabla \cdot (\delta\mathbf{D})] \phi dV,$$

where $\delta\mathbf{D}$ is the resultant change in electric displacement. The integral is taken over all space. Integrating by parts,

$$\begin{aligned} \delta W &= \int \nabla \cdot [(\delta\mathbf{D})\phi] dV - \int (\delta\mathbf{D}) \cdot (\nabla\phi) dV \\ &= \int_{\partial V} [(\delta\mathbf{D})\phi] \cdot d\mathbf{a} + \int (\delta\mathbf{D}) \cdot \mathbf{E} dV \\ &= \int (\delta\mathbf{D}) \cdot \mathbf{E} dV. \end{aligned}$$

In the second line we used the divergence theorem and the fact that this surface integral vanishes when integrating over all space. Note that these results work for any material. If the medium is a linear dielectric, $\mathbf{D} = \epsilon\mathbf{E}$, so

$$(\delta\mathbf{D}) \cdot \mathbf{E} = \epsilon(\delta\mathbf{E}) \cdot \mathbf{E} = \frac{1}{2} \delta(\epsilon\mathbf{E}^2) = \frac{1}{2} \delta(\mathbf{D} \cdot \mathbf{E}).$$

Thus, the infinitesimal work is

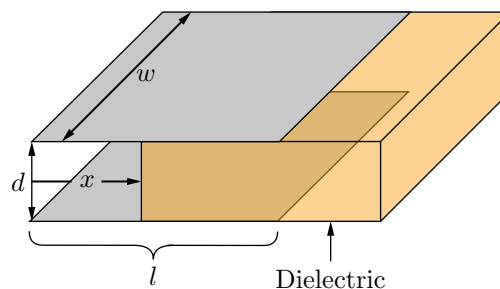
$$\delta W = \delta \left(\frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dV \right),$$

which gives the total work

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dV.$$

4.4.4 Forces on Dielectrics

Take a slab of linear dielectric material for example. It is to be inserted between a parallel-plate capacitor.



It is always assumed that the field is uniform inside the capacitor, and zero outside. If this is the case there will be no force on the dielectric. The actual field of a parallel-plate capacitor has some **fringing fields** around the edges. It is these fringing fields that pulls the dielectric inside. We already know that higher portion of dielectric inside the plates, the higher energy is stored. Thus the electrostatic force on the dielectric must be responsible for the increase in energy:

$$dW = -F dx \implies F = -\frac{dW}{dx}.$$

The negative sign is to ensure that the physics work properly. Assuming that the charge Q on the plates is held constant, the force on the dielectric is given by

$$F = -\frac{dW}{dx} = -\frac{d}{dx} \left(\frac{1}{2} \frac{Q^2}{C} \right) = \frac{1}{2} \frac{Q^2}{C^2} \frac{dC}{dx} = \frac{1}{2} \phi^2 \frac{dC}{dx}.$$

This is general for all capacitors. For parallel plate capacitors, the capacitance (with dielectric partially inserted according to the diagram) is

$$C = \frac{\epsilon_0}{d} [wx + \epsilon_r w(l - x)] = \frac{\epsilon_0 w}{d} (\epsilon_r l - \chi_e x).$$

Hence the force is

$$F = -\frac{\epsilon_0 \chi_e w}{2d} \phi^2.$$

5 MAGNETOSTATICS

In previous chapters about electrostatics, we are confined to situations where the *source* charges are not moving. In the following chapters, we will talk about forces between charges in motion.

5.1 The Lorentz Force Law

The question arises from wires in which there are currents present. When the currents are set up in the same direction, the wires repel each other; when the currents are in opposite direction, the wires attract. You may think that the attraction (or repulsion) is due to electric fields generated by the currents, but this is wrong. A wire with currents does not have a net charge. It is true that there are negative charges flowing, but there are also stationary positive charges inside whatever metal the wire is made of. A stationary test charge placed near a wire indeed does not get attracted or repelled. There must be a new field arising. We call this field, produced by moving charges, the **magnetic field $\mathbf{B}$** .

5.1.1 Magnetic Fields

The units of magnetic fields is **Tesla** (T). Another useful unit is **Gauss** (Gs), and $1 \text{ T} = 10^4 \text{ Gs}$. Magnetic fields can be detected much more easily than electric fields. All you need to do is to place a compass in space, and the needle will point in the direction of magnetic field. If you do this near a wire, you will find that the magnetic field circles around the wire according to the *right hand rule*. The thumb indicates the direction of current flow, and other fingers curl around the magnetic field.

5.1.2 Magnetic Forces

The force on a moving charge q with velocity $\mathbf{v}$ due to a magnetic field $\mathbf{B}$ is given by the **Lorentz force law**:

$$\boxed{\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}.} \quad (5.1)$$

This law is an axiom of electromagnetism (just like Coulomb's law), which is found by experiments. The cross product form is due to Oliver Heaviside. If there are also electric field present, the total force on the charge is the sum of them,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

From the Lorentz force law (5.1), we see that magnetic field does no work in *any* cases,

$$dW_B = \mathbf{F}_B \cdot d\boldsymbol{\ell} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0$$

because $\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$ (and $\mathbf{B}$), and the dot product vanishes. No work done means the speed of a particle inside a magnetic field does not change, but its direction does.

Example 5.1. Cyclotron motion

A charge q is in a circular motion in a uniform magnetic field $\mathbf{B}$. The magnetic force works as a centripetal force on the charge,

$$qvB = \frac{mv^2}{R}.$$

We can work out the radius of such circular trajectory,

$$R = \frac{mv}{Bq}. \quad (5.2)$$

This radius is called the **Larmor radius**, particularly useful in plasma physics. The frequency of the motion depends only on the charge-to-mass ratio and the strength of the magnetic field,

$$f = \frac{1}{T} = \frac{v}{2\pi R} = \frac{vBq}{2\pi mv} = \frac{Bq}{2\pi m}. \quad (5.3)$$

It has a name called the **cyclotron frequency**. The earliest particle accelerators were built based on this property of magnetic force.

5.1.3 Currents

The **current** in a wire is the charge per unit time passing a given point. By this definition, negative charges moving to the left is the same as positive ones to the right. Even though it is really electrons (negative charges) moving, we define the current as the flow of positive charges by convention. The current has units of Coulombs/second, or **amperes** (A). Apparently currents have some direction, so it should be a vector.

Line Current A line charge λ moving down a wire has a current given by

$$\mathbf{I} = \frac{dq}{dt} = \frac{\lambda d\ell}{dt} = \frac{\lambda(\mathbf{v} dt)}{dt} = \lambda\mathbf{v}. \quad (5.4)$$

Here λ refers only to moving charges. The force on the wire due to a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_\lambda = \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \lambda d\ell = \int (\mathbf{I} \times \mathbf{B}) d\ell,$$

We can assume that $\mathbf{I}$ and $d\ell$ are in the same direction. In this way,

$$\mathbf{F}_\lambda = \int I(d\ell \times \mathbf{B}). \quad (5.5)$$

If the current is constant in magnitude,

$$\mathbf{F}_\lambda = I \int (d\ell \times \mathbf{B}).$$

Surface Current When charges flow on a surface, the **surface current density** $\mathbf{K}$ is defined as

$$\mathbf{K} \equiv \frac{d\mathbf{I}}{d\ell_\perp}. \quad (5.6)$$

K is seen as the current per unit width. If the surface charge density σ is flowing in a velocity $\mathbf{v}$, then the surface current density is also

$$\mathbf{K} = \sigma \mathbf{v}. \quad (5.7)$$

The magnetic force on the surface current is

$$\mathbf{F}_\sigma = \int (\mathbf{v} \times \mathbf{B}) \sigma da = \int (\mathbf{K} \times \mathbf{B}) da. \quad (5.8)$$

Volume Current The most general current is a volume current with **volume current density** $\mathbf{J}$ (or simply “current density”), which is defined as

$$\mathbf{J} \equiv \frac{d\mathbf{I}}{da_\perp}. \quad (5.9)$$

Like line current and surface current, the volume current density is related to the usual volume charge density

$$\mathbf{J} = \rho \mathbf{v} \quad (5.10)$$

and the force on the current is

$$\mathbf{F}_\rho = \int (\mathbf{v} \times \mathbf{B}) \rho dV = \int (\mathbf{J} \times \mathbf{B}) dV. \quad (5.11)$$

5.1.4 The Continuity Equation

For a closed surface S , we can derive the total current through it by the definition of current density (5.9),

$$I = \oint_S J da = \mathbf{J} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{J}) dV,$$

where V is the volume enclosed by S . We know that charge is conserved: rate of charge flowing out of a volume must be accompanied by a decrease in charge inside the volume per unit time, so

$$\int_V (\nabla \cdot \mathbf{J}) dV = -\frac{d}{dt} \int_V \rho dV = -\int_V \left(\frac{\partial \rho}{\partial t} \right) dV.$$

This applies to any volume, so we have that

$$\boxed{\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0.} \quad (5.12)$$

This is a mathematical statement of local charge conservation. It is called the **continuity equation**.

5.2 The Biot-Savart Law

5.2.1 Steady Currents

It is found that steady currents produce magnetic fields that are constant in time. The theory of steady currents is called **magnetostatics**. Mathematically, electrostatics and magnetostatics means

$$\frac{\partial \rho}{\partial t} = 0, \quad \frac{\partial \mathbf{J}}{\partial t} = 0$$

at all space and all times. The continuity equation (5.12) in electro/magnetostatics becomes

$$\nabla \cdot \mathbf{J} = 0.$$

5.2.2 The Magnetic Field of a Steady Current

The magnetic field of a steady line current is given by the **Biot-Savart law**:

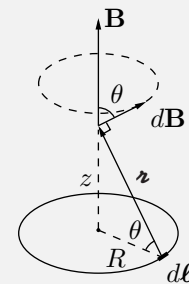
$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\boldsymbol{\ell}' \times \hat{\mathbf{z}}}{z^2}. \quad (5.13)$$

$d\boldsymbol{\ell}'$ is the line element along the current wire, and $\mathbf{z}$ is the displacement vector from the source $\mathbf{r}'$ to $\mathbf{r}$. The (exact) constant $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$ is called the **permeability of free space**. The Biot-Savart law is the magnetostatic version of Coulomb's law, both carrying a $1/r^2$ factor.

Example 5.2. Magnetic field above the center of a circular current loop

Find the magnetic field a distance z above the center of a circular loop of radius R , which carries a steady current I .

Set up a cylindrical coordinates such that the center of the loop is at the origin and the loop lies in the xy -plane. The line element $d\boldsymbol{\ell}$ is in the $\hat{\phi}$ direction. Taking a cross product with $\hat{\mathbf{z}}$ gives the magnetic field element $d\mathbf{B}$, in angle θ with the z -axis. As the line element goes over the loop, $d\mathbf{B}$ sweeps out a cone. All $\hat{\mathbf{r}}$ components cancel and $\hat{\mathbf{z}}$ components add up. By azimuthal symmetry, the total $\hat{\mathbf{z}}$ component is



$$B(z) = \frac{\mu_0 I}{4\pi} \int \frac{d\ell'}{z^2} \cos \theta = \frac{\mu_0 I}{4\pi} \frac{R \cos \theta}{z^2} \int_0^{2\pi} d\phi = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}.$$

5.3 The Divergence and Curl of Magnetic Field

We will derive the divergence and curl of magnetic field directly from the Biot-Savart law for a most general volume current:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{z}}}{z^2} dV'.$$

Since we are interested in the divergence and curl of $\mathbf{B}(\mathbf{r})$, all ∇ 's are taken with respect to $\mathbf{r}$ coordinates, not $\mathbf{r}'$ coordinates, unless otherwise specified.

5.3.1 The Divergence

Taking the divergence of $\mathbf{B}$:

$$\nabla \cdot \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{z^2} \right) dV'.$$

By the product rule of divergence (1.6),

$$\nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{z^2} \right) = \frac{\hat{\mathbf{z}}}{z^2} \cdot (\nabla \times \mathbf{J}) - \mathbf{J} \cdot \left(\nabla \times \frac{\hat{\mathbf{z}}}{z^2} \right) = 0.$$

The first term vanishes because $\nabla \times \mathbf{J} = 0$ ($\mathbf{J}$ is a function of $\mathbf{r}'$, not $\mathbf{r}$). The second term vanishes as $\nabla \times (\hat{\mathbf{z}}/z^2) = 0$. In conclusion,

$$\boxed{\nabla \cdot \mathbf{B} = 0.} \quad (5.14)$$

All magnetic fields are divergence free.

5.3.2 The Curl

Taking the curl of $\mathbf{B}$:

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{z^2} \right) dV'.$$

By the product rule of curl (1.8),

$$\nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{z}}}{z^2} \right) = \mathbf{J} \left(\nabla \cdot \frac{\hat{\mathbf{z}}}{z^2} \right) - (\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{z^2},$$

where again all derivatives of $\mathbf{J}$ are dropped. Now let us look at the first term,

$$\frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \left(\nabla \cdot \frac{\hat{\mathbf{z}}}{z^2} \right) dV = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') 4\pi \delta^3(\mathbf{r} - \mathbf{r}') dV = \mu_0 \mathbf{J}(\mathbf{r}).$$

by $\nabla \cdot (\hat{\mathbf{z}}/z^2) = 4\pi \delta^3(\mathbf{r})$. The second term involves a subtle change of differentiation variable. The trick is to use ∇' instead of ∇ with a cost of a minus sign:

$$-(\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{z^2} = (\mathbf{J} \cdot \nabla') \frac{\hat{\mathbf{z}}}{z^2}.$$

We can do this because $\mathbf{J}(\mathbf{r}')$ is not inside the differentiation, and $\partial_x f(x - x') = -\partial_{x'} f(x - x')$. For convenience, we will do this for the x component,

$$(\mathbf{J} \cdot \nabla') \left(\frac{x - x'}{z^3} \right) = \nabla' \cdot \left(\frac{x - x'}{z^3} \mathbf{J} \right) - \left(\frac{x - x'}{z^3} \right) (\nabla' \cdot \mathbf{J}),$$

using product rule of divergence (1.5). For steady current, $\nabla' \cdot \mathbf{J} = 0$, so

$$\left[-(\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{z}}}{z^2} \right]_x = \nabla' \cdot \left(\frac{x - x'}{z^3} \mathbf{J} \right).$$

Invoking the divergence theorem,

$$\int_V \nabla' \cdot \left[\frac{(x - x')}{z^3} \mathbf{J} \right] dV' = \oint_{\partial V} \frac{x - x'}{z^3} \mathbf{J} \cdot d\mathbf{a}'.$$

This V in the original Biot-Savart law encloses all currents, so $\mathbf{J} = 0$ on ∂V . This integral contributes nothing to $\nabla \times \mathbf{B}$. The same works for y and z components. In conclusion,

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J}.} \quad (5.15)$$

This equation is called [Ampère's law in differential form](#).

5.3.3 Ampère's Law

Using Stokes' theorem,

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}.$$

We arrive at [Ampère's law in integral form](#):

$$\boxed{\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}} \quad (5.16)$$

where I_{enc} is the current enclosed by the **Amperian loop**. Ampère's law says that the circulation of $\mathbf{B}$ around a loop is proportional to the current enclosed by the loop. This is the analogous to Gauss's law where the flux of electric field through a Gaussian surface is proportional to the charge enclosed by the surface.

Example 5.3. Infinite Straight Wire

Find the magnetic field around a infinite straight wire carrying a current I .

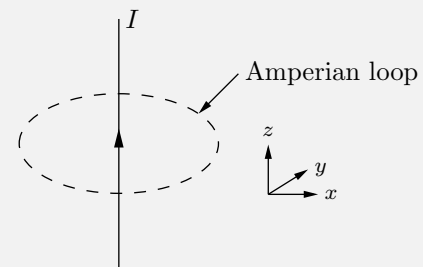
Let the wire lie on the z -axis. First let us make some symmetry arguments. By translational symmetry along z -direction, $B_z = 0$. What about radial components? Suppose the radial component of $\mathbf{B}$ is pointing outward. We know that B is linearly proportional to the current, so if we flip the direction of current, the radial component of $\mathbf{B}$ will point inward. Now rotate the whole configuration by 180° around the y -axis. The current returns to its original state, but now $\mathbf{B}$ has inward radial component. The only way to resolve this is that $B_r = 0$ over all space, so the only nontrivial component is B_ϕ . Consider a circular Amperian loop of radius r centered at the wire and with its normal parallel to the wire.

By Ampère's law,

$$\mu_0 I_{\text{enc}} = \oint_{\mathbf{B}} \cdot d\boldsymbol{\ell} = \int_0^{2\pi} (B_\phi \hat{\boldsymbol{\phi}}) \cdot (d\ell \hat{\boldsymbol{\phi}}) = 2\pi r B_\phi.$$

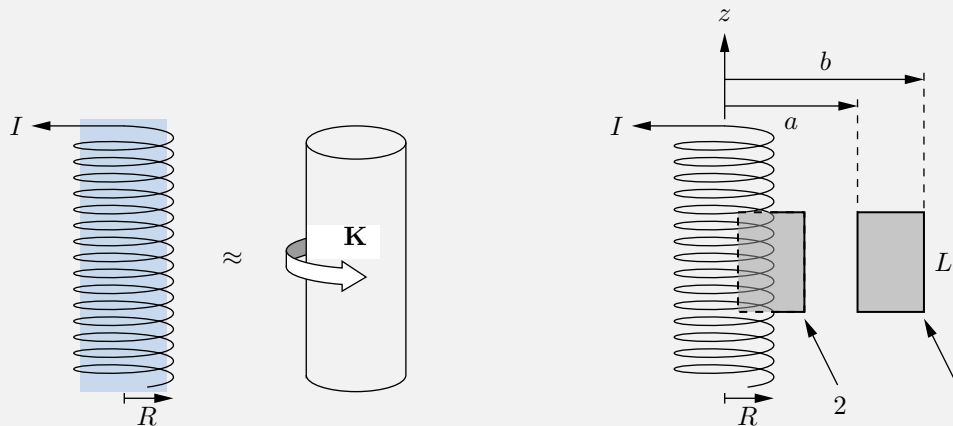
The magnetic field over all space is

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\boldsymbol{\phi}}.$$



Example 5.4. Solenoids

Find the magnetic field of an infinite solenoid, consisting of n closely wound turns per unit length on a cylinder of radius R , each carrying a steady current I . You may approximate the solenoid as a surface current running through a metal sheet foil wrapped around the cylinder (left figure below). The surface current density is $\mathbf{K} = nI\hat{\boldsymbol{\phi}}$.



Same as the last example, we will first make some analysis on $\mathbf{B}$. Similar arguments from infinite wire apply here: first, $\mathbf{B}$ cannot have a radial component. Second, the azimuthal symmetry requires that $\mathbf{B}$ is independent of ϕ , and translational symmetry along $\hat{\mathbf{z}}$ makes it independent of z . We can gain one more information directly from the Biot-Savart law: if we were to use it, the integrand is some cross product of $\mathbf{K}$. But $\mathbf{K}$ is entirely in the $\hat{\boldsymbol{\phi}}$ direction, so $B_\phi = 0$. To sum up, $\mathbf{B}$ is along the $\hat{\mathbf{z}}$ direction, and is only a function of r . Moreover, $\mathbf{B}$ had better be zero as $r \rightarrow \infty$.

Consider the Amperian loop 1. It encloses no current, so by Ampère’s law,

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = [B_{\text{out}}(a) - B_{\text{out}}(b)]L = \mu_0 I_{\text{enc}} = 0 \implies B_{\text{out}}(a) = B_{\text{out}}(b).$$

Since a and b are arbitrary radius, the field outside is constant. If the field at infinity is zero, then it means the field outside the solenoid is zero everywhere. Same logic, the field inside the solenoid is also constant, but this time not zero. Consider the Amperian loop 2. The outer half of the loop contributes nothing to the line integral, so

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = B_{\text{in}}L = \mu_0 I_{\text{enc}} = \mu_0 nIL \implies B_{\text{in}} = \mu_0 nI.$$

In conclusion,

$$\mathbf{B} = \begin{cases} \mu_0 nI \hat{\mathbf{z}} & \text{inside the solenoid,} \\ 0 & \text{outside the solenoid.} \end{cases} \quad (5.17)$$

Solenoids are like parallel-plate capacitors in electrostatics. It can produce a strong magnetic field in a confined region (in the blue region above) and zero elsewhere.

5.3.4 Comparison of Magnetostatics and Electrostatics

For now, we have acquired [Maxwell’s equations](#) for electrostatics and magnetostatics in very simple but elegant forms:

$$\begin{cases} \nabla \cdot \mathbf{E} = \rho/\epsilon_0, \\ \nabla \times \mathbf{E} = 0, \\ \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \end{cases}$$

They are consequences of Coulomb’s law, Biot-Savart law, and principle of superposition. These equations, plus the force law $\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ constitutes the whole formulation of electrostatics and magnetostatics. Let us compare these two fields.

- Electric fields have “sources” because they have a nonzero divergence when charges are present. Magnetic fields curls around a current but does not have [magnetic charges](#) ([magnetic monopoles](#)). It is not proved, though, that magnetic monopoles does not exist, we just have not found one.
- Electric forces are typically much larger than magnetic ones—just compare $1/4\pi\epsilon_0$ with $\mu_0/4\pi$ —they are off by $\sim 10^{16}$. It is only when both the source charges and test charges are moving at velocities comparable to the speed of light that the magnetic force approaches the electric force in strength. Then how to we produce magnetic fields but ensuring that electric fields are small? Just make huge amount of charges move while keeping the wire neutral.
- It is clear that a magnetic field does not have a scalar potential because it has a nonzero curl. But there is still one alternative: the divergence of $\mathbf{B}$ vanishes. It turns out that we can define a “vector potential” for magnetic fields. However, this potential still contains three components, so this is making magnetic fields harder to manipulate than electric fields.

5.4 Magnetic Vector Potential

5.4.1 The Vector Potential

Recall that defining the potential ϕ as $\mathbf{E} = -\nabla\phi$ ensures that $\nabla \times \mathbf{E} = 0$ automatically, because the curl of a gradient is always zero. For magnetostatics, what can we say about $\mathbf{B}$ such that $\nabla \cdot \mathbf{B}$ always? From vector calculus, the divergence of a curl is always zero, so we will define the **vector potential** $\mathbf{A}$ as

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \quad (5.18)$$

Ampère's law says that

$$\nabla \times \mathbf{B} = \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$$

It would be nice if we can throw away $\nabla(\nabla \cdot \mathbf{A})$ such that $-\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$. This is then a Poisson equation and we know how to solve it. Note that there is one more degree of freedom that have not been used in magnetostatics. You can always add some kind of “constant” to a potential that does not affect the physical quantity (like $\mathbf{E}$ or $\mathbf{B}$) in order to make calculations easy. In electrostatics for potential ϕ , the constant is a scalar because the gradient of a scalar is zero, and then we can set the condition $\phi(\infty) = 0$. Likewise, one can always exploit the **gauge freedom** to add a gradient to the vector potential without changing the magnetic field:

$$\mathbf{B}' = \nabla \times (\mathbf{A} + \nabla\lambda) = \nabla \times \mathbf{A} + \nabla \times (\nabla\lambda) = \nabla \times \mathbf{A} = \mathbf{B}.$$

Using this freedom, it turns out that there is always an $\mathbf{A}$ such that

$$\nabla \cdot \mathbf{A} = 0$$

by adding a proper $\nabla\lambda$. This is called the **Coulomb gauge**.

Proof. Suppose the vector potential $\mathbf{A}_0$ is not divergenceless. Now add a $\nabla\lambda$ to $\mathbf{A}_0$ to get a new vector potential $\mathbf{A}$:

$$\mathbf{A} = \mathbf{A}_0 + \nabla\lambda \implies \nabla \cdot \mathbf{A} = \nabla \cdot \mathbf{A}_0 + \nabla^2 \lambda.$$

To make $\nabla \cdot \mathbf{A} = 0$, it requires that

$$\nabla^2 \lambda = -\nabla \cdot \mathbf{A}_0.$$

This is again Poisson's equation. Now $\nabla \cdot \mathbf{A}_0$ is like ρ/ϵ_0 in Poisson's equation for potential ϕ . If $\nabla \cdot \mathbf{A}_0$ goes to zero at infinity, we already know the answer,

$$\lambda = \frac{1}{4\pi} \int \frac{\nabla \cdot \mathbf{A}_0}{z} dV'.$$

If $\nabla \cdot \mathbf{A}_0$ does not go to zero at infinity, λ is harder to find, but there still exist a solution for a particular boundary condition. The whole point is: it is always possible to make the vector potential divergenceless. $\square$

As desired, with divergenceless $\mathbf{A}$, Ampère's law becomes

$$\boxed{\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}.} \quad (5.19)$$

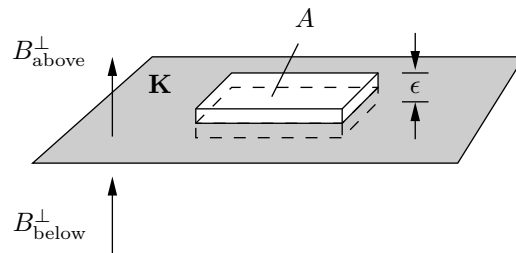
This is three Poisson's equations because the Laplacian is on a vector. Assuming $\mathbf{J}(\infty) = 0$, the solution is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{z} dV' = \frac{\mu_0 I}{4\pi} \int \frac{d\ell'}{z} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}')}{z} da'. \quad (5.20)$$

The vector potential $\mathbf{A}$ is not as useful as ϕ . First, it is still a vector with three components. Second, while potential has a close connection to potential energy, vector potential does not have a simple physical meaning because magnetic fields do no work.

5.4.2 Boundary Conditions

This section is very similar to 2.3.4. Imagine a small sheet with a surface current $\mathbf{K}$ on it. We want to know the behavior of the magnetic field just above and below the sheet.



Using $\nabla \cdot \mathbf{B}$, we will find that the normal component of the magnetic field is continuous.

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}.$$

By $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$, the parallel component of the magnetic field satisfies $B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$. The jump condition of the magnetic field can be summarized as

$$\mathbf{B}_{\text{above}} - \mathbf{B}_{\text{below}} = \mu_0 (\mathbf{K} \times \hat{\mathbf{n}}). \quad (5.21)$$

Like scalar potential in electrostatics, the vector potential is continuous across any surface:

$$\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}}.$$

Since $\mathbf{B}$ is not continuous, we can work out component by component using (5.21) and $\nabla \times \mathbf{A} = \mathbf{B}$ to get the boundary condition for $\mathbf{A}$:

$$\frac{\partial \mathbf{A}_{\text{above}}}{\partial n} - \frac{\partial \mathbf{A}_{\text{below}}}{\partial n} = -\mu_0 \mathbf{K}. \quad (5.22)$$

where $\partial \mathbf{A} / \partial n = \nabla \mathbf{A} \cdot \hat{\mathbf{n}}$.

5.4.3 Multipole Expansion of the Vector Potential

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{d\ell'}{r} = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\ell'.$$

The magnetic monopole term vanishes because $\oint d\ell' = 0$. The dipole term is

$$\boxed{\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}}, \quad (5.23)$$

where $\mathbf{m}$ is the **magnetic dipole moment**:

$$\boxed{m \equiv I \int d\mathbf{a} = I\mathbf{S}}. \quad (5.24)$$

Here $\mathbf{S}$ is the “vector area” of the loop. If the loop is flat, $\mathbf{S}$ is the ordinary area enclosed with direction according to the right-hand rule.

To be written.

6 MAGNETIC FIELDS IN MATTER

To be written.

7 ELECTRODYNAMICS

7.1 Electromotive Force

7.1.1 Ohm's Law

For most substances, the current density $\mathbf{J}$ is proportional to the force per unit charge $\mathbf{f}$:

$$\mathbf{J} = \sigma \mathbf{f}. \quad (7.1)$$

The constant of proportionality σ is experimentally determined and is called the **conductivity** of the substance. A **perfect conductor** has $\sigma = \infty$, while a **perfect insulator** has $\sigma = 0$. Here the force per unit charge $\mathbf{f}$ can be driven by anything: gravitational, chemical, electromagnetic, etc. For our purpose, we will only consider electromagnetic forces, so (7.1) becomes

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

In general, the velocity of charges is small that the second term is negligible, so

$$\boxed{\mathbf{J} = \sigma \mathbf{E}}. \quad (7.2)$$

This is called **Ohm's law**, a special case of (7.1). Materials that obey Ohm's law are called **ohmic materials**. For steady currents, charge cannot build up along a wire. By the continuity equation (5.12), this means

$$\nabla \cdot \mathbf{E} = \frac{1}{\sigma} \nabla \cdot \mathbf{J} = 0$$

for uniform conductivity inside the conductor. Once we know that $\nabla \cdot \mathbf{E} = \nabla^2 \phi / \epsilon_0 = 0$, we can use all tools developed in Chapter 3 to solve Laplace's equation.

Theorem 7.1. Uniform electric field inside a wire

Consider a uniform wire of cross-sectional area A , length L , and a uniform conductivity σ . Let the potential be constant over each, and the potential difference between the ends be ϕ_0 . The electric field inside the wire is uniform.

Proof. Inside the wire, the potential ϕ obeys Laplace's equation. We can let one end of the potential to be $\phi = 0$ and the other end $\phi = \phi_0$. On the cylindrical surface of the wire, $\mathbf{J} \cdot \hat{\mathbf{n}} = 0$, so $\mathbf{E} \cdot \hat{\mathbf{n}} = \partial\phi/\partial n = 0$. Now the boundary conditions (ϕ or $\partial\phi/\partial n$) are specified on all surfaces. By the third uniqueness theorem 3.4, the electric field inside is uniquely determined.

We guess the solution to be

$$\phi(z) = \frac{\phi_0 z}{L} \implies \mathbf{E} = -\nabla\phi = -\frac{V_0}{L} \hat{\mathbf{z}}.$$

where z is measured along the wire axis. All boundary conditions are satisfied, and this electric field is uniform throughout the

wire. □

Example 7.1. Ohm's law

We know that the electric field is uniform throughout the wire. By (7.2), the current density is also uniform. The current I is easily calculated,

$$I = JA = \sigma EA = \frac{\sigma A}{L} \phi_0.$$

This is the more familiar Ohm's law, where we identify the quantity $\sigma A/L$ as the reciprocal of resistance, $1/R$, of the wire.

7.1.2 Electromotive Force

Now consider a closed loop of wire of cross-sectional area A , length L , and uniform conductivity σ . The average current inside the wire (over the total length L) is, by Ohm's law (7.2),

$$I_{\text{ave}} = \frac{A}{L} \oint_C \mathbf{J} \cdot d\boldsymbol{\ell} = \frac{A\sigma}{L} \int_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0.$$

This means if we only consider electrostatic force, there will be no net current around the loop at all! There must be some kind of source that produce the current. Let the force per unit charge be

$$\mathbf{f} = \mathbf{E} + \mathbf{f}_s,$$

where $\mathbf{f}_s$ is the force provided by the source. Such a source can be many things: the most common one is a battery (a chemical force). It can also be mechanical pressure, temperature gradient, photoelectric, etc. The net effect of a source is to produce an average current

$$I_{\text{ave}} = \frac{A\sigma}{L} \oint_C \mathbf{f}_s \cdot d\boldsymbol{\ell}.$$

We define the line integral as the **electromotive force** (EMF),

$$\boxed{\mathcal{E} = \oint_C \mathbf{f}_s \cdot d\boldsymbol{\ell}.} \tag{7.3}$$

Within an ideal source of emf with $\sigma = \infty$, such as a resistanceless battery, the net force on the charges is zero because of (7.1). Thus, the electric field cancels the source force, $\mathbf{E} = -\mathbf{f}_s$. The potential difference between the terminals is

$$\phi = - \int_a^b \mathbf{E} \cdot d\boldsymbol{\ell} = \oint \mathbf{f}_s \cdot d\boldsymbol{\ell} = \mathcal{E}.$$

7.1.3 Motional EMF

The most common way to produce an EMF is to move a wire through a magnetic field. Such an EMF is called a **motional EMF**.

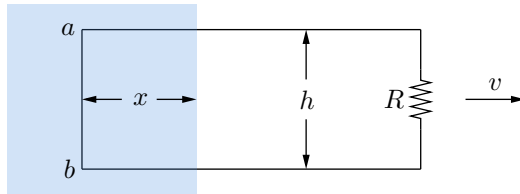


Figure 7.1: A circuit moving through a uniform magnetic field $\mathbf{B}$.

As shown in Figure 7.1, a circuit is moving through a uniform magnetic field $\mathbf{B}$ (blue region) with speed v . Because of this motion, the charges in the segment ab is moving to the right with speed v and hence experience a magnetic force of magnitude $F_B = qvB$. In this case, the magnetic force produces an EMF

$$\mathcal{E} = \oint_C \mathbf{f}_B \cdot d\boldsymbol{\ell} = \int_a^b (vB) d\ell = vBh.$$

Other segments do not produce any EMF. Charges in horizontal segments inside the magnetic field experience a vertical force, perpendicular to the wire, so $\mathbf{f}_{\text{hor}} \cdot d\boldsymbol{\ell} = 0$. Charges in other segments experience no forces. Note that magnetic force always does no work, which means all the work is provided by the person pulling the circuit.

A more elegant way to find motional EMF is to use the magnetic flux

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{a}.$$

The flux inside the circuit loop in Figure 7.1 is $\Phi = Bhx$. As the loop is moving, the flux decreases at a rate

$$\frac{d\Phi_B}{dt} = Bh \frac{dx}{dt} = -vBh = \mathcal{E}.$$

It can be shown that the **flux rule** for motional EMF is independent of the geometry of the loop. The EMF is only related to the rate of change of the flux:

$$\boxed{\mathcal{E} = -\frac{d\Phi_B}{dt}.} \quad (7.4)$$

Proof. Suppose the circuit loop C is in arbitrary shape in same magnetic field $\mathbf{B}(\mathbf{r})$. Each infinitesimal line element $d\boldsymbol{\ell}$ of the loop is moving with some velocity $\mathbf{v}(\mathbf{r}, t)$. (The shape of the loop is not constrained in its original one in this case. It can be stretched or shrunk by external forces.) The Lorentz force law $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ gives the charge inside the loop some motion. Let the velocity of the charge at each $d\boldsymbol{\ell}$ have velocity $\mathbf{u}(\mathbf{r}, t)$ relative to the wire. The total motional EMF is

$$\mathcal{E} = \oint_C \mathbf{f}_B \cdot d\boldsymbol{\ell} = \oint_C d\boldsymbol{\ell} \cdot [(\mathbf{v} + \mathbf{u}) \times \mathbf{B}].$$

Using the scalar triple product formula (1.1), $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$,

$$\mathcal{E} = \oint_C [d\boldsymbol{\ell} \times (\mathbf{v} + \mathbf{u})] \cdot \mathbf{B} = - \oint_C \mathbf{B} \cdot [(\mathbf{v} + \mathbf{u}) \times d\boldsymbol{\ell}].$$

The motion of charges is always parallel to the wire, so $\mathbf{u} \times d\boldsymbol{\ell} = 0$. We are left with

$$\mathcal{E} = - \oint_C \mathbf{B} \cdot (\mathbf{v} \times d\boldsymbol{\ell}) = -\frac{1}{dt} \oint_C \mathbf{B} \cdot (\mathbf{v} dt \times d\boldsymbol{\ell}).$$

Note that this cross product is between two vectors with dimensions of length. The geometric meaning of a cross product is the area of the parallelogram formed by its sides $\mathbf{v} dt$ and $d\boldsymbol{\ell}$, with a direction normal to both vectors. Since $\mathbf{v} dt$ is the displacement of $d\boldsymbol{\ell}$

within time dt , the cross product is the area swiped out by $d\ell$ in dt . Then the line integral can be written as a difference of two surface integrals

$$\mathcal{E} = -\frac{1}{dt} \int_{S(t+dt)-S(t)} \mathbf{B} \cdot d\mathbf{a} = -\frac{1}{dt} \left[\int_{S(t+dt)} \mathbf{B} \cdot d\mathbf{a} - \int_{S(t)} \mathbf{B} \cdot d\mathbf{a} \right],$$

where $S(t)$ is the surface bounded by the loop at time t , and $S(t+dt)$ is the surface after dt . We recognize the two surface integrals as magnetic fluxes, so

$$\mathcal{E} = -\frac{1}{dt} [\Phi_B(t+dt) - \Phi_B(t)] = -\frac{d\Phi_B}{dt}.$$

□

7.2 Electromagnetic Induction

7.2.1 Faraday's Law

The development of electromagnetism happened prior to the existence of special relativity. In 1831, Michael Faraday performed three experiments and reported as follows:

1. Pulling a loop of wire to the right through a magnetic field produces a current in the loop.
2. Moving the magnet to the left to the loop produces a current in the loop.
3. Changing the strength of the magnetic field when both the magnet and the loop are at rest also produces a current in the loop.

The first conclusion is exact the motional EMF. The second conclusion should be expected because the relative motion is the same as the first one. However, relativity was not formulated during the era of classical electrodynamics. If the loop is stationary, there is no charge moving inside the magnetic field, so no current should be produced. The only (EM) interaction that can move stationary charges is the electric field. Faraday claimed that *a changing magnetic field induces an electric field*. He found empirically that the EMF is equal to the rate of change of the magnetic flux,

$$\mathcal{E} = \oint \mathbf{E} \cdot d\ell = -\frac{d\Phi}{dt}.$$

If we write out the flux integral, we get **Faraday's law in integral form**:

$$\oint_C \mathbf{E} \cdot d\ell = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}. \quad (7.5)$$

By Stokes' theorem, (7.5) can be converted to **Faraday's law in differential form**:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (7.6)$$

Now we are in the realm of electrodynamics, where $\nabla \times \mathbf{E}$ no longer vanishes. Faraday's law reduces to $\nabla \times \mathbf{E} = 0$ in magnetostatics (constant $\mathbf{B}$). It turns out that Faraday's law works for all three of his experiments, though their mechanism are different when viewing from classical electromagnetism. This suggests that there are some theory that explains all three mechanisms in Faraday's law, and that requires special relativity.

The negative sign in Faraday's law is important. It tells us that whenever there is a change in flux, an electric field is induced to counter that change. This phenomenon is known as **Lenz's law**. For example, consider a circular conducting loop in the xy -plane

and let $d\mathbf{a} = \hat{\mathbf{z}} da$. Now turn on a magnetic field $d\mathbf{B}$ in the $\hat{\mathbf{z}}$ direction. An induced electric field will flow in the $-\hat{\phi}$ direction, producing a magnetic field in the $-\hat{\mathbf{z}}$ direction according to the right hand rule, opposite to $d\mathbf{B}$.

In the absence of charge density, $\rho = 0$, we have

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

They have the same form as the two equations in magnetostatics,

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

Hence we can use the analog to Biot-Savart law (5.13) to determine the induced electric field,

$$\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B} / \partial t) \times \hat{\mathbf{z}}}{z^2} dV = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\mathbf{z}}}{z^2} dV.$$

If there is symmetry, the integral form of Faraday's law can use the same trick as Ampère's law,

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt} \iff \oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I.$$

7.2.2 Inductance

Consider two loops of wire at rest. Running a current I_1 in loop 1 produces a magnetic field $\mathbf{B}_1$. Through loop 2 there will be some magnetic flux of $\mathbf{B}_1$. By Biot-Savart law (5.13), we know that $\mathbf{B}_1$ is proportional to I_1 ,

$$\mathbf{B}_1 = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\boldsymbol{\ell}_1 \times \hat{\mathbf{z}}}{z^2} \propto I_1.$$

The flux through loop 2 is then also proportional to I_1 ,

$$\Phi_2 = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{a}_2 = M_{21} I_1.$$

We call the constant of proportionality M_{21} the **mutual inductance** of the two loops. By Stokes' theorem, the flux through loop 2 can be written in terms of the vector potential,

$$\Phi_2 = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{a}_2 = \int_{S_2} (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 = \oint_{C_2} \mathbf{A}_1 \cdot d\boldsymbol{\ell}_2.$$

According to (5.20),

$$\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\boldsymbol{\ell}_1}{z},$$

so we can write the flux and mutual inductance in a double line integral form

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \left(\oint_{C_1} \frac{d\boldsymbol{\ell}_1}{z} \right) \cdot d\boldsymbol{\ell}_2 \implies M_{21} = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\boldsymbol{\ell}_1 \cdot d\boldsymbol{\ell}_2}{z}. \quad (7.7)$$

This double line integral is called **Neumann formula**. Here z represent the distance between each line element $d\boldsymbol{\ell}_1$ to $d\boldsymbol{\ell}_2$. In general, such a double line integral is hard to solve. Nonetheless, we can extract some useful information about mutual inductance. First, it depends only on the geometry of the two loops, similar to capacitance. Second, if we exchange the role of loop 1 and 2, the mutual inductance is unchanged; the flux through loop 1. This means if we run a current around loop 2 instead of loop 1 while keeping the same geometry, the flux through loop 1 will be the same as Φ_2 when we run a current around loop 1. We will just call

$$M_{21} = M_{12} = M.$$

Now forget about loop 2. When we turn on the current I_1 inside loop 1, the magnetic field strength within loop 1 itself also increases, proportional to I_1 . The flux through loop 1 is then proportional to the current

$$\Phi = LI, \quad (7.8)$$

where L is called the **self-inductance** or simply **inductance**. The units of inductance is henries ($H = V \cdot s/A$). It depends on the geometry of loop 1 only. By Faraday's law, a change in current will induce an EMF inside the loop,

$$\mathcal{E} = -L \frac{dI}{dt}.$$

The inductance is always a positive quantity. Lenz's law says that if you change the current inside the loop, there is an induced **back EMF** that oppose the change in current. The greater L is, the harder to change the current.

7.2.3 Energy in Magnetic Fields

To keep the current going, one must do work against the back EMF. This fixed amount of work is recoverable—when the current is turned off, the energy is back. The work done per unit charge against the back EMF in one trip around the circuit is $-\mathcal{E}$. The current I represents the charge per unit time passing through the wire, so the power against the back EMF is

$$\frac{dW}{dt} = \frac{dW}{dQ} \frac{dQ}{dt} = -\mathcal{E}I = LI \frac{dI}{dt}.$$

To start with no current at $t = 0$ and reach the final current I , the work done on the charges integrated over time is

$$W = \frac{1}{2} LI^2. \quad (7.9)$$

It turns out that the work done can be interpreted as the energy stored in magnetic fields. The flux through the loop is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{a} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}.$$

Meanwhile, the definition of inductance (7.8) says $\Phi = LI$, so

$$LI = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}.$$

Then (7.9) becomes

$$W = \frac{1}{2} I \oint_C \mathbf{A} \cdot d\boldsymbol{\ell} - \frac{1}{2} \oint_C (\mathbf{A} \cdot \mathbf{I}) d\ell.$$

In this form, it is convenient to generalize the work to volume currents $\mathbf{J}$ and relate it to magnetic field $\mathbf{B}$,

$$W = \frac{1}{2} \int_V (\mathbf{A} \cdot \mathbf{J}) dV = \frac{1}{2\mu_0} \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) dV.$$

Using the product rule (1.6),

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \implies \mathbf{A} \cdot (\nabla \times \mathbf{B}) = \mathbf{B} \cdot \mathbf{B} - \nabla \cdot (\mathbf{A} \times \mathbf{B}).$$

The work done, in terms of $\mathbf{A}$ and $\mathbf{B}$, is

$$W = \frac{1}{2\mu_0} \left[\int_V \mathbf{B}^2 dV - \int_V \nabla \cdot (\mathbf{A} \times \mathbf{B}) dV \right] = \frac{1}{2\mu_0} \left[\int_V \mathbf{B}^2 dV - \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a} \right].$$

The work is fixed no matter how we choose the volume V (as long as it encloses all currents). $\mathbf{B}^2$ is non-negative, so choosing a larger volume increases the first integral while decreases the second. We can extend the volume to enclose over all space. For a finite current distribution, $B \sim 1/r^2$ and $A \sim 1/r$ but $S \sim r^2$ at large r , so the second integral vanishes when evaluated at infinity. The energy stored in magnetic field is

$$\boxed{W = \frac{1}{2\mu_0} \int \mathbf{B}^2 dV.} \quad [\text{all space}]. \quad (7.10)$$

Comparing (7.10) to (2.15), we find that they are very similar:

$$\begin{aligned} W_E &= \frac{1}{2} \int (\rho\phi) dV = \frac{\epsilon_0}{2} \int \mathbf{E}^2 dV, \\ W_B &= \frac{1}{2} \int (\mathbf{A} \cdot \mathbf{J}) dV = \frac{1}{2\mu_0} \int \mathbf{B}^2 dV. \end{aligned}$$

7.3 Maxwell's Equations

Currently we know four equations of electromagnetic theory:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}. \end{aligned}$$

All of them are recorded as Maxwell's equations except Ampère's law. When we take the divergence on both sides,

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J}).$$

The divergence of curl is always zero, but the RHS may not be zero. The continuity equation (5.12) says that $\nabla \cdot \mathbf{J} = -\partial\rho/\partial t$. In magnetostatics, we assumed that $\nabla \cdot \mathbf{J} = 0$, but it works only for steady currents. Since the LHS being zero is a pure mathematical result, the problem has to do with the RHS.

7.3.1 Fixing Ampère's Law

The conclusion found by Maxwell was that the continuity equation was right and Ampère's law was incomplete. By the continuity equation and Gauss's law,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot \mathbf{E}) = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).$$

To cancel with $\nabla \cdot \mathbf{J}$ in Ampère's law, we arrive at

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},} \quad (7.11)$$

known as **Maxwell-Ampère's law**. Like Faraday's law, it means that *a changing electric field induces a magnetic field*. Maxwell called the extra term **displacement current**:

$$\mathbf{J}_d \equiv \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (7.12)$$

This extra term plays a crucial role in electromagnetic waves.

7.3.2 Maxwell's Equations

We now conclude the full glory of classical electromagnetism—Maxwell's equations. In differential form,

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \quad (7.13)$$

In this form, we know that an electric field can be produced either by charges ρ or changing magnetic fields $\partial \mathbf{B} / \partial t$; a magnetic field can be produced either by currents $\mathbf{J}$ or changing electric fields $\partial \mathbf{E} / \partial t$. However, $\partial \mathbf{B} / \partial t$ and $\partial \mathbf{E} / \partial t$ themselves are due to charges and currents, so a more physical way to write Maxwell's equations is

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, \\ \nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J}. \end{aligned}$$

The charges and currents are all on one side. Note that even the continuity equation is encoded in Maxwell's law (specifically, in Maxwell-Ampère's law). The only additional equation needed to describe classical electromagnetism is the Lorentz force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Maxwell's equations tells how charges produce fields; the Lorentz force law tells how fields affect charges.

8 CONSERVATION LAWS

The first conservation law in electromagnetism is the *local* conservation of charge. Within any region, the current flowing out must account for the change in charge inside the region. It is summarized in the continuity equation,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

This chapter deals with other conservation laws such as local conservation of energy and momentum. There are new concepts introduced, such as the energy density, momentum density, energy current, and momentum current, analogous to ρ and $\mathbf{J}$.

8.1 Poynting's Theorem

We already know the energy stored in electric fields $\mathbf{E}$ and magnetic fields $\mathbf{B}$ as

$$W_E = \frac{\epsilon_0}{2} \int \mathbf{E}^2 dV, \quad W_B = \frac{1}{2\mu_0} \int \mathbf{B}^2 dV.$$

The total energy stored in electromagnetic fields per unit volume should be

$$u = \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2 \right). \quad (8.1)$$

Consider a charge and current configuration that produces $\mathbf{E}$ and $\mathbf{B}$ at time t . The charges will move subject to these fields. According to the Lorentz force law, the work done on a charge q within time dt is

$$dW = \mathbf{F} \cdot d\boldsymbol{\ell} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = q\mathbf{E} \cdot \mathbf{v} dt.$$

We can replace $q \rightarrow \int \rho dV$ and $\rho\mathbf{v} \rightarrow \mathbf{J}$ so that

$$\frac{dW}{dt} = \frac{1}{dt} \int_V (\rho \mathbf{E} \cdot \mathbf{v} dt) dV = \int_V (\mathbf{E} \cdot \mathbf{J}) dV.$$

The integrand $\mathbf{E} \cdot \mathbf{J}$ is interpreted as the power delivered per unit volume. Using Maxwell-Ampère's law (7.11), write the integrand all in terms of fields,

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}.$$

By the produce rule (1.6) and Faraday's law (7.6),

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \nabla \cdot (\mathbf{E} \times \mathbf{B}) = -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}).$$

Plugging this into $\mathbf{E} \cdot \mathbf{J}$,

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \left[-\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}) \right] - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} = -\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}).$$

Then the work per unit time becomes

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \left[\frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2 \right) + \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) \right] dV.$$

By the divergence theorem,

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2 \right) dV - \frac{1}{\mu_0} \oint_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}.$$

This equation is known as **Poynting's theorem**. We recognize the first integral as the total energy within the volume V , so the second integral should represent the energy transported out of V by electromagnetic fields. The second integrand, the energy per unit time, per unit area, transported by the fields is called the **Poynting vector**:

$$\mathbf{S} \equiv \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}). \quad (8.2)$$

The Poynting vector is also the **energy flux density** across a surface. In terms of the Poynting vector and energy density, Poynting's theorem can be written compactly as

$$\boxed{\frac{dW}{dt} = -\frac{d}{dt} \int_V u dV - \oint_S \mathbf{S} \cdot d\mathbf{a}.} \quad (8.3)$$

If there is no work done on the charges in V , or if V does not contain any charges,

$$\int_V \frac{\partial u}{\partial t} dV = -\oint_S \mathbf{S} \cdot d\mathbf{a} = -\int_V (\nabla \cdot \mathbf{S}) dV,$$

or in differential form,

$$\nabla \cdot \mathbf{S} + \frac{\partial u}{\partial t} = 0. \quad (8.4)$$

This is the continuity equation for energy density. It infers the local conservation of electromagnetic energy. In general, electromagnetic energy is not conserved because moving charges carry away kinetic energy (mechanical instead of electromagnetic). Only when there is no work on charges, (8.4) holds true.

8.2 Momentum

8.2.1 Maxwell's Stress Tensor

The total electromagnetic force on the charges in a volume V is

$$\mathbf{F} = \int_V \rho(\mathbf{E} + \mathbf{v} \times \mathbf{B}) dV = \int_V (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) dV.$$

The force per unit volume is $\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}$. Again, to write all quantities in terms of fields, we use Maxwell's equations,

$$\begin{aligned} \mathbf{f} &= \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B} \quad \leftarrow (\text{Gauss's law \& Maxwell-Ampère's law}) \\ &= \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \times \mathbf{B} - \epsilon_0 \left[\frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) - \mathbf{E} \times \frac{\partial \mathbf{B}}{\partial t} \right] \\ &= \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} - \frac{1}{\mu_0} \mathbf{B} \times (\nabla \times \mathbf{B}) - \epsilon_0 \left[\frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) + \mathbf{E} \times (\nabla \times \mathbf{E}) \right] \quad \leftarrow (\text{Faraday's law}) \\ &= \epsilon_0 [(\nabla \cdot \mathbf{E}) \mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] - \frac{1}{\mu_0} [\mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) \\ &= \epsilon_0 [(\nabla \cdot \mathbf{E}) \mathbf{E} - \mathbf{E} \times (\nabla \times \mathbf{E})] - \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B}) \mathbf{B} - \mathbf{B} \times (\nabla \times \mathbf{B})] - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) \quad \leftarrow (\nabla \cdot \mathbf{B} = 0). \end{aligned}$$

The last step is to make things symmetrical. The product rule (1.4) says that

$$\nabla(\mathbf{E}^2) = 2(\mathbf{E} \cdot \nabla)\mathbf{E} + 2\mathbf{E} \times (\nabla \times \mathbf{E}) \implies \mathbf{E} \times (\nabla \times \mathbf{E}) = \frac{1}{2}\nabla(\mathbf{E}^2) - (\mathbf{E} \cdot \nabla)\mathbf{E}$$

and similar for $\mathbf{B} \times (\nabla \times \mathbf{B})$. Thus, the force per unit volume becomes

$$\mathbf{f} = \epsilon_0[(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0}[(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] - \frac{1}{2}\nabla\left(\epsilon_0\mathbf{E}^2 + \frac{1}{\mu_0}\mathbf{B}^2\right) - \epsilon_0\frac{\partial}{\partial t}(\mathbf{E} \times \mathbf{B}).$$

To simplify this expression further, we define the **Maxwell stress tensor**:

$$T_{ij} \equiv \epsilon_0\left(E_i E_j - \frac{1}{2}\delta_{ij}\mathbf{E}^2\right) + \frac{1}{\mu_0}\left(B_i B_j - \frac{1}{2}\delta_{ij}\mathbf{B}^2\right). \quad (8.5)$$

Here i, j can be x, y, z . There are in total nine components in Maxwell stress tensor $\hat{\mathbf{T}}$. The divergence of the Maxwell stress tensor has three components labeled by j ,

$$(\nabla \cdot \hat{\mathbf{T}})_j = \epsilon_0\left[(\nabla \cdot \mathbf{E})E_j + (\mathbf{E} \cdot \nabla)E_j - \frac{1}{2}\nabla_j(\mathbf{E}^2)\right] + \frac{1}{\mu_0}\left[(\nabla \cdot \mathbf{B})B_j + (\mathbf{B} \cdot \nabla)B_j - \frac{1}{2}\nabla_j(\mathbf{B}^2)\right].$$

It is more convenient to show this in component form,

$$\begin{aligned} (\nabla \cdot \hat{\mathbf{T}})_j &= \sum_i \partial_i T_{ij} = \epsilon_0 \sum_i \left[\partial_i (E_i E_j) - \frac{1}{2} \delta_{ij} \partial_i (\mathbf{E}^2) \right] + \frac{1}{\mu_0} \sum_i \left[\partial_i (B_i B_j) - \frac{1}{2} \delta_{ij} \partial_i (\mathbf{B}^2) \right] \\ &= \epsilon_0 \sum_i \left[(\partial_i E_i) E_j - (E_i \partial_i) E_j - \frac{1}{2} \partial_j (\mathbf{E}^2) \right] + \frac{1}{\mu_0} \sum_i \left[(\partial_i B_i) B_j - (B_i \partial_i) B_j - \frac{1}{2} \partial_j (\mathbf{B}^2) \right] \\ &= \epsilon_0 \left[(\nabla \cdot \mathbf{E}) E_j + (\mathbf{E} \cdot \nabla) E_j - \frac{1}{2} \nabla_j (\mathbf{E}^2) \right] + \frac{1}{\mu_0} \left[(\nabla \cdot \mathbf{B}) B_j + (\mathbf{B} \cdot \nabla) B_j - \frac{1}{2} \nabla_j (\mathbf{B}^2) \right]. \end{aligned}$$

In terms of Maxwell stress tensor, the force per unit volume is

$$\mathbf{f} = \nabla \cdot \hat{\mathbf{T}} - \mu_0 \epsilon_0 \frac{\partial \mathbf{S}}{\partial t},$$

where $\mathbf{S}$ is the Poynting vector (8.2). The total electromagnetic force on the charges in V , is the integral

$$\boxed{\mathbf{F} = \oint_S \hat{\mathbf{T}} \cdot d\mathbf{a} - \mu_0 \epsilon_0 \frac{d}{dt} \int_V \mathbf{S} dV.} \quad (8.6)$$

8.2.2 Conservation of Momentum

Applying Newton's second law, $\mathbf{F} = d\mathbf{p}_m/dt$, (8.6) can be written in the form

$$\frac{d\mathbf{p}_m}{dt} = -\mu_0 \epsilon_0 \frac{d}{dt} \int_V \mathbf{S} dV + \oint_S \hat{\mathbf{T}} \cdot d\mathbf{a},$$

where $\mathbf{p}_m$ is the mechanical momentum of the particles in V . This is the conservation of momentum in electromagnetism. Compare this to Poynting's theorem (8.3), the first integral represents the momentum stored in the fields, while the second integral is the momentum per unit time flowing in through the surface. Whenever there is a decrease in field momentum or fields that carries momentum into the volume, mechanical momentum increases. We define the **momentum density** of the fields as

$$\mathbf{g} \equiv \mu_0 \epsilon_0 \mathbf{S} = \epsilon_0 (\mathbf{E} \times \mathbf{B}). \quad (8.7)$$

If the mechanical momentum in V is constant, then

$$\nabla \cdot \hat{\mathbf{T}} - \frac{\partial \mathbf{g}}{\partial t} = 0. \quad (8.8)$$

This is the continuity equation for electromagnetic momentum. We can further define the angular momentum density,

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g} = \epsilon_0 [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})]. \quad (8.9)$$

Note that even static fields can have momentum or angular momentum as long as $\mathbf{E} \times \mathbf{B}$ is nonzero.

9 ELECTROMAGNETIC WAVES

9.1 Waves in One Dimension

9.1.1 The Wave Equation

A wave, mathematically, is any function f that satisfies the wave equation. The one-dimensional wave equation is

$$\frac{\partial^2 f}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}. \quad (9.1)$$

The constant v is the speed of the wave. It is easy to obtain two kinds of solution: $f(x, t) = g(x - vt)$ and $f(x, t) = h(x + vt)$, where g and h can be arbitrary functions. The wave equation is a linear differential equation, so the most general solution to the wave equation can be expressed as a linear combination of $g(x - vt)$ and $h(x + vt)$:

$$f(x, t) = C_1 g(x - vt) + C_2 h(x + vt),$$

for some constants C_1 and C_2 . Physically, $g(x - vt)$ represents a wave that travels in the $+x$ -direction, while $h(x + vt)$ represents a wave that travels in the $-x$ -direction. A typical wave can be sinusoidal,

$$f(x, t) = A \cos[k(x - vt) + \delta],$$

where $A > 0$ is the **amplitude**, and $0 \leq \delta < 2\pi$ (or sometimes $-\pi < \delta \leq \pi$) is the **phase constant**. The argument of the cosine is called the **phase**. The quantity k is called the **wavenumber**, related to the wavelength λ by

$$k = \frac{2\pi}{\lambda}.$$

The **period** T of the wave is defined as the time for it to travel over one wavelength, and the inverse of it is the **frequency** ν ,

$$T = \frac{\lambda}{v} = \frac{2\pi}{kv}, \quad \nu = \frac{1}{T} = \frac{v}{\lambda}.$$

In physics, a more useful quantity is the angular frequency $\omega = 2\pi\nu = kv$. In terms of ω , the sinusoidal wave can be expressed as

$$f(x, t) = A \cos(kx - \omega t + \delta).$$

Since complex exponentials are more easy to handle with, a sinusoidal wave can also be written as $\tilde{f}(x, t) = \tilde{A}e^{i(kx - \omega t)}$, where $\tilde{A} = Ae^{i\delta}$ is the complex amplitude. The actual wave will be the real part $f(x, t) = \text{Re}[\tilde{f}(x, t)]$.

9.1.2 Boundary Conditions

Suppose the wave does not travel forever but hit some boundary. Consider two strings with different tension (and hence different velocity or wavenumber) attached together with a knot at $x = 0$. Let the incident wave come from the left in string 1:

$$\tilde{f}_i(x, t) = \tilde{A}_i e^{i(k_1 x - \omega t)}, \quad [x < 0].$$

As it reaches the boundary at $x = 0$, there will be a reflected wave moving to the left,

$$\tilde{f}_r(x, t) = \tilde{A}_r e^{i(-k_1 x - \omega t)}, \quad [x < 0].$$

and a transmitted wave to the right that continues in string 2,

$$\tilde{f}_t(x, t) = \tilde{A}_t e^{i(k_2 x - \omega t)}, \quad [x > 0].$$

The frequency is determined by the one shaking the string. The net effect on the string will be

$$\tilde{f}(x, t) = \begin{cases} \tilde{A}_i e^{i(k_1 x - \omega t)} + \tilde{A}_r e^{i(-k_1 x - \omega t)}, & x < 0, \\ \tilde{A}_t e^{i(k_2 x - \omega t)}, & x > 0. \end{cases}$$

The boundary conditions are the following:

1. $f(x, t)$ is continuous at $x = 0$,

$$f(0^-, t) = f(0^+, t).$$

Physically, it means the string does not break at the knot when the wave is present.

2. df/dx is also continuous at $x = 0$,

$$\left. \frac{\partial f}{\partial x} \right|_{0^-} = \left. \frac{\partial f}{\partial x} \right|_{0^+}.$$

Physically, if the knot has negligible mass, there should be no net force on it. Otherwise, it will have infinite acceleration. Then if df/dx is not continuous at $x = 0$, the tension in the strings (which is always tangent to them) will yield a nonzero net force. Only when the two tensions are parallel but opposite, or mathematically with the same derivative at the knot, there is no net force.

Since complex exponentials are linear combinations of sines and cosines, the real boundary condition also applies to complex wave functions:

$$\tilde{f}(0^-, t) = \tilde{f}(0^+, t), \quad \left. \frac{\partial \tilde{f}}{\partial x} \right|_{0^-} = \left. \frac{\partial \tilde{f}}{\partial x} \right|_{0^+}.$$

Applying the boundary conditions, we can find

$$\tilde{A}_r = \left(\frac{k_1 - k_2}{k_1 + k_2} \right) \tilde{A}_i, \quad \tilde{A}_t = \left(\frac{2k_1}{k_1 + k_2} \right) \tilde{A}_i.$$

Note that for fixed frequency $\omega = kv$, the wavenumber is inversely related to the speed, so

$$\tilde{A}_r = \left(\frac{v_2 - v_1}{v_2 + v_1} \right) \tilde{A}_i, \quad \tilde{A}_t = \left(\frac{2v_2}{v_2 + v_1} \right) \tilde{A}_i.$$

The real amplitudes and phases satisfy

$$A_r e^{i\delta_r} = \left(\frac{v_2 - v_1}{v_2 + v_1} \right) A_i e^{i\delta_i}, \quad A_t e^{i\delta_t} = \left(\frac{2v_2}{v_2 + v_1} \right) A_i e^{i\delta_i}.$$

The amplitudes are determined only by the ratio of speeds but not the phase because all $e^{i\delta}$ has amplitude 1. On the other hand, to match both real and imaginary part of $e^{i\delta}$, all phases should either be the same or off by π . For example, if $v_2 > v_1$ (which occurs when the second string is lighter), then

$$e^{i\delta_r} = e^{i\delta_i}, \quad e^{i\delta_t} = e^{i\delta_i} \implies \delta_r = \delta_t = \delta_i.$$

If $v_2 < v_1$ (when the second string is heavier), then

$$e^{i\delta_r} = -e^{i\delta_i}, \quad e^{i\delta_t} = e^{i\delta_i} \implies \delta_r + \pi = \delta_t = \delta_i.$$

9.1.3 Polarization

Polarization is a characteristic of transverse waves that oscillates perpendicular to the direction of propagation. In general, the wave can exist in any direction: it can be vertical $\tilde{f}_v(x, t) = \tilde{A}e^{i(kx - \omega t)}\hat{\mathbf{z}}$, horizontal $\tilde{f}_h(x, t) = \tilde{A}e^{i(kx - \omega t)}\hat{\mathbf{y}}$, or in any direction $\tilde{f}(x, t) = \tilde{A}e^{i(kx - \omega t)}\hat{\mathbf{n}}$. The direction $\hat{\mathbf{n}}$ is called the **polarization vector**. It defines the plane of vibration. Transverse waves require $\hat{\mathbf{n}} \cdot \hat{\mathbf{x}} = 0$ because $\hat{\mathbf{x}}$ is the propagation direction. The polarization vector can also be expressed in terms of the polarization angle θ (from the z -axis),

$$\hat{\mathbf{n}} = \hat{\mathbf{y}} \sin \theta + \hat{\mathbf{z}} \cos \theta.$$

In particular, the wave $\tilde{f}(x, t) = \tilde{A}e^{i(kx - \omega t)}\hat{\mathbf{n}}$ can be seen as a superposition of two waves,

$$\tilde{f}(x, t) = \tilde{A}\hat{\mathbf{z}} \cos \theta e^{i(kx - \omega t)} + \tilde{A}\hat{\mathbf{y}} \sin \theta e^{i(kx - \omega t)}.$$

9.2 Electromagnetic Waves in Vacuum

9.2.1 The Wave Equation for $\mathbf{E}$ and $\mathbf{B}$

In vacuum, there is no charge or current. Maxwell's equations in vacuum are

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{aligned}$$

Applying the curl to the last two equations and using $\nabla \cdot \mathbf{E} = 0$, $\nabla \cdot \mathbf{B} = 0$,

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}, \quad \nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = -\nabla^2 \mathbf{B}.$$

Meanwhile,

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla \times (\nabla \times \mathbf{B}) = \nabla \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.$$

Hence we get two wave equations for $\mathbf{E}$ and $\mathbf{B}$ in three dimension:

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.} \quad (9.2)$$

This means electromagnetic (EM) waves can propagate in vacuum with speed $v = 1/\sqrt{\mu_0 \epsilon_0} = 3.00 \times 10^8 \text{ m/s} \equiv c$, the speed of light.

9.2.2 Monochromatic Plane Waves

Different frequencies of EM waves correspond to different colors (in the visible range), so sinusoidal waves of single frequency ω are called **monochromatic**. **Plane waves** travels in a single direction. For example, plane waves traveling in z -direction does not have x or y dependence. Then we are interested in

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}, \quad \tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(kz - \omega t)},$$

where $\omega = ck$. Not all solutions to the wave equation obey Maxwell's equations because there are constraints to EM waves $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Taking the divergence to the solutions $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{B}}$ forces that

$$(\tilde{E}_0)_z = 0, \quad (\tilde{B}_0)_z = 0.$$

This means electromagnetic waves are transverse. Taking the curl of $\mathbf{E}$ and apply Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ gives a relation between wave amplitudes:

$$-k(\tilde{E}_0)_y = \omega(\tilde{B}_0)_x, \quad k(\tilde{E}_0)_x = \omega(\tilde{B}_0)_y \implies \tilde{\mathbf{B}}_0 = \frac{k}{\omega}(\hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0).$$

This means $\mathbf{E}$ and $\mathbf{B}$ are mutually perpendicular, and perpendicular to the propagation direction. The real amplitudes obeys the simple relation

$$B_0 = \frac{E_0}{c}. \quad (9.3)$$

Since the z -direction is arbitrary, the general monochromatic plane waves can have the **propagation wave vector** $\mathbf{k}$ in the direction of propagation. Given the polarization vector $\hat{\mathbf{n}}$, the plane waves can be expressed as

$$\begin{aligned} \tilde{\mathbf{E}}(\mathbf{r}, t) &= \tilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{\mathbf{n}}, \\ \tilde{\mathbf{B}}(\mathbf{r}, t) &= \frac{1}{c} \tilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{n}}) = \frac{1}{c} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}, \end{aligned} \quad (9.4)$$

with $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = 0$ for transverse waves. The real plane wave solutions are

$$\begin{aligned} \mathbf{E}(\mathbf{r}, t) &= E_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta) \hat{\mathbf{n}}, \\ \mathbf{B}(\mathbf{r}, t) &= \frac{E_0}{c} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \delta) (\hat{\mathbf{k}} \times \hat{\mathbf{n}}). \end{aligned}$$

9.2.3 Energy and Momentum in Electromagnetic Waves

The energy density in electromagnetic fields is given by (8.1). For monochromatic plane waves,

$$u = \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2 \right) = \frac{1}{2} \left(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0 c^2} \mathbf{E}^2 \right) = \frac{1}{2} (\epsilon_0 \mathbf{E}^2 + \epsilon_0 \mathbf{E}^2) = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta).$$

Note that the electric field and magnetic field have the same contribution. The energy flux transported by the fields is given by the Poynting vector (8.2):

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{\mathbf{z}} = cu\hat{\mathbf{z}}.$$

The momentum density in the fields (8.7) is

$$\mathbf{g} = \mu_0\epsilon_0\mathbf{S} = \frac{1}{c^2}\mathbf{S} = \frac{1}{c}u\hat{\mathbf{z}}.$$

The period of electromagnetic waves are typically short compared to macroscopic measurement. Hence it is more useful to know the average value of energy and momentum densities. The average of $\cos^2$ over one period is $1/2$, so

$$\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2, \quad \langle \mathbf{S} \rangle = \frac{1}{2}c\epsilon_0 E_0^2\hat{\mathbf{z}}, \quad \langle \mathbf{g} \rangle = \frac{1}{2c}\epsilon_0 E_0^2\hat{\mathbf{z}}.$$

The average power per unit area transported by the electromagnetic wave is the **intensity**, given by

$$I \equiv \langle \mathbf{S} \rangle = \frac{1}{2}c\epsilon_0 E_0^2. \quad (9.5)$$

For a perfect absorber of surface area A , when a light shines on it, the fields transfer momentum. Within time δt , the momentum transferred is the volume integral of $\langle \mathbf{g} \rangle$, or $\delta \mathbf{p} = \langle \mathbf{g} \rangle A c \delta t$. The **radiation pressure**, given by the average force per unit area, is

$$P = \frac{1}{A} \frac{dp}{dt} = c\langle g \rangle = \frac{1}{2}\epsilon_0 E_0^2 = \frac{I}{c}.$$

A perfect reflector has twice the pressure because the momentum switches direction instead of being absorbed.

9.3 Electromagnetic Waves in Matter

9.3.1 Propagation in Linear Media

Inside matter where there is no free charge or free current, Maxwell's equations read

$$\begin{aligned} \nabla \cdot \mathbf{D} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{H} &= \frac{\partial \mathbf{D}}{\partial t}. \end{aligned}$$

If the medium is linear and homogeneous ($\mathbf{D} = \epsilon\mathbf{E}$ and $\mathbf{H} = \mathbf{B}/\mu$ with constant ϵ and μ), Maxwell's equations reduce to

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu\epsilon\frac{\partial \mathbf{E}}{\partial t}. \end{aligned}$$

This is almost identical to Maxwell's equations in vacuum, except that $\mu_0 \rightarrow \mu$ and $\epsilon_0 \rightarrow \epsilon$. The speed of EM waves inside a linear homogeneous medium is

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}, \quad \text{where } n = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} \quad (9.6)$$

is the **index of refraction** of the medium. Most materials have $\mu \approx \mu_0$, so their indices of refraction are $n \approx \sqrt{\epsilon_r}$, where ϵ_r is the dielectric constant. In most cases $\epsilon_r > \epsilon$, so light moves more slowly in matter. All results from the last section works in matter, with $\mu_0 \rightarrow \mu$, $\epsilon_0 \rightarrow \epsilon$, and $c \rightarrow v$. The results are

$$u = \frac{1}{2} \left(\epsilon \mathbf{E}^2 + \frac{1}{\mu} \mathbf{B}^2 \right), \quad \mathbf{S} = \frac{1}{\mu} (\mathbf{E} \times \mathbf{B}), \quad I = \frac{1}{2} \epsilon v E_0^2.$$

Physics becomes more interesting when light travels from one medium into another. Like the example of strings, we need boundary conditions (in electromagnetic version):

$$\epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp, \quad \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel, \quad B_1^\perp = B_2^\perp, \quad \frac{1}{\mu_1} \mathbf{B}_1^\parallel = \frac{1}{\mu_2} \mathbf{B}_2^\parallel. \quad (9.7)$$

9.3.2 Reflection and Transmission at Normal Incidence

Consider two linear homogeneous media, labeled 1 and 2, with the boundary at the xy -plane. A plane wave of frequency ω travels in the z -direction and is polarized in the x -direction. The incident wave is

$$\tilde{\mathbf{E}}_i(z, t) = \tilde{E}_{0,i} e^{i(k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_i(z, t) = \frac{E_{0,i}}{v_1} e^{i(k_1 z - \omega t)} \hat{\mathbf{y}},$$

where $v_1 = 1/\sqrt{\mu_1 \epsilon_1}$. When it hits the boundary, there is a reflected wave and transmitted wave

$$\begin{aligned} \tilde{\mathbf{E}}_r(z, t) &= \tilde{E}_{0,r} e^{i(-k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_r(z, t) = -\frac{\tilde{E}_{0,r}}{v_1} e^{i(-k_1 z - \omega t)} \hat{\mathbf{y}}, \\ \tilde{\mathbf{E}}_t(z, t) &= \tilde{E}_{0,t} e^{i(k_2 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_t(z, t) = \frac{\tilde{E}_{0,t}}{v_2} e^{i(k_2 z - \omega t)} \hat{\mathbf{y}}. \end{aligned}$$

(The negative sign in $\tilde{\mathbf{B}}_r$ is to ensure that $\tilde{\mathbf{E}}_r \times \tilde{\mathbf{B}}_r$ is in the $-z$ -direction.) The combined fields, $\tilde{\mathbf{E}}_i + \tilde{\mathbf{E}}_r$ and $\tilde{\mathbf{B}}_i + \tilde{\mathbf{B}}_r$, in medium 1 and $\tilde{\mathbf{E}}_t$ and $\tilde{\mathbf{B}}_t$ in medium 2 should satisfy boundary conditions (9.7). There are no perpendicular components at the boundary. The parallel components of $\mathbf{E}$ need to satisfy

$$\tilde{E}_{0,i} + \tilde{E}_{0,r} = \tilde{E}_{0,t} \quad \text{and,}$$

while the parallel components of $\mathbf{B}$ says

$$\frac{1}{\mu_1} \left(\frac{\tilde{E}_{0,i}}{v_1} - \frac{\tilde{E}_{0,r}}{v_2} \right) = \frac{1}{\mu_2} \left(\frac{\tilde{E}_{0,t}}{v_2} \right) \implies \tilde{E}_{0,i} - \tilde{E}_{0,r} = \beta \tilde{E}_{0,t},$$

where $\beta \equiv \mu_1 v_1 / \mu_2 v_2 = \mu_1 n_2 / \mu_2 n_1$. This is a linear system of equations. The outgoing amplitudes $E_{0,r}$ and $E_{0,t}$ can be solved in terms of $E_{0,i}$:

$$\tilde{E}_{0,r} = \left(\frac{1 - \beta}{1 + \beta} \right) \tilde{E}_{0,i}, \quad \tilde{E}_{0,t} = \left(\frac{2}{1 + \beta} \right) \tilde{E}_{0,i}.$$

If $\mu_1, \mu_2 \approx \mu_0$ so that $\beta = v_1 / v_2$, we get a result identical to our string example:

$$\tilde{E}_{0,r} = \left(\frac{v_2 - v_1}{v_2 + v_1} \right) \tilde{E}_{0,i}, \quad \tilde{E}_{0,t} = \frac{2v_2}{v_2 + v_1} \tilde{E}_{0,i}.$$

Again, the reflected wave is in phase if $v_2 > v_1$ and out of phase if $v_2 < v_1$. The real amplitudes are

$$E_{0,r} = \left| \frac{v_2 - v_1}{v_2 + v_1} \right| E_{0,i} = \left| \frac{n_1 - n_2}{n_1 + n_2} \right| E_{0,i}, \quad E_{0,t} = \left(\frac{2v_2}{v_2 + v_1} \right) E_{0,i} = \left(\frac{2n_1}{n_1 + n_2} \right) E_{0,i}.$$

To find the energy transmitted and reflected, we use $I = \frac{1}{2}\epsilon v E_0^2$. The **reflection coefficient** R and the **transmission coefficient** are

$$R \equiv \frac{I_r}{I_i} = \left(\frac{E_{0,r}}{E_{0,i}} \right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2, \quad T \equiv \frac{I_t}{I_i} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \left(\frac{E_{0,t}}{E_{0,i}} \right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2}.$$

Note that $R + T = 1$. (In fact, we can check that $R + T = 1$ holds even if $\mu_1, \mu_2 \neq \mu_0$, as required by the conservation of energy.)

9.3.3 Reflection and Transmission at Oblique Incidence

Reflection and transmission at oblique incidence is a generalization of normal incidence. The setup is shown in Figure 9.1:

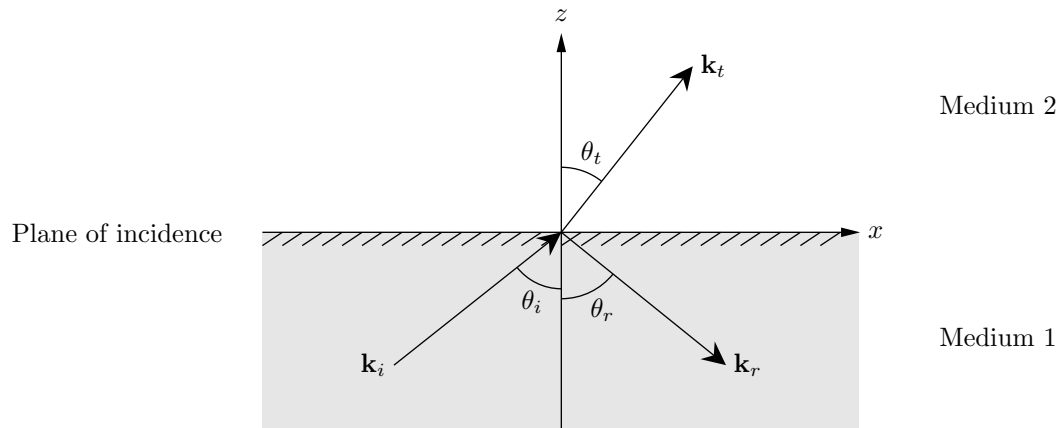


Figure 9.1: Reflection and transmission of electromagnetic wave at oblique incidence.

Consider a monochromatic plane wave

$$\tilde{\mathbf{E}}_i(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0,i} e^{i(\mathbf{k}_i \cdot \mathbf{r} - \omega t)}, \quad \tilde{\mathbf{B}}_i(\mathbf{r}, t) = \frac{1}{v_1} (\hat{\mathbf{k}}_i \times \tilde{\mathbf{E}}_i).$$

The reflected wave and transmitted wave are

$$\begin{aligned} \tilde{\mathbf{E}}_r(\mathbf{r}, t) &= \tilde{\mathbf{E}}_{0,r} e^{i(\mathbf{k}_r \cdot \mathbf{r} - \omega t)}, & \tilde{\mathbf{B}}_r(\mathbf{r}, t) &= \frac{1}{v_1} (\hat{\mathbf{k}}_r \times \tilde{\mathbf{E}}_r), \\ \tilde{\mathbf{E}}_t(\mathbf{r}, t) &= \tilde{\mathbf{E}}_{0,t} e^{i(\mathbf{k}_t \cdot \mathbf{r} - \omega t)}, & \tilde{\mathbf{B}}_t(\mathbf{r}, t) &= \frac{1}{v_2} (\hat{\mathbf{k}}_t \times \tilde{\mathbf{E}}_t). \end{aligned}$$

Linear media do not change the frequency of the light, so the three wave numbers are related by

$$k_i v_1 = k_r v_1 = k_t v_2 = \omega \quad \implies \quad k_i = k_r = \frac{v_2}{v_1} k_t = \frac{n_1}{n_2} k_t.$$

The boundary conditions have the generic form

$$[\quad] e^{i(\mathbf{k}_i \cdot \mathbf{r} - \omega t)} + [\quad] e^{i(\mathbf{k}_r \cdot \mathbf{r} - \omega t)} = [\quad] e^{i(\mathbf{k}_t \cdot \mathbf{r} - \omega t)}.$$

Note that it holds for the plane $z = 0$ at all times, or for all values of x , y , and t . It can be proved that if $Ae^{iax} + Be^{ibx} = Ce^{icx}$ for some nonzero constants A, B, C, a, b, c and for all x , then $a = b = c$ and $A + B = C$. This means for $z = 0$,

$$\mathbf{k}_i \cdot \mathbf{r} = \mathbf{k}_r \cdot \mathbf{r} = \mathbf{k}_t \cdot \mathbf{r} \quad \implies \quad k_{i,x} = k_{r,x} = k_{t,x}, \quad k_{i,y} = k_{r,y} = k_{t,y}.$$

The two equations of wave number give us three laws:

1. **Plane of incidence.** The incident, reflected, and transmitted wave vectors form a plane called the **plane of incidence** that contains the z -axis. This means we can just choose the xz -plane such that all wave vectors lie in this plane. Then $k_{i,x} = k_{r,x} = k_{t,x}$ implies (see Figure 9.1)

$$k_i \sin \theta_i = k_r \sin \theta_r = k_t \sin \theta_t,$$

where θ_i is the **angle of incidence**, θ_r is the **angle of reflection**, and θ_t is the **angle of refraction**.

2. **Law of reflection.** Because $k_i = k_r$, the angle of incidence is equal to the angle of reflection:

$$\theta_i = \theta_r. \quad (9.8)$$

3. **Law of refraction (Snell's law).** Because $k_i = n_1 k_t / n_2$, the angle of refraction and the angle of incidence obey

$$n_1 \sin \theta_i = n_2 \sin \theta_t \quad (9.9)$$

These laws hold for all kinds of waves because we haven't used specific electromagnetic boundary conditions (9.7) yet. We only accounted for the exponentials (i.e. waves). Now we know the exponential factors cancel, so we can focus on the field amplitudes using (9.7):

$$\begin{cases} \text{(i)} & \epsilon_1 (\tilde{\mathbf{E}}_{0,i} + \tilde{\mathbf{E}}_{0,r})_z = \epsilon_2 (\tilde{\mathbf{E}}_{0,t})_z, \\ \text{(ii)} & (\tilde{\mathbf{E}}_{0,i} + \tilde{\mathbf{E}}_{0,r})_{x,y} = (\tilde{\mathbf{E}}_{0,t})_{x,y}, \\ \text{(iii)} & (\tilde{\mathbf{B}}_{0,i} + \tilde{\mathbf{B}}_{0,r})_z = (\tilde{\mathbf{B}}_{0,t})_z, \\ \text{(iv)} & \frac{1}{\mu_1} (\tilde{\mathbf{B}}_{0,i} + \tilde{\mathbf{B}}_{0,r})_{x,y} = \frac{1}{\mu_2} (\tilde{\mathbf{B}}_{0,t})_{x,y}. \end{cases}$$

Consider the case where the polarization vector $\hat{\mathbf{n}}$ is in the xz -plane (see Figure 9.2). We claim that the reflected and transmitted wave are also polarized in this plane.

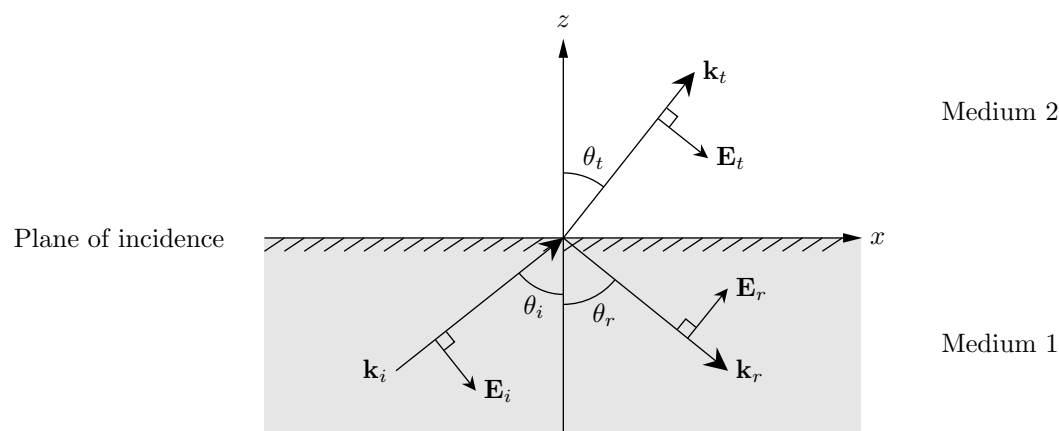


Figure 9.2: Polarization of incident, reflected, and transmitted waves.

Proof. It is assumed that $(\tilde{\mathbf{E}}_{0,i})_y = 0$. Since the magnetic field $\tilde{\mathbf{B}}_{0,i}$ is perpendicular to $\tilde{\mathbf{E}}_{0,t}$, we must have $(\tilde{\mathbf{B}}_{0,i})_x = 0$. Then the y -component of boundary condition (ii) reads $(\tilde{\mathbf{E}}_{0,r})_y = (\tilde{\mathbf{E}}_{0,t})_y$, and the x -component of boundary condition (iv) reads

$$\frac{1}{\mu_1} (\tilde{\mathbf{B}}_{0,r})_x = \frac{1}{\mu_2} (\tilde{\mathbf{B}}_{0,t})_x \implies \frac{1}{\mu_1 v_1} (\hat{\mathbf{k}}_r \times \tilde{\mathbf{E}}_{0,r})_x = \frac{1}{\mu_2 v_2} (\hat{\mathbf{k}}_t \times \tilde{\mathbf{E}}_{0,t})_x.$$

From Figure 9.2, the geometry says $\hat{\mathbf{k}}_r = \hat{\mathbf{x}} \sin \theta_r - \hat{\mathbf{z}} \cos \theta_r$ and $\hat{\mathbf{k}}_t = \hat{\mathbf{x}} \sin \theta_t + \hat{\mathbf{z}} \cos \theta_t$. Computing the cross product, we find that

$$(\hat{\mathbf{k}}_r \times \tilde{\mathbf{E}}_{0,r})_z = -(\tilde{\mathbf{E}}_{0,r})_y \sin \theta_r, \quad (\hat{\mathbf{k}}_t \times \tilde{\mathbf{E}}_{0,t})_z = (\tilde{\mathbf{E}}_{0,t})_y \sin \theta_t.$$

Boundary condition (iv) becomes

$$-\frac{1}{\mu_1 v_1}(\tilde{\mathbf{E}}_{0,r})_y \sin \theta_r = \frac{1}{\mu_2 v_2}(\tilde{\mathbf{E}}_{0,t})_y \sin \theta_t \implies -\frac{1}{\mu_1}(\tilde{\mathbf{E}}_{0,r})_y (n_1 \sin \theta_r) = \frac{1}{\mu_2}(\tilde{\mathbf{E}}_{0,t})_y (n_2 \sin \theta_t).$$

Snell's law cancels $n_1 \sin \theta_r = n_2 \sin \theta_t$, so $-(\tilde{\mathbf{E}}_{0,r})_y/\mu_1 = (\tilde{\mathbf{E}}_{0,t})_y/\mu_2$. But we know $(\tilde{\mathbf{E}}_{0,r})_y = (\tilde{\mathbf{E}}_{0,t})_y$, so both of them should be zero because permeabilities only take positive values. The electric fields does not have y -components, which means all waves are polarized in the xz -plane. $\square$

Now we can drop the component subscripts and call $(\tilde{\mathbf{E}}_{0,i})_x = E_{0,i}$ and so on. Boundary conditions become

$$\begin{cases} \text{(i)} & \epsilon_1(-\tilde{E}_{0,i} \sin \theta_i + E_{0,r} \sin \theta_r) = \epsilon_2(-\tilde{E}_{0,t}) \sin \theta_t, \\ \text{(ii)} & \tilde{E}_{0,i} \cos \theta_i + \tilde{E}_{0,r} \cos \theta_r = \tilde{E}_{0,t} \cos \theta_t, \\ \text{(iv)} & \frac{1}{\mu_1 v_1}(\tilde{E}_{0,i} - \tilde{E}_{0,r}) = \frac{1}{\mu_2 v_2} \tilde{E}_{0,t}. \end{cases}$$

(Boundary condition [iii] is trivial because magnetic fields only have y -components.) It seems like there are three equations for three unknowns $E_{0,i}$, $E_{0,r}$, and $E_{0,t}$, but (i) and (iv) are the same condition, related by Snell's law $n_1 \sin \theta_r = n_2 \sin \theta_t$ and law of reflection $\theta_i = \theta_r$. They can be written simply as

$$\tilde{E}_{0,i} - \tilde{E}_{0,r} = \beta \tilde{E}_{0,t}, \quad \text{where} \quad \beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}.$$

Boundary condition (ii) reduces to

$$\tilde{E}_{0,i} + \tilde{E}_{0,r} = \alpha \tilde{E}_{0,t}, \quad \text{where} \quad \alpha \equiv \frac{\cos \theta_t}{\cos \theta_i}.$$

Solving this system of equations gives the reflected and transmitted amplitudes:

$$\boxed{\tilde{E}_{0,r} = \left(\frac{\alpha - \beta}{\alpha + \beta} \right) \tilde{E}_{0,i}, \quad \tilde{E}_{0,t} = \left(\frac{2}{\alpha + \beta} \right) \tilde{E}_{0,i}.} \quad (9.10)$$

Equations (9.10) are known as **Fresnel's equations**. Two other Fresnel's equations can be obtained when we assume the polarization is perpendicular to the plane of incidence (see Appendix A.2.1). In a typical geometric optics experiment, we can control the angle of incidence θ_i . Indeed, both reflected and transmitted amplitudes depend on this single variable when we write α as

$$\alpha = \frac{\sqrt{1 - \sin^2 \theta_t}}{\cos \theta_i} = \frac{\sqrt{1 - [(n_1/n_2) \sin \theta_i]^2}}{\cos \theta_i}.$$

If the incident wave strikes at an angle called **Brewster's angle**, θ_B , there will be no reflected wave. According to (9.10), this occurs at $\alpha = \beta$, or

$$\sin^2 \theta_B = \frac{1 - \beta^2}{(n_1/n_2)^2 - \beta^2}. \quad (9.11)$$

The power per unit area at the interface is $\mathbf{S} \cdot \hat{\mathbf{z}}$. The incident, reflected, and transmitted intensities are

$$I_i = \frac{1}{2} \epsilon_1 v_1 E_{0,i}^2 \cos \theta_i, \quad I_r = \frac{1}{2} \epsilon_1 v_1 E_{0,r}^2 \cos \theta_r, \quad I_t = \frac{1}{2} \epsilon_2 v_2 E_{0,t}^2 \cos \theta_t.$$

The reflection and transmission coefficients are

$$R \equiv \frac{I_r}{I_i} = \left(\frac{E_{0,r}}{E_{0,i}} \right)^2 = \left(\frac{\alpha - \beta}{\alpha + \beta} \right)^2, \quad T \equiv \frac{I_t}{I_i} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \left(\frac{E_{0,t}}{E_{0,i}} \right)^2 \frac{\cos \theta_t}{\cos \theta_i} = \alpha \beta \left(\frac{2}{\alpha + \beta} \right)^2.$$

Again, it is expected that $R + T = 1$.

9.4 Absorption and Dispersion

9.4.1 Electromagnetic Waves in Conductors

In previous sections, we assumed that there is no free charge density ρ_f or free current density $\mathbf{J}_f$. This assumption works for insulators. In conductors, charges are allowed to move freely, so $\mathbf{J}_f$ is nonzero in general. According to Ohm's law, the free current density in a conductor is proportional to the electric field, $\mathbf{J}_f = \sigma \mathbf{E}$. With this assumption, Maxwell's equations read

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho_f}{\epsilon}, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu\sigma \mathbf{E} + \mu\epsilon \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}$$

By the continuity equation for free charge, Ohm's law, and Gauss's law,

$$\nabla \cdot \mathbf{J}_f = -\frac{\partial \rho_f}{\partial t} \implies \frac{\partial \rho_f}{\partial t} = -\sigma(\nabla \cdot \mathbf{E}) = -\frac{\sigma}{\epsilon} \rho_f.$$

The solution to this differential equation is $\rho_f(t) = e^{-(\sigma/\epsilon)t} \rho_f(0)$. Any initial free charge $\rho_f(0)$ will flow to the edges (surface of the conductor) in a characteristic time $\tau \equiv \epsilon/\sigma$. A perfect conductor has $\sigma = \infty$, and hence $\tau = 0$. We will assume that the media considered are good conductors, so $\rho_f = 0$. In this case, only the free current density is present, and Maxwell's equations become

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu\epsilon \frac{\partial \mathbf{E}}{\partial t} + \mu\sigma \mathbf{E}.\end{aligned}$$

The additional term is $\mu\sigma \mathbf{E}$ in Maxwell-Ampère's law. Applying the curl to Faraday's law and Maxwell-Ampère's law gives to equations:

$$\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla^2 \mathbf{B} = \mu\epsilon \frac{\partial^2 \mathbf{B}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{B}}{\partial t}.$$

These equations also admit plane wave solutions, but with a complex wave number $\tilde{k}$,

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(\tilde{k}z - \omega t)}, \quad \tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(\tilde{k}z - \omega t)}, \quad \tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega.$$

Even the square of $\tilde{k}$ is complex. Taking the square root and separate the real and imaginary part:

$$\tilde{k} = k + i\kappa, \quad \text{where} \quad k \equiv \omega \sqrt{\frac{\epsilon\mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} + 1 \right]^{1/2}, \quad \kappa \equiv \omega \sqrt{\frac{\epsilon\mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} - 1 \right]^{1/2}. \quad (9.12)$$

The imaginary part of $\tilde{k}$ creates a damping in the wave with increasing z ,

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{-\kappa z} e^{i(kz - \omega t)}, \quad \tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{-\kappa z} e^{i(kz - \omega t)}.$$

The characteristic measurement of distance an electromagnetic wave can travel inside a conductor is called the **skin depth**:

$$d \equiv \frac{1}{\kappa}. \quad (9.13)$$

The real part of $\tilde{k}$ is related to the wavelength, speed, and index of refraction by $\lambda = 2\pi/k$, $v = \omega/k$, and $n = ck/\omega$. Maxwell's equations still require that $\mathbf{E}$ and $\mathbf{B}$ have no z -component (i.e. they are transverse) as they are divergence-free. Thus, we can let $\mathbf{E}$ to be polarized in the x -direction,

$$\tilde{\mathbf{E}}(z, t) = \tilde{E}_0 e^{-\kappa z} e^{i(kz - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}(z, t) = \frac{\tilde{k}}{\omega} \tilde{E}_0 e^{-\kappa z} e^{i(kz - \omega t)} \hat{\mathbf{y}}.$$

The fields are again mutually perpendicular. However, the electric and magnetic fields are no longer in phase. To see this, write $\tilde{k}$ in polar form,

$$\tilde{k} = K e^{i\phi}, \quad \text{where} \quad K \equiv \sqrt{k^2 + \kappa^2} = \omega \left[\epsilon \mu \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega} \right)^2} \right]^2, \quad \phi \equiv \tan^{-1}(\kappa/k).$$

The complex amplitudes $\tilde{E}_0 = E_0 e^{i\delta_E}$ and $\tilde{B}_0 = B_0 e^{i\delta_B}$ are then related by

$$B_0 e^{i\delta_B} = \frac{\tilde{k}}{\omega} E_0 e^{i\delta_E} = \frac{K}{\omega} E_0 e^{i(\delta_E + \phi)}.$$

The magnetic field lags behind the electric field, with a phase difference $\delta_B - \delta_E = \phi$. The real amplitudes are related by $B_0/E_0 = K/\omega$. Lastly, the real electric and magnetic fields are

$$\mathbf{E}(z, t) = E_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E) \hat{\mathbf{x}}, \quad \mathbf{B}(z, t) = B_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E + \phi) \hat{\mathbf{y}}. \quad (9.14)$$

9.4.2 Reflection at a Conducting Surface

In the presence of free charges, a generalized boundary condition is

$$\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp = \sigma_f, \quad \mathbf{E}_1^\parallel - \mathbf{E}_2^\parallel = 0, \quad B_1^\perp - B_2^\perp = 0, \quad \frac{1}{\mu_1} \mathbf{B}_1^\parallel - \frac{1}{\mu_2} \mathbf{B}_2^\parallel = \mathbf{K}_f \times \hat{\mathbf{n}}. \quad (9.15)$$

Here σ_f is the free surface charges, $\mathbf{K}_f$ is the free surface current, and $\hat{\mathbf{n}}$ is the unit normal vector to the surface pointing from medium 2 to 1. The free surface current can be written as $\mathbf{J}_f$ with a delta function at the surface. Ohmic conductors, with $\mathbf{J}_f = \sigma \mathbf{E}$, cannot have free surface current if $\sigma \neq \infty$. Otherwise, it requires $\mathbf{E} \rightarrow \infty$ to build the delta function.

Consider a non-conducting linear medium 1 and a conductor 2, with the xy -plane being the boundary. Assume a monochromatic plane wave travels in the z -direction and is polarized in the x -direction:

$$\tilde{\mathbf{E}}_i(z, t) = \tilde{E}_{0,i} e^{i(k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_i(z, t) = \frac{1}{v_1} \tilde{E}_{0,i} e^{i(k_1 z - \omega t)} \hat{\mathbf{y}}.$$

The reflected and transmitted waves are

$$\begin{aligned} \tilde{\mathbf{E}}_r(z, t) &= \tilde{E}_{0,r} e^{i(-k_1 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_r(z, t) = -\frac{1}{v_1} \tilde{E}_{0,r} e^{i(-k_1 z - \omega t)} \hat{\mathbf{y}}, \\ \tilde{\mathbf{E}}_t(z, t) &= \tilde{E}_{0,t} e^{i(\tilde{k}_2 z - \omega t)} \hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_t(z, t) = \frac{\tilde{k}_2}{\omega} \tilde{E}_{0,t} e^{i(\tilde{k}_2 z - \omega t)} \hat{\mathbf{y}}. \end{aligned}$$

According to boundary conditions (9.15), there is no $E^\perp$ on both sides, so $\sigma_f = 0$. $B^\perp$ condition is trivial. The parallel components $E^\parallel$ and $B^\parallel$ give

$$\tilde{E}_{0,i} + \tilde{E}_{0,r} = \tilde{E}_{0,t}, \quad \tilde{E}_{0,i} - \tilde{E}_{0,r} = \tilde{\beta} \tilde{E}_{0,t}, \quad \text{where} \quad \tilde{\beta} \equiv \frac{\mu_1 v_1}{\mu_2 \omega} \tilde{k}_2.$$

where $\mathbf{K}_f = 0$ for Ohmic conductors. Solving the system of equations in terms of $\tilde{E}_{0,i}$,

$$\tilde{E}_{0,r} = \left(\frac{1 - \tilde{\beta}}{1 + \tilde{\beta}} \right) \tilde{E}_{0,i}, \quad \tilde{E}_{0,t} = \left(\frac{2}{1 + \tilde{\beta}} \right) \tilde{E}_{0,i}.$$

This is identical to the case where both media are non-conducting, except that $\tilde{\beta}$ is now a complex number. If media 2 is a perfect conductor ($\sigma = \infty$), then (9.12) says $k_2 \rightarrow \infty$, and $\tilde{\beta} = \infty$. The solution becomes

$$\tilde{E}_{0,r} = -\tilde{E}_{0,i}, \quad \tilde{E}_{0,t} = 0.$$

It is a total reflection, with a π phase in the reflected wave. In real life, it means excellent conductors make good mirrors. That's why mirrors are often made of silver.

9.4.3 Dispersion

It is well known that the index of refraction of a medium is a function of frequency of light. For example, a rainbow of colors can be created by a prism or raindrop because blue lights bend more than red ones do. This phenomenon is known as **dispersion**. Media with frequency dependent index of refraction are **dispersive**.

A wave packet made of sinusoidal components with different frequencies will change shape when propagates through dispersive media. Individual sinusoidal components travels at the **phase velocity**

$$v = \frac{\omega}{k}, \tag{9.16}$$

but the wave packet as a whole travels at the **group velocity**

$$v_g = \frac{d\omega}{dk}. \tag{9.17}$$

To be written.

9.5 Guided Waves

In this section we consider electromagnetic waves confined to a hollow pipe, or **wave guide**, instead of plane waves over all space. We assume that the wave guide is a perfect conductor, so $\mathbf{E} = 0$. By Faraday's law, $\partial \mathbf{B} / \partial t = 0$. If the initial magnetic field is $\mathbf{B}_i = 0$, then it remains zero forever ($\mathbf{B} = 0$ inside perfect conductors).

9.5.1 Wave Guides

The boundary conditions (9.15) at the inner wall are $\mathbf{E}^{\parallel} = 0$ and $B^{\perp} = 0$. We still consider monochromatic waves traveling in the z -direction in the form,

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \tilde{\mathbf{E}}_0(x, y) e^{i(kz - \omega t)}, \quad \tilde{\mathbf{B}}(\mathbf{r}, t) = \tilde{\mathbf{B}}_0(x, y) e^{i(kz - \omega t)}.$$

The goal is to find the electric and magnetic fields that satisfy Maxwell's equations inside the wave guide,

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}$$

with boundary conditions $\mathbf{E}^\parallel = 0$ and $B^\perp = 0$. We first separate the components of $\tilde{\mathbf{E}}_0$ and $\tilde{\mathbf{B}}_0$ in Cartesian coordinates:

$$\tilde{\mathbf{E}}_0 = E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}} + E_z \hat{\mathbf{z}}, \quad \tilde{\mathbf{B}}_0 = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}},$$

all of which are functions of x and y . Substituting these into $\nabla \times \mathbf{E}$ and $\nabla \times \mathbf{B}$, we get (see Appendix A.2.2)

$$\begin{aligned}\text{(i)} \quad \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= i\omega B_z, & \text{(iv)} \quad \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= -\frac{i\omega}{c^2} E_z, \\ \text{(ii)} \quad \frac{\partial E_z}{\partial y} - ikE_y &= i\omega B_x, & \text{(v)} \quad \frac{\partial B_z}{\partial y} - ikB_y &= -\frac{i\omega}{c^2} E_x, \\ \text{(iii)} \quad ikE_x - \frac{\partial E_z}{\partial x} &= i\omega B_y, & \text{(vi)} \quad ikB_x - \frac{\partial B_z}{\partial x} &= -\frac{i\omega}{c^2} E_y.\end{aligned}$$

Using (ii), (iii), (v), and (vi), we can solve for E_x , E_y , B_x , and B_y in terms of E_z and B_z ,

$$\begin{aligned}E_x &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial x} + \omega \frac{\partial B_z}{\partial y} \right), & B_x &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial x} - \frac{\omega}{c^2} \frac{\partial E_z}{\partial y} \right), \\ E_y &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial y} - \omega \frac{\partial B_z}{\partial x} \right), & B_y &= \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial B_z}{\partial y} + \frac{\omega}{c^2} \frac{\partial E_z}{\partial x} \right).\end{aligned}\tag{9.18}$$

Substituting E_x , E_y , B_x , B_y into constraints $\nabla \cdot \tilde{\mathbf{E}} = 0$ and $\nabla \cdot \tilde{\mathbf{B}} = 0$ gives equations for E_z and B_z :

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] E_z = 0, \quad \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] B_z = 0.$$

$E_z = 0$ waves are called **transverse electric (TE) waves**, while $B_z = 0$ waves are called **transverse magnetic (TM) waves**. It turns out that **TEM waves** (both $E_z = 0$ and $B_z = 0$) cannot occur in a hollow wave guide.

Proof. If $E_z = 0$ and $B_z = 0$, Gauss's law and Faraday's law (z -component) says

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} = 0, \quad \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0.$$

In other words, the transverse part of $\tilde{\mathbf{E}}_0$ is divergence-free and curl-free. It can thus be written as a gradient of ϕ that satisfies Laplace's equation $\nabla^2 \phi$. We can use all results from electrostatics. In particular, the surface of the conductor is an equipotential, and Laplace's equation requires ϕ have no local maxima or minima. This means the potential is a constant, and $\mathbf{E} = \nabla \phi = 0$. That is, there is nothing inside the wave guide; TEM wave does not exist. $\square$

Note that we cannot prove $E_x = E_y = 0$ using equations like

$$E_x = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial x} + \omega \frac{\partial B_z}{\partial y} \right), \quad E_y = \frac{i}{(\omega/c)^2 - k^2} \left(k \frac{\partial E_z}{\partial y} - \omega \frac{\partial B_z}{\partial x} \right)$$

because it is possible that $\omega/c = k$ in the denominator. In fact, when the wave guide is not hollow, we can find that $\omega = ck$, and TEM waves can exist. Hence the boundary conditions for Laplace's equation in the above proof is important.

9.5.2 TE Waves in a Rectangular Wave Guide

Consider the propagation of TE waves in a rectangular wave guide of dimensions $[0, a] \times [0, b]$ and extending to infinities in the z -direction. The goal is to solve

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \left(\frac{\omega}{c} \right)^2 - k^2 \right] B_z = 0 \quad \text{with} \quad B^\perp = 0.$$

We first consider separable solutions $B_z(x, y) = X(x)Y(y)$, so

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} + \left[\left(\frac{\omega}{c} \right)^2 - k^2 \right] XY = 0 \quad \Rightarrow \quad \frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \left(\frac{\omega}{c} \right)^2 - k^2 = 0.$$

Both x - and y -dependent terms must be constants,

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -k_x^2, \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2, \quad \text{and} \quad -k_x^2 - k_y^2 + (\omega/c)^2 - k^2 = 0.$$

The general solution to the X -equation is $X(x) = A \sin(k_x x) + B \cos(k_x x)$. The boundary condition says $B_x(0, y) = B_x(a, y) = 0$. Then by the B_x -equation in (9.18), $dX/dx = 0$ at $x = 0$ and $x = a$, so

$$X(x) = B \cos(k_x x), \quad \text{where} \quad k_x = \frac{m\pi}{a}, \quad m = 0, 1, 2, \dots$$

The Y -equation is the same,

$$Y(y) = D \cos(k_y y), \quad \text{where} \quad k_y = \frac{n\pi}{b}, \quad n = 0, 1, 2, \dots$$

Therefore, the solution to B_z (called the TE_{mn} mode) is

$$B_z = B_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right), \quad k = \sqrt{\left(\frac{\omega}{c}\right)^2 - \pi^2 \left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 \right]}. \quad (9.19)$$

At least one of the indices need to be nonzero. There exist a **cutoff frequency** ω_{mn} where the wave number becomes imaginary,

$$\omega_{mn} \equiv c\pi \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}. \quad (9.20)$$

The traveling wave becomes an exponential decay of fields. The wave number can then be written as $k = \sqrt{\omega^2/\omega_{mn}^2}/c$. The phase velocity and group velocity are given by

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}} > c, \quad v_g = \frac{1}{dk/d\omega} = c\sqrt{1 - (\omega_{mn}/\omega)^2} < c.$$

(Many velocities are identified in wave mechanics. Physical quantities in TE_{mn} mode, such as energy, travels at the group velocity, which is less than the speed of light. See Appendix A.2.3.)

9.5.3 The Coaxial Transmission Line

To be written.

10 POTENTIALS AND FIELDS

10.1 The Potential Formulation

10.1.1 Scalar and Vector Potentials

In electrostatics and magnetostatics, we set up differential equations of scalar potential ϕ and vector potential $\mathbf{A}$ to find electric and magnetic fields. The motivations are $\nabla \times \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$, so we can set $\mathbf{E} = -\nabla\phi$ and $\mathbf{B} = \nabla \times \mathbf{A}$. In electrodynamics, the magnetic field is still divergence-free, so we can write $\mathbf{B} = \nabla \times \mathbf{A}$. According to Faraday's law, the curl of electric field becomes

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}).$$

In other words,

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0 \implies \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla\phi$$

for some scalar potential ϕ . In terms of ϕ and $\mathbf{A}$, the electric field can be expressed as

$$\boxed{\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}.} \quad (10.1)$$

This comes from $\nabla \times \mathbf{B} = 0$ and Faraday's law. By Gauss's law, the first equation that ϕ and $\mathbf{A}$ should satisfy is

$$\nabla^2\phi + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{\rho}{\epsilon_0}. \quad (10.2)$$

By Maxwell-Ampère's law,

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} - \mu_0 \epsilon_0 \nabla \frac{\partial \phi}{\partial t} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2}.$$

Using the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, we obtain the second equation

$$\left(\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} \right) - \nabla \left(\nabla \cdot \mathbf{A} + \mu_0 \epsilon_0 \frac{\partial \phi}{\partial t} \right) = -\mu_0 \mathbf{J}. \quad (10.3)$$

Equations (10.2) and (10.3) relate potentials (LHS) to charge and current distribution (RHS). They satisfy all four of Maxwell's equations. The goal is to solve for ϕ and $\mathbf{A}$.

10.1.2 Gauge Transformations

The fields $\mathbf{E}$ and $\mathbf{B}$ are the physical quantities. They can be obtained by differentiating potentials ϕ and $\mathbf{A}$. We can impose more conditions on ϕ and $\mathbf{A}$ to simplify (10.2) and (10.3) as long as $\mathbf{E}$ and $\mathbf{B}$ are unchanged.

Consider two sets of potentials, $(\phi, \mathbf{A})$ and $(\phi', \mathbf{A}')$, that produce the same electric and magnetic fields. Let $\mathbf{A}' = \mathbf{A} + \boldsymbol{\alpha}$, and $\phi' = \phi + \beta$, where $\boldsymbol{\alpha}$ is a vector field and β is a scalar field. To produce the same magnetic field,

$$\mathbf{B} = \nabla \times \mathbf{A}' = \nabla \times \mathbf{A} \implies \nabla \times \boldsymbol{\alpha} = 0.$$

The curl of a gradient is zero, so we let $\boldsymbol{\alpha} = \nabla\lambda$ for some scalar field λ . To produce the same electric field,

$$\mathbf{E} = -\nabla\phi' - \frac{\partial\mathbf{A}'}{\partial t} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} \implies \nabla\beta + \frac{\partial\boldsymbol{\alpha}}{\partial t} = \nabla\left(\beta + \frac{\partial\lambda}{\partial t}\right) = 0.$$

This means the term in the bracket is independent of position, so we let

$$\beta = -\frac{\partial\lambda}{\partial t} + k(t)$$

for some $k(t)$ that is only a function of time. In fact, we can absorb $k(t)$ into λ by defining λ' such that

$$\lambda' = \lambda - \int_0^t k(t') dt' \quad \text{and then} \quad \beta = -\frac{\partial\lambda'}{\partial t}.$$

In conclusion, we can transform ϕ and $\mathbf{A}$ to new ones by **gauge transformations**

$$\mathbf{A}' = \mathbf{A} + \nabla\lambda, \quad \phi' = \phi - \frac{\partial\lambda}{\partial t}$$

for any scalar function $\lambda(\mathbf{r}, t)$. Such gauge transformations are not unique, and the choice of gauge depends on the problems we want to solve.

10.1.3 Coulomb Gauge and Lorenz Gauge

The Coulomb gauge is encountered in magnetostatics. It is always possible to find a $\lambda(\mathbf{r}, t)$ such that $\nabla \cdot \mathbf{A} = 0$. In the Coulomb gauge, the potential equations (10.2) and (10.3) become

$$\nabla^2\phi = -\frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla^2\mathbf{A} - \mu_0\epsilon_0\frac{\partial^2\mathbf{A}}{\partial t^2} = -\mu_0\mathbf{J} + \mu_0\epsilon_0\nabla\left(\frac{\partial\phi}{\partial t}\right).$$

The scalar potential equation is just Poisson's equation, with the solution (and $\phi(\infty) = 0$)

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t)}{r} dV'.$$

However, the vector potential equation is not easy to solve in Coulomb gauge. Another useful gauge is called the **Lorenz gauge** (after L. V. Lorenz, not H. A. Lorentz). It is always possible to find a $\lambda(\mathbf{r}, t)$ such that (see proof in Appendix A.3.1)

$$\nabla \cdot \mathbf{A} = -\mu_0\epsilon_0\frac{\partial\phi}{\partial t} \iff \nabla \cdot \mathbf{A} + \frac{1}{c^2}\frac{\partial\phi}{\partial t} = 0. \quad (10.4)$$

In the Lorenz gauge, the potential equations read

$$-\frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} + \nabla^2\phi = -\frac{\rho}{\epsilon_0} \quad \text{and} \quad -\frac{1}{c^2}\frac{\partial^2\mathbf{A}}{\partial t^2} + \nabla^2\mathbf{A} = -\mu_0\mathbf{J}. \quad (10.5)$$

The two equations are treated equally, and they are in good forms in special relativity. The operator (called the **d'Alembertian**)

$$-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2 = \partial_\mu\partial^\mu \equiv \square^2$$

can be seen as the four-dimensional spacetime version of the Laplacian. In the Lorenz gauge, (10.5) are in the form of the **inhomogeneous wave equation**, with the source term being charge and current distributions. In this chapter, we shall always work in the Lorenz gauge.

10.1.4 Lorentz Force Law in Potential Form

The Lorentz force law in terms of potentials is

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q \left[-\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} + \mathbf{v} \times (\nabla \times \mathbf{A}) \right], \quad (10.6)$$

where $\mathbf{p} = m\mathbf{v}$ is the momentum of the particle. Since the velocity $\mathbf{v}$ is a function of time but not position, we can write (1.4) as

$$\nabla(\mathbf{v} \cdot \mathbf{A}) = \mathbf{v} \times (\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla \times \mathbf{v}) + (\mathbf{v} \cdot \nabla)\mathbf{A} + (\mathbf{A} \cdot \nabla)\mathbf{v} = \mathbf{v} \times (\nabla \times \mathbf{A}) + (\mathbf{v} \cdot \nabla)\mathbf{A}.$$

Substituting $\mathbf{v} \times (\nabla \times \mathbf{A}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}$ into (10.6),

$$\frac{d\mathbf{p}}{dt} = -q \left[\frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A} + \nabla(\phi - \mathbf{v} \cdot \mathbf{A}) \right] = -q \left[\frac{d\mathbf{A}}{dt} + \nabla(\phi - \mathbf{v} \cdot \mathbf{A}) \right].$$

In this equation, the total derivative

$$\frac{d\mathbf{A}}{dt} = \frac{\partial \mathbf{A}}{\partial t} + \frac{\partial \mathbf{A}}{\partial x} \frac{dx}{dt} + \frac{\partial \mathbf{A}}{\partial y} \frac{dy}{dt} + \frac{\partial \mathbf{A}}{\partial z} \frac{dz}{dt} = \frac{\partial \mathbf{A}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{A} \quad (10.7)$$

is called the **convective derivative** of $\mathbf{A}$. The convective derivative consists of two rates of change of the particle's potential $\mathbf{A}(\mathbf{r}, t)$. When the particle moves to a new position with potential $\mathbf{A}(\mathbf{r} + \mathbf{v} dt, t + dt)$, the change in $\mathbf{A}$ is due to both the time-dependence of it and the change in location of the particle. In terms of the convective derivative, the Lorentz force law becomes

$$\frac{d}{dt}(\mathbf{p} + q\mathbf{A}) = -\nabla[q(\phi - \mathbf{v} \cdot \mathbf{A})]. \quad (10.8)$$

We may be familiar with this equation in classical mechanics in the form $d\mathbf{p}/dt = -\nabla V$, where V is the potential energy of the particle. In electromagnetism, we recognize $\mathbf{p}_A = \mathbf{p} + q\mathbf{A}$ as the **canonical momentum**, and $U = q(\phi - \mathbf{v} \cdot \mathbf{A})$. Another similar relation can be obtained (see Appendix A.3.2) for the energy of the particle,

$$\frac{d}{dt}(T + q\phi) = q \left(\frac{\partial \phi}{\partial t} - \mathbf{v} \cdot \frac{\partial \mathbf{A}}{\partial t} \right),$$

where T is the kinetic energy of the particle and $q\phi$ is its potential energy. Since ϕ is the potential energy per unit charge, it suggests us to interpret $\mathbf{A}$ as a “potential momentum” per unit charge.

10.2 Continuous Distributions

In static electromagnetism, we have four Poisson's equation (1 for ϕ and 3 for $\mathbf{A}$),

$$\begin{aligned} \nabla^2 \phi &= -\frac{\rho}{\epsilon_0} \implies \phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} dV', \\ \nabla^2 \mathbf{A} &= -\mu_0 \mathbf{J} \implies \mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{z} dV'. \end{aligned}$$

Nowadays we know that electromagnetic interaction travels at the speed of light. Hence the particle is affected by the potential at $(\mathbf{r}, t_r)$ at some earlier time $t_r \equiv t - z/c$, where t_r is called the **retarded time**. This is because it takes time z/c for the change in potential to “travel” from $\mathbf{r}'$ to $\mathbf{r}$. The nonstatic case thus have solutions

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{z} dV', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{z} dV'. \quad (10.9)$$

These effective time-dependent potentials at the particle's position are called **retarded potentials**. Note that t_r is not a particular time for all charge distribution. Evidently, farther charge distributions with larger z are evaluated at earlier retarded times. Also, the reasoning above works only in the potential formulation; similar arguments does not work for fields $\mathbf{E}$ or $\mathbf{B}$. Now we shall justify our reasoning by proving that (10.9) obeys the inhomogeneous wave equation (10.5) in the Lorenz gauge.

Proof. First, $t_r = t - z/c$, so $\partial/\partial t_r = \partial/\partial t$. We can write the gradient of ρ as

$$\nabla \rho = \frac{\partial \rho}{\partial t_r} \nabla t_r = \dot{\rho} \nabla t_r = -\frac{1}{c} \dot{\rho} \nabla z = -\frac{\dot{\rho}}{c} \hat{\mathbf{z}}.$$

The gradient of ϕ is

$$\nabla \phi = \frac{1}{4\pi\epsilon_0} \int \left[(\nabla \rho) \frac{1}{z} + \rho \nabla \left(\frac{1}{z} \right) \right] dV' = \frac{1}{4\pi\epsilon_0} \int \left[-\frac{\dot{\rho}}{c} \frac{\hat{\mathbf{z}}}{z} - \rho \frac{\hat{\mathbf{z}}}{z^2} \right] dV'.$$

Taking the divergence by using the product rule (1.5),

$$\nabla^2 \phi = \frac{1}{4\pi\epsilon_0} \int \left\{ -\frac{1}{c} \left[\frac{\hat{\mathbf{z}}}{z} \cdot \nabla \dot{\rho} + \dot{\rho} \nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z} \right) \right] - \left[\frac{\hat{\mathbf{z}}}{z^2} \cdot \nabla \rho + \rho \nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) \right] \right\} dV'.$$

Here are more derivatives for reference:

$$\nabla \dot{\rho} = -\frac{1}{c} \ddot{\rho} \nabla z = -\frac{\ddot{\rho}}{c} \hat{\mathbf{z}}, \quad \nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z} \right) = \frac{1}{z^2}, \quad \nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) = 4\pi \delta^3(\mathbf{z}).$$

Then $\nabla^2 \phi$ becomes

$$\nabla^2 \phi = \frac{1}{4\pi\epsilon_0} \int \left\{ -\frac{1}{c} \left[-\frac{\ddot{\rho}}{cz} + \frac{\dot{\rho}}{z^2} \right] - \left[-\frac{\dot{\rho}}{cz^2} + 4\pi \delta^3(\mathbf{z}) \rho \right] \right\} dV' = \frac{1}{4\pi\epsilon_0} \int \left[\frac{1}{c^2} \frac{\ddot{\rho}}{z} - 4\pi \delta^3(\mathbf{z}) \rho \right] dV'.$$

The first integral term is the second time-derivative of ϕ according to (10.9), while the second integral term is just $-\rho(\mathbf{r}, t)/\epsilon_0$. Hence provided that (10.9) is true, we confirm that

$$\nabla^2 \phi = \frac{1}{c^2} \left[\frac{1}{4\pi\epsilon_0} \int \frac{\ddot{\rho}}{z} dV' \right] - \frac{1}{\epsilon_0} \int \delta^3(\mathbf{r} - \mathbf{r}') d^3\mathbf{r}' = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\rho}{\epsilon_0} \implies -\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} + \nabla^2 \phi = -\frac{\rho}{\epsilon_0}.$$

Because all we used are vector calculus identities, the inhomogeneous equation for the vector potential follows exactly the same logic. Just replace every ϕ by $\mathbf{A}$ and every ρ by $\mathbf{J}$.

The second task is to prove that (10.9) satisfies the Lorenz gauge condition (10.4),

$$\nabla \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{z} \right] dV' \stackrel{?}{=} -\mu_0 \epsilon_0 \frac{\partial \phi}{\partial t}.$$

Let ∇' denotes the gradient operator with respect to primed coordinates. We have

$$\nabla \cdot \left(\frac{\mathbf{J}}{z} \right) = \frac{1}{z} (\nabla \cdot \mathbf{J}) + \mathbf{J} \cdot \nabla \left(\frac{1}{z} \right), \quad \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right) = \frac{1}{z} (\nabla' \cdot \mathbf{J}) + \mathbf{J} \cdot \nabla' \left(\frac{1}{z} \right).$$

Since $\mathbf{z} = \mathbf{r} - \mathbf{r}'$, $\nabla(1/z) = -\nabla'(1/z)$, so

$$\nabla \cdot \left(\frac{\mathbf{J}}{z} \right) = \frac{1}{z} (\nabla \cdot \mathbf{J}) - \mathbf{J} \cdot \nabla' \left(\frac{1}{z} \right) = \frac{1}{z} (\nabla \cdot \mathbf{J}) + \frac{1}{z} (\nabla' \cdot \mathbf{J}) - \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right).$$

The divergence $\nabla \cdot \mathbf{J}$ and $\nabla' \cdot \mathbf{J}$ are

$$\nabla \cdot \mathbf{J}(\mathbf{r}', t_r) = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = \frac{\partial J_x}{\partial t_r} \frac{\partial t_r}{\partial x} + \frac{\partial J_y}{\partial t_r} \frac{\partial t_r}{\partial y} + \frac{\partial J_z}{\partial t_r} \frac{\partial t_r}{\partial z} = -\frac{1}{c} \left(\frac{\partial J_x}{\partial t_r} \frac{\partial z}{\partial x} + \frac{\partial J_y}{\partial t_r} \frac{\partial z}{\partial y} + \frac{\partial J_z}{\partial t_r} \frac{\partial z}{\partial z} \right) = -\frac{1}{c} \mathbf{j} \cdot \nabla z.$$

$$\nabla' \cdot \mathbf{J}(\mathbf{r}', t_r) = \nabla' \cdot \mathbf{J}(\mathbf{r}') + \left(\frac{\partial J_x}{\partial t_r} \frac{\partial t_r}{\partial x} + \frac{\partial J_y}{\partial t_r} \frac{\partial t_r}{\partial y} + \frac{\partial J_z}{\partial t_r} \frac{\partial t_r}{\partial z} \right) = -\dot{\rho} - \frac{1}{c} \dot{\mathbf{J}} \cdot \nabla' \mathbf{z}.$$

(The two divergences are different because $\mathbf{J}(\mathbf{r}', t_r)$ depends only explicitly on primed coordinates $\mathbf{r}'$ but not unprimed ones. Thus, all unprimed-spatial dependence in $\nabla \cdot \mathbf{J}$ comes from $t_r(\mathbf{r}, \mathbf{r}')$. But in $\nabla' \cdot \mathbf{J}$, ∇' differentiates $\mathbf{J}$ explicitly on primed coordinates as well as implicitly on $t_r(\mathbf{r}, \mathbf{r}')$. The explicit part are written as $\nabla' \cdot \mathbf{J}(\mathbf{r}')$, and it is replaced by $-\dot{\rho}$ via the continuity equation.) Substituting $\nabla \cdot \mathbf{J}$ and $\nabla' \cdot \mathbf{J}$ into $\nabla \cdot (\mathbf{J}/z)$:

$$\nabla \cdot \left(\frac{\mathbf{J}}{z} \right) = \frac{1}{z} \left(-\frac{1}{c} \dot{\mathbf{J}} \cdot \nabla \mathbf{z} \right) + \frac{1}{z} \left(-\dot{\rho} - \frac{1}{c} \dot{\mathbf{J}} \cdot \nabla' \mathbf{z} \right) - \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right) = -\frac{1}{z} \dot{\rho} - \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right),$$

where we used the fact that $\nabla' \mathbf{z} = -\nabla \mathbf{z}$ to cancel the first two terms. The divergence of the vector potential is

$$\nabla \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \left[\frac{\dot{\rho}}{z} - \nabla' \cdot \left(\frac{\mathbf{J}}{z} \right) \right] dV' = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{1}{4\pi \epsilon_0} \int \frac{\rho}{z} dV' \right) - \frac{\mu_0}{4\pi} \oint \frac{\mathbf{J}}{z} \cdot d\mathbf{a}' = -\mu_0 \epsilon_0 \frac{\partial \phi}{\partial t}.$$

The surface integral vanishes because $\mathbf{J} = 0$ at infinity for finite charge distribution. The Lorenz gauge condition is confirmed. $\square$

It is straightforward to compute $\mathbf{E}$ and $\mathbf{B}$ by taking the derivatives,

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

The result is known as **Jefimenko's equations**:

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi \epsilon_0} \int \left[\frac{\rho(\mathbf{r}', t_r)}{z^2} \hat{\mathbf{z}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{cz} \hat{\mathbf{z}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2 z} \right] dV', \quad \mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{z^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{cz} \right] \times \hat{\mathbf{z}} dV'.} \quad (10.10)$$

Note that $\mathbf{E}$ and $\mathbf{B}$ involves time-derivatives of ρ and $\mathbf{J}$. The fields are not obtained by replacing t with t_r directly, unlike ϕ and $\mathbf{A}$.

10.3 Point Charges

10.3.1 Liénard-Wiechert Potentials

Consider a point charge q moving on a trajectory $\mathbf{w}(t)$. We want to find the retarded potentials $\phi(\mathbf{r}, t)$ and $\mathbf{A}(\mathbf{r}, t)$ generated by this point charge. The potential is not simply $\phi = q/4\pi\epsilon_0 z$. According to (10.9),

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{z} dV'.$$

The charge density depends on t_r , *even though it is a point charge*. The charge density of a moving point charge at time t is given by

$$\rho(\mathbf{r}', t) = q \delta^3(\mathbf{r}' - \mathbf{w}(t)).$$

Since we need the charge density at the retarded time t_r , we can write it with another delta function to pick out $t' = t_r$:

$$\rho(\mathbf{r}', t_r) = q \int \delta^3(\mathbf{r}' - \mathbf{w}(t')) \delta(t' - t_r) dt'.$$

The potential is

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \frac{\delta(t' - t_r)}{|\mathbf{r} - \mathbf{r}'|} \delta^3(\mathbf{r}' - \mathbf{w}(t')) d^3\mathbf{r}' dt' = \frac{q}{4\pi\epsilon_0} \int \frac{\delta(t' - t_r)}{|\mathbf{r} - \mathbf{w}(t')|} dt'.$$

Note that t_r is also a function of $\mathbf{r}'$ before spatial integration: $t_r = t - |\mathbf{r} - \mathbf{r}'|/c$. After spatial integration, $t'_r = t - |\mathbf{r} - \mathbf{w}(t')|/c$. In other words, t'_r is now a function of t , t' , and $\mathbf{r}$. (The prime of $t'_r = t - |\mathbf{r} - \mathbf{w}(t')|/c$ is to indicate that $\mathbf{w}(t')$ is a function of the dummy variable t' instead of t .) The delta function $\delta(t' - t'_r)$ then has a function of t' , $f(t')$, as the argument. Let $\delta(f(t')) \equiv \delta(t' - t'_r)$. We have the identity

$$\delta(f(t')) = \sum_i \frac{\delta(t' - t_i)}{|f'(t_i)|}, \quad \text{where} \quad f'(t_i) = \left. \frac{\partial f}{\partial t'} \right|_{t_i},$$

and each t_i is a zero of f (i.e. $f(t_i) = 0$). With this identity, the delta function reduces to

$$\delta(t' - t'_r) = \frac{\delta(t' - t_r)}{\partial_{t'}[t' - (t - |\mathbf{r} - \mathbf{w}(t')|/c)]_{t'=t_r}} = \frac{\delta(t' - t_r)}{1 + [-\mathbf{v}(t')/c] \cdot [\mathbf{r} - \mathbf{w}(t')]/|\mathbf{r} - \mathbf{w}(t')|_{t'=t_r}} = \frac{\delta(t' - t_r)}{1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c},$$

where $\mathbf{v}(t_r) = \dot{\mathbf{w}} = \dot{\mathbf{z}}$ and $\hat{\mathbf{z}} = [\mathbf{r} - \mathbf{w}(t_r)]/|\mathbf{r} - \mathbf{w}(t_r)|$. The derivative of the absolute value is $\partial_{t'}|\mathbf{z}| = \hat{\mathbf{z}} \cdot \mathbf{v}$. Plugging $\delta(t' - t'_r)$ back into ϕ ,

$$\phi(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \int \frac{1}{|\mathbf{r} - \mathbf{w}(t')|} \frac{\delta(t' - t_r)}{(1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c)} dt' = \left(\frac{1}{1 - \hat{\mathbf{z}} \cdot \mathbf{v}/c} \right) \frac{q}{4\pi\epsilon_0 z}.$$

This reduces to the static case $\phi = q/4\pi\epsilon_0 z$ when $\mathbf{v}(t_r) = 0$. In terms of the retarded position $\mathbf{z} = \mathbf{r} - \mathbf{w}(t_r)$, the scalar potential is

$$\boxed{\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(zc - \mathbf{z} \cdot \mathbf{v})}.} \quad (10.11)$$

The current density at the retarded time is $\mathbf{J}(\mathbf{r}, t_r) = \rho(\mathbf{r}, t_r)\mathbf{v}(t_r)$. The vector potential integral (10.9) is

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{q\mathbf{v}(t')\delta(t' - t'_r)}{|\mathbf{r} - \mathbf{w}(t')|} dt'.$$

Following the same procedure, the vector potential is

$$\boxed{\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \frac{qc\mathbf{v}}{(zc - \mathbf{z} \cdot \mathbf{v})} = \frac{\mathbf{v}}{c^2} \phi(\mathbf{r}, t).} \quad (10.12)$$

The potentials for a moving charge (10.11) and (10.12) are known as **Liénard-Wiechert potentials**.

10.3.2 The Fields of a Moving Charge

The electric and magnetic fields of a moving point charge can be obtained by differentiating the Liénard-Wiechert potentials:

$$\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}, \quad \text{where} \quad \phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(zc - \mathbf{z} \cdot \mathbf{v})}, \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mathbf{v}}{c^2} \phi(\mathbf{r}, t).$$

The differentiation is not easy because the potentials involve $\mathbf{z} = \mathbf{r} - \mathbf{w}(t_r)$ and $\mathbf{v} = \dot{\mathbf{w}}(t_r)$, while $t_r = t - |\mathbf{r} - \mathbf{w}(t_r)|/c$ is itself a function of $\mathbf{r}$ and t .

Gradient of ϕ . The gradient of ϕ is

$$\nabla\phi = -\frac{qc}{4\pi\epsilon_0} \frac{\nabla(zc - \mathbf{z} \cdot \mathbf{v})}{(zc - \mathbf{z} \cdot \mathbf{v})^2} = \frac{qc}{4\pi\epsilon_0} \frac{(c^2\nabla t_r) + (\mathbf{z} \cdot \nabla)\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{z} + \mathbf{z} \times (\nabla \times \mathbf{v}) + \mathbf{v} \times (\nabla \times \mathbf{z})}{(zc - \mathbf{z} \cdot \mathbf{v})^2}.$$

The first term comes from $\nabla(zc)$ with $z = c(t - t_r)$, and the other four terms come from the product rule of $\nabla(\mathbf{z} \cdot \mathbf{v})$. We shall compute them one by one:

$$(\mathbf{z} \cdot \nabla)\mathbf{v} = \left(z_x \frac{\partial}{\partial x} + z_y \frac{\partial}{\partial y} + z_z \frac{\partial}{\partial z} \right) \mathbf{v}(t_r) = z_x \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial x} + z_y \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial y} + z_z \frac{d\mathbf{v}}{dt_r} \frac{\partial t_r}{\partial z} = \mathbf{a}(\mathbf{z} \cdot \nabla t_r),$$

where $\mathbf{a} \equiv \dot{\mathbf{v}}$ is the acceleration of the particle at t_r ;

$$(\mathbf{v} \cdot \nabla) \mathbf{z} = (\mathbf{v} \cdot \nabla) \mathbf{r} - (\mathbf{v} \cdot \nabla) \mathbf{w} = \left(v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \right) (x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + z \hat{\mathbf{z}}) - \mathbf{v}(\mathbf{v} \cdot \nabla t_r) = \mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \nabla t_r),$$

where we used the fact that $(\mathbf{v} \cdot \nabla) \mathbf{w} = \mathbf{v}(\mathbf{v} \cdot \nabla t_r)$ for the same reasoning as $(\mathbf{z} \cdot \nabla) \mathbf{v} = \mathbf{a}(\mathbf{z} \cdot \nabla t_r)$;

$$\nabla \times \mathbf{v} = \left(\frac{dv_z}{dt_r} \frac{\partial t_r}{\partial y} - \frac{dv_y}{dt_r} \frac{\partial t_r}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{dv_x}{dt_r} \frac{\partial t_r}{\partial z} - \frac{dv_z}{dt_r} \frac{\partial t_r}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{dv_y}{dt_r} \frac{\partial t_r}{\partial x} - \frac{dv_x}{dt_r} \frac{\partial t_r}{\partial y} \right) \hat{\mathbf{z}} = -\mathbf{a} \times \nabla t_r;$$

$$\nabla \times \mathbf{z} = \nabla \times \mathbf{r} - \nabla \times \mathbf{w} = \mathbf{v} \times \nabla t_r.$$

Substitute these terms into $\nabla \phi$:

$$\begin{aligned} \nabla \phi &= \frac{qc}{4\pi\epsilon_0} \frac{(c^2 \nabla t_r) + \mathbf{a}(\mathbf{z} \cdot \nabla t_r) + [\mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \nabla t_r)] - \mathbf{z} \times (\mathbf{a} \times \nabla t_r) + \mathbf{v} \times (\mathbf{v} \times \nabla t_r)}{(\mathbf{z}c + \mathbf{z} \cdot \mathbf{v})^2} \\ &= \frac{qc}{4\pi\epsilon_0} \frac{(c^2 \nabla t_r) + \mathbf{a}(\mathbf{z} \cdot \nabla t_r) + \mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \nabla t_r) - \mathbf{a}(\mathbf{z} \cdot \nabla t_r) + (\mathbf{z} \times \mathbf{a}) \nabla t_r + \mathbf{v}(\mathbf{v} \cdot \nabla t_r) - (\mathbf{v} \cdot \mathbf{v}) \nabla t_r}{(\mathbf{z}c + \mathbf{z} \cdot \mathbf{v})^2} \\ &= \frac{qc}{4\pi\epsilon_0} \frac{\mathbf{v} + (c^2 - v^2 + \mathbf{z} \cdot \mathbf{a}) \nabla t_r}{(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})^2}. \end{aligned}$$

In the second equality, we used the triple vector product rule (1.2). Now all we need to know is ∇t_r . Going back to its definition, we have

$$-c \nabla t_r = \nabla z = \nabla \sqrt{\mathbf{z} \cdot \mathbf{z}} = \frac{\nabla(\mathbf{z} \cdot \mathbf{z})}{2\sqrt{\mathbf{z} \cdot \mathbf{z}}} = \frac{1}{z} [(\mathbf{z} \cdot \nabla) \mathbf{z} + \mathbf{z} \times (\nabla \times \mathbf{z})]$$

It is known that $\nabla \times \mathbf{z} = \mathbf{v} \times \nabla t_r$, and similar to $(\mathbf{v} \cdot \nabla) \mathbf{z} = \mathbf{v} - \mathbf{v}(\mathbf{v} \cdot \nabla t_r)$, we know that $(\mathbf{z} \cdot \nabla) \mathbf{z} = \mathbf{z} - \mathbf{v}(\mathbf{z} \cdot \nabla t_r)$. Hence we get an equation for ∇t_r :

$$-c \nabla t_r = \frac{1}{z} [\mathbf{z} - \mathbf{v}(\mathbf{z} \cdot \nabla t_r) + \mathbf{z} \times (\mathbf{v} \times \nabla t_r)] = \frac{1}{z} [\mathbf{z} - (\mathbf{z} \cdot \mathbf{v}) \nabla t_r].$$

The solution is

$$\nabla t_r = -\frac{\mathbf{z}}{\mathbf{z}c - \mathbf{z} \cdot \mathbf{v}}. \quad (10.13)$$

In conclusion, the gradient of ϕ is

$$\nabla \phi = \frac{qc}{4\pi\epsilon_0(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})} [(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v}) \mathbf{v} - (c^2 - v^2 + \mathbf{z} \cdot \mathbf{a}) \mathbf{z}]. \quad (10.14)$$

Time-Derivative of $\mathbf{A}$. The time-derivative of $\mathbf{A}$ is

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{\partial}{\partial t} \left[\frac{\mathbf{v}}{c^2} \phi(\mathbf{r}, t) \right] = \frac{\partial}{\partial t} \left[\frac{\mathbf{v}}{c^2} \frac{qc}{4\pi\epsilon_0(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})} \right] = \frac{q}{4\pi\epsilon_0 c} \left[\frac{1}{(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})} \frac{\partial \mathbf{v}}{\partial t} - \frac{\mathbf{v}}{(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})^2} \frac{\partial}{\partial t} (\mathbf{z}c - \mathbf{z} \cdot \mathbf{v}) \right].$$

There are several time-derivatives to compute:

$$\frac{\partial \mathbf{v}}{\partial t} = \frac{\partial \mathbf{v}}{\partial t_r} \frac{\partial t_r}{\partial t} = \mathbf{a} \frac{\partial t_r}{\partial t}, \quad \frac{\partial z}{\partial t} = c \frac{\partial}{\partial t} (t - t_r) = c \left(1 - \frac{\partial t_r}{\partial t} \right),$$

$$\frac{\partial}{\partial t} (\mathbf{z} \cdot \mathbf{v}) = \frac{\partial \mathbf{z}}{\partial t} \cdot \mathbf{v} + \mathbf{z} \cdot \frac{\partial \mathbf{v}}{\partial t} = \frac{\partial}{\partial t} [\mathbf{r} - \mathbf{w}(t_r)] + \mathbf{z} \cdot \mathbf{a} \frac{\partial t_r}{\partial t} = (-v^2 + \mathbf{z} \cdot \mathbf{a}) \frac{\partial t_r}{\partial t}.$$

Plug these into $\partial \mathbf{A} / \partial t$,

$$\begin{aligned} \frac{\partial \mathbf{A}}{\partial t} &= \frac{q}{4\pi\epsilon_0 c} \left\{ \frac{\mathbf{a}}{(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})} \frac{\partial t_r}{\partial t} - \frac{\mathbf{v}}{(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})^2} \left[c^2 + (-c^2 + v^2 - \mathbf{z} \cdot \mathbf{a}) \frac{\partial t_r}{\partial t} \right] \right\} \\ &= \frac{q}{4\pi\epsilon_0 c (\mathbf{z}c - \mathbf{z} \cdot \mathbf{v})^2} \left\{ -c^2 \mathbf{v} + [(\mathbf{z}c - \mathbf{z} \cdot \mathbf{v}) \mathbf{a} + (c^2 - v^2 + \mathbf{z} \cdot \mathbf{a}) \mathbf{v}] \frac{\partial t_r}{\partial t} \right\}. \end{aligned}$$

Now we should determine $\partial t_r / \partial t$:

$$\frac{\partial t_r}{\partial t} = \frac{\partial}{\partial t} \left[t - \frac{|\mathbf{r} - \mathbf{w}(t_r)|}{c} \right] = 1 - \frac{\mathbf{r} - \mathbf{w}(t_r)}{|\mathbf{r} - \mathbf{w}(t_r)|} \left(-\frac{\mathbf{v}}{c} \right) \frac{\partial t_r}{\partial t} = 1 + \frac{\hat{\mathbf{z}} \cdot \mathbf{v}}{c} \frac{\partial t_r}{\partial t}.$$

The solution of this equation is

$$\frac{\partial t_r}{\partial t} = \frac{zc}{zc - \mathbf{z} \cdot \mathbf{v}}. \quad (10.15)$$

Substitute this back into $\partial \mathbf{A} / \partial t$,

$$\frac{\partial \mathbf{A}}{\partial t} = \frac{q}{4\pi\epsilon_0(zc - \mathbf{z} \cdot \mathbf{v})^3} [(zc - \mathbf{z} \cdot \mathbf{v})(z\mathbf{a} - c\mathbf{v}) + z(c^2 - v^2 + \mathbf{z} \cdot \mathbf{a})\mathbf{v}]. \quad (10.16)$$

Combining (10.14) and (10.16), we find the electric field to be

$$\boxed{\mathbf{E}(\mathbf{r}, t) = \frac{qz}{4\pi\epsilon_0(\mathbf{z} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})],} \quad (10.17)$$

where $\mathbf{u} \equiv c\hat{\mathbf{z}} - \mathbf{v}$. The first term, containing $(c^2 - v^2)\mathbf{u}$, falls off like $1/z^2$. It is sometimes called the **generalized Coulomb field** or **velocity field**. The second term, containing $\mathbf{z} \times (\mathbf{u} \times \mathbf{a})$, falls off like $1/z$, so it dominates at large distance. It is called the **radiation field** (because it is responsible for electromagnetic radiation) or **acceleration field**.

Curl of $\mathbf{A}$. To find the magnetic field, we compute the curl of $\mathbf{A}$,

$$\nabla \times \mathbf{A} = \frac{1}{c^2} \nabla(\phi\mathbf{v}) = \frac{1}{c^2} [\phi(\nabla \times \mathbf{v}) - \mathbf{v} \times (\nabla\phi)].$$

In the second equality we used the product rule (1.7). Both $\nabla \times \mathbf{v}$ and $\nabla\phi$ are known:

$$\nabla \times \mathbf{v} = -\mathbf{a} \times \nabla t_r = \frac{\mathbf{a} \times \mathbf{z}}{zc - \mathbf{z} \cdot \mathbf{v}}, \quad \nabla\phi = \frac{qc}{4\pi\epsilon_0(zc - \mathbf{z} \cdot \mathbf{v})} [(zc - \mathbf{z} \cdot \mathbf{v})\mathbf{v} - (c^2 - v^2 + \mathbf{z} \cdot \mathbf{a})\mathbf{z}].$$

Combining these with ϕ (10.11), the curl of $\mathbf{A}$ is

$$\nabla \times \mathbf{A} = -\frac{q}{4\pi\epsilon_0 c(zc - \mathbf{z} \cdot \mathbf{v})^3} \mathbf{z} \times [(c^2 - v^2)\mathbf{v} + (\mathbf{z} \cdot \mathbf{a})\mathbf{v} + (\mathbf{z} \cdot \mathbf{u})\mathbf{a}].$$

Since $\mathbf{z} \times \hat{\mathbf{z}} = 0$, the cross product $\mathbf{z} \times \mathbf{v}$ is the same as $-\mathbf{z} \times \mathbf{u}$, with $\mathbf{u} \equiv c\hat{\mathbf{z}} - \mathbf{v}$. Thus, every $\mathbf{v}$ in the square bracket can be replaced by $\mathbf{u}$ with a cost of a minus sign:

$$\nabla \times \mathbf{A} = +\frac{q}{4\pi\epsilon_0(\mathbf{z} \cdot \mathbf{u})^3} \mathbf{z} \times [(c^2 - v^2)\mathbf{u} + (\mathbf{z} \cdot \mathbf{a})\mathbf{u} - (\mathbf{z} \cdot \mathbf{u})\mathbf{a}].$$

With the vector triple product rule (1.2), we find that $\nabla \times \mathbf{A}$ has nearly the same form as $\mathbf{E}$,

$$\nabla \times \mathbf{A} = \frac{qz}{4\pi\epsilon_0 c(\mathbf{z} \cdot \mathbf{u})^3} \times [(c^2 - v^2)\mathbf{v} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})].$$

In other words, the magnetic field is

$$\boxed{\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E}(\mathbf{r}, t).} \quad (10.18)$$

The magnetic field of a point charge is always perpendicular to the electric field, and to the vector from the retarded point. The behavior of the velocity field and acceleration field works for $\mathbf{B}$ as well.

We have obtained the electric and magnetic fields of a moving point charge. Finally, we shall conclude the entire theory of classical

electrodynamics by writing out the force law. Then with the principle of superposition, we can write out the net force on a charged particle due to any moving charge configuration. According to Lorentz force law, this is the resulting force on a charged particle Q with velocity $\mathbf{V}$ by a moving point charge,

$$\mathbf{F} = \frac{Qq}{4\pi\epsilon_0} \frac{z}{(\mathbf{z} \cdot \mathbf{u})} \left\{ [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})] + \frac{\mathbf{V}}{c} \times [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})] \right\}. \quad (10.19)$$

All of $\mathbf{z}$, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{a}$ are evaluated at the retarded time t_r . Only the velocity $\mathbf{V}$ is evaluated at the coordinate time t .

11 RADIATION

11.1 Power Radiated by a Point Charge

11.1.1 What is Radiation?

Radiation is a process where charges accelerate and transport energy irreversibly out to infinity by their fields. We assume that the charge configuration is localized (not extending to infinity). Consider a large sphere of radius r centered at the origin. The power passing through the surface is

$$P(r, t) = \oint \mathbf{S} \cdot d\mathbf{a} = \frac{1}{\mu_0} \oint (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}.$$

This is the power measured at (r, t) . Because the energy carried by the fields travels at the speed of light, it leaves the source at the retarded time $t_0 = t - r/c$. Hence the power radiated to infinity is

$$P_{\text{rad}}(t_0) = \lim_{r \rightarrow \infty} P\left(r, t_0 + \frac{r}{c}\right).$$

In the static case, both electric and magnetic fields fall off like $1/r^2$ (or faster, if the charge configuration is neutral) according to Coulomb's law and Biot-Savart law. This means the Poynting vector goes like $1/r^4$. Because the surface area of the sphere is $4\pi r^2$, when integrating the surface integral at infinity, the total power radiated is zero. Therefore, static charge configurations do not radiate. Radiation occurs only in electrodynamics. Indeed, according to Jefimenko's equations (10.10), only the terms involving $\dot{\rho}$ and $\dot{\mathbf{J}}$ fall off like $1/r$. They should account for electromagnetic radiation.

11.1.2 The Larmor Formula

From the Liénard-Wiechert potentials, we derived the electric (10.17) and magnetic fields (10.18) of a point charge in arbitrary motion,

$$\mathbf{E}(\mathbf{r}, t) = \frac{q\mathbf{z}}{4\pi\epsilon_0(\mathbf{z} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})], \quad \mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E}(\mathbf{r}, t),$$

where $\mathbf{u} = c\hat{\mathbf{z}} - \mathbf{v}$. The Poynting vector is

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0 c} [\mathbf{E} \times (\hat{\mathbf{z}} \times \mathbf{E})] = \frac{1}{\mu_0 c} [\mathbf{E}^2 \hat{\mathbf{z}} - (\hat{\mathbf{z}} \cdot \mathbf{E}) \mathbf{E}].$$

Not all energy is radiated by the fields to infinity; there are some fields surrounding the moving point charge. Suppose we would like to know the total power radiated by the particle at time t . Draw a large sphere of radius $\mathcal{r}$ centered at $\mathbf{w}(t)$. At some later time, $t_s \equiv t + \mathcal{r}/c$, the radiation reaches the sphere, and then we integrate $\mathbf{S}$ over the surface. For all points on the sphere, $t = t_s - \mathcal{r}/c$ is the retarded time. The electric field takes the form

$$\mathbf{E}(\mathbf{r}, t) = \frac{q\mathbf{z}}{4\pi\epsilon_0(\mathbf{z} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})].$$

Since $\mathbf{S}$ depends on the square of the field, terms that falls off faster than $1/z^2$ in $\mathbf{E}^2$ does not contribute to the flux at infinity. Hence we keep only $1/z^2$ terms in $\mathbf{E}^2$, or equivalently, $1/z$ terms in $\mathbf{E}$ (the radiation field):

$$\mathbf{E}_{\text{rad}} = \frac{qz}{4\pi\epsilon_0(\mathbf{z} \cdot \mathbf{u})^3} [\mathbf{z} \times (\mathbf{u} \times \mathbf{a})], \quad \text{and} \quad \mathbf{S}_{\text{rad}} = \frac{1}{\mu_0 c} \mathbf{E}_{\text{rad}}^2 \hat{\mathbf{z}}. \quad (11.1)$$

Consider the case where the charge is instantaneously at rest at time t but experiences acceleration $\mathbf{a}$. Then $\mathbf{u} = c\hat{\mathbf{z}}$,

$$\mathbf{E}_{\text{rad}} = \frac{q}{4\pi\epsilon_0 c^2 z} [\hat{\mathbf{z}} \times (\hat{\mathbf{z}} \times \mathbf{a})] = \frac{\mu_0 q}{4\pi z} [(\hat{\mathbf{z}} \cdot \mathbf{a})\hat{\mathbf{z}} - \mathbf{a}],$$

and the Poynting vector is

$$\mathbf{S}_{\text{rad}} = \frac{1}{\mu_0 c} \left(\frac{\mu_0 q}{4\pi z} \right)^2 [a^2 - (\hat{\mathbf{z}} \cdot \mathbf{a})^2] \hat{\mathbf{z}} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \left(\frac{\sin^2 \theta}{z^2} \right) \hat{\mathbf{z}},$$

where θ is the angle between $\hat{\mathbf{z}}$ and $\mathbf{a}$. The total power radiated is

$$P(t) = \oint \mathbf{S}_{\text{rad}} \cdot d\mathbf{a} = \frac{\mu_0 q^2 a^2}{16\pi^2 c} \int \frac{\sin^2 \theta}{z^2} z^2 \sin \theta d\theta d\phi = \frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.2)$$

This is called the **Larmor formula**. It is the power radiated by a moving point particle when $\mathbf{v}(t) = 0$. Note that the Larmor formula does not work for general extended objects.

11.1.3 The Liénard Formula

Now we consider the general case where $\mathbf{v} \neq 0$. In this case, the rate at which energy passes through the sphere is not the same as the rate at which that energy left the particle. If dW/dt_s is the rate at which energy passes through the sphere at radius z , then the rate at which energy left the charge is

$$\frac{dW}{dt} = \frac{dW}{dt_s} \frac{\partial t_s}{\partial t}.$$

When we calculated $\partial \mathbf{A}/\partial t$ to get $\mathbf{E}$ in the last Chapter, we found (10.15)

$$\frac{\partial t_r}{\partial t} = \frac{zc}{zc - \mathbf{z} \cdot \mathbf{v}} \implies \frac{\partial t}{\partial t_r} = \frac{\mathbf{z} \cdot \mathbf{u}}{zc}.$$

As mentioned in the last section, we see $t = t_s - z/c$ as the retarded time t_r for radiation, while t_s plays the role of t in (10.15), so

$$\frac{\partial t_s}{\partial t} = \frac{\mathbf{z} \cdot \mathbf{u}}{zc} \implies \frac{dW}{dt} = \frac{\mathbf{z} \cdot \mathbf{u}}{zc} \frac{dW}{dt_s}.$$

From the radiation field (11.1), the power radiated to a patch of area $z^2 d\Omega$, with solid angle $d\Omega = \sin \theta d\theta d\phi$, is

$$\frac{dP}{d\Omega} = \left(\frac{\mathbf{z} \cdot \mathbf{u}}{zc} \right) \frac{1}{\mu_0 c} \mathbf{E}_{\text{rad}}^2 z^2 = \frac{q^2}{16\pi^2 \epsilon_0} \frac{|\hat{\mathbf{z}} \times (\mathbf{u} \times \mathbf{a})|^2}{(\hat{\mathbf{z}} \cdot \mathbf{u})^5}.$$

The total power radiated is

$$P = \int \frac{dP}{d\Omega} \sin \theta d\theta d\phi = \frac{\mu_0 q^2 \gamma^6}{6\pi c} \left(a^2 - \left| \frac{\mathbf{v} \times \mathbf{a}}{c^2} \right|^2 \right), \quad (11.3)$$

where $\gamma \equiv 1/\sqrt{1 - v^2/c^2}$. Equation (11.3) is **Liénard's generalization** of the Larmor formula. It reduces to the Larmor formula when $v \ll c$. A detailed integration process is shown in Appendix A.4.

11.2 The Radiation Reaction

A THEOREMS, PROOFS AND DERIVATIONS

A.1 Theorems

To be written.

A.2 Electromagnetic Waves

A.2.1 Fresnel's Equations

To be written.

A.2.2 Guided Waves

To be written.

A.2.3 TE_{mn} Mode

To be written.

A.3 The Potential Formulation

A.3.1 The Lorenz Gauge

To be written.

A.3.2 Rate of Change of Energy in Potential Form

To be written.

A.4 Liénard's Generalization

To be written.